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Statistical Inference for Linear Stochastic Approximation with Richardson–Romberg Extrapolation

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Abstract

This thesis studies statistical inference for constant-stepsize linear stochastic approximation with Markovian noise and Richardson–Romberg extrapolation. The goal of the work is to describe how confidence statements can be constructed for Polyak–Ruppert averaged iterates when the observations are generated by a dependent Markov chain and the algorithm is run with non-vanishing step sizes. The main tasks are to formulate the linear stochastic approximation model, separate the deterministic and stochastic components of the averaged error, analyze the effect of Richardson–Romberg coupling of the step sizes α and 2α, and connect the resulting distributional approximation with long-run variance estimation.

The methodology combines finite-sample decompositions of linear recursions, Lyapunov stability arguments, Poisson-equation tools for geometrically ergodic Markov chains, martingale approximation, Berry–Esseen type bounds, and overlapping batch-means variance estimation. The empirical part is based on simulations for representative linear stochastic approximation models. These experiments compare ordinary averaging with the Richardson–Romberg estimator and examine the interaction between bias reduction and covariance estimation.

The main result is a non-asymptotic normal approximation for scalar projections of the Richardson–Romberg averaged estimator, together with the identification of the covariance target $\Sigma_\infty = \overline{A}^{-1}\Sigma_\varepsilon^{(M)}\overline{A}^{-\top}$. The analysis shows which terms are cancelled by extrapolation, which residual terms are controlled by mixing and stability, and why burn-in and variance estimation are essential for confidence intervals. The numerical results support the theoretical conclusion that Richardson–Romberg extrapolation can substantially reduce step-size bias while preserving the standard inference target when the covariance estimator is calibrated appropriately.

Contents

Chapter 1

Introduction

The topic of this thesis is statistical inference for Richardson–Romberg averaged constant-stepsize linear stochastic approximation with Markovian noise. The object of study is the Polyak–Ruppert averaged LSA trajectory, while the subject is the distribution of scalar projections of the RR estimator and its covariance normalization. The methods used in the work are deterministic error decompositions, Lyapunov contraction estimates, Markov-chain Poisson equations, martingale Berry–Esseen inequalities, Richardson–Romberg weight analysis, and finite-start burn-in transfer. The problem is relevant because constant-stepsize stochastic approximation is widely used in reinforcement learning and stochastic optimization, but its steady-state bias complicates confidence interval construction under dependent data. The goal is to obtain a non-asymptotic normal approximation for the averaged RR statistic. The main tasks are to identify the leading linearized term, control RR weights and misadjustment terms, transfer the stationary theorem to deterministic starts, and check the practical behavior of the resulting confidence intervals. The main result is a balanced-scale Berry–Esseen bound with covariance target Σ_∞. The thesis is organized as follows: the first chapters develop the deterministic and stationary proof ingredients, the burn-in chapter transfers the result to finite starts, the experiment chapter records numerical diagnostics, and the appendices collect key objects and external inputs.

Stochastic approximation (SA) algorithms are a cornerstone of modern computational statistics, optimization, and reinforcement learning. Introduced by Robbins and Monro (1951), these iterative procedures provide a principled way to find roots of equations or optimize objectives when only noisy observations are available. A particularly important subclass is the *linear stochastic approximation* (LSA) algorithm, which arises naturally in temporal-difference (TD) learning, policy evaluation, and stochastic gradient descent for linear models.

In this work, we study the LSA recursion with a *constant step size* $\alpha > 0$:

$$\theta_k^{(\alpha)} = \theta_{k-1}^{(\alpha)} - \alpha\big\{A(Z_k)\theta_{k-1}^{(\alpha)} - b(Z_k)\big\}, \qquad k \geq 1. \tag{1.1}$$

Here $\{Z_k\}_{k\in\mathbb{N}}$ is a time-homogeneous Markov chain on a measurable space $(\mathsf{Z}, \mathcal{Z})$ with transition kernel Q and unique invariant distribution π. The mappings $A\colon \mathsf{Z} \to \mathbb{R}^{d\times d}$ and $b\colon \mathsf{Z} \to \mathbb{R}^d$ are measurable functions satisfying

$$\overline{A} := \int_{\mathsf{Z}} A(z)\,\mathrm{d}\pi(z), \qquad \bar{b} := \int_{\mathsf{Z}} b(z)\,\mathrm{d}\pi(z).$$

We use the stability convention that $-\overline{A}$ is Hurwitz, equivalently all eigenvalues of $\overline{A}$ have strictly positive real parts. Then the target parameter $\theta^* = \overline{A}^{-1}\bar{b}$ is uniquely defined.

The *Polyak–Ruppert averaging* procedure (Polyak, 1990; Ruppert, 1988) provides an effective variance reduction technique. Given a burn-in period $n_0 \geq 0$, the averaged iterate is defined as

$$\bar{\theta}_n^{(\alpha)} = \frac{1}{n - n_0} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}. \tag{1.2}$$

1.1 Bias of constant step-size iterates

The use of a constant step size $\alpha > 0$ offers several practical advantages: it enables geometrically fast forgetting of the initial condition (Dieuleveut, Durmus, and Bach, 2020) and simplifies hyperparameter tuning compared to diminishing step-size schedules. However, unlike the classical regime $\alpha_k \to 0$ with $\sum_k \alpha_k = \infty$ and $\sum_k \alpha_k^2 < \infty$, a constant step size produces iterates that converge only *in distribution* to a stationary measure Π_α, rather than almost surely to θ^*. The stationary expectation $\mathbb{E}[\theta_\infty^{(\alpha)}]$ is generally *biased* with respect to θ^*, and this bias cannot be eliminated by Polyak–Ruppert averaging alone.

As shown in Levin, Naumov, and Samsonov (2025), the stationary bias has a leading linear term in α:

$$\lim_{n\to\infty} \mathbb{E}[\theta_n^{(\alpha)}] = \theta^* + \alpha\Delta + O(\alpha^{3/2}), \tag{1.3}$$

where

$$\Delta = \overline{A}^{-1} \sum_{k=1}^{\infty} \mathbb{E}\Big[\big\{\mathsf{Q}^k \widetilde{A}(Z_\infty)\big\}\varepsilon(Z_\infty)\Big]$$

depends on the correlation structure of the Markov chain, and $\widetilde{A}(z) = A(z) - \overline{A}$ is the centered matrix-valued function. Under stronger expansion assumptions, the power-series approach of Huo, Chen, and Xie (2024) gives higher-order bias expansions in integer powers of α; in the Levin decomposition, the first misadjustment bias component itself has an $O(\alpha^2)$ remainder after the leading $\alpha\Delta$ term.

1.2 Richardson–Romberg extrapolation

To eliminate the leading $O(\alpha)$ bias term, we employ the *Richardson–Romberg* (RR) *extrapolation* procedure. Two LSA sequences are run *on the same Markov chain trajectory* $\{Z_k\}$ with step sizes α and 2α, and the RR iterate is formed as

$$\overline{\theta}_n^{(\alpha,\mathrm{RR})} = 2\overline{\theta}_n^{(\alpha)} - \overline{\theta}_n^{(2\alpha)}. \tag{1.4}$$

Since both sequences share the same noise realization, the leading bias term $\alpha\Delta$ cancels, leaving a residual bias of order $O(\alpha^{3/2})$ or higher (Levin et al., 2025).

More generally, one can consider the multi-level extrapolation with M step sizes $\mathcal{A} = \{\alpha_1, \ldots, \alpha_M\}$ and coefficients $\{h_m\}$ determined by the Vandermonde system (Huo et al., 2024):

$$\sum_{m=1}^{M} h_m = 1, \qquad \sum_{m=1}^{M} h_m \alpha_m^l = 0, \qquad l = 1, \ldots, M-1,$$

which cancels successive powers in settings where such a power-series expansion is available.

1.3 Problem statement and goals

The high-order moment bounds for the PR-averaged RR iterate $\overline{\theta}_n^{(\alpha,\mathrm{RR})}$ established in Levin et al. (2025) show that the leading error term scales as $\sqrt{\operatorname{Tr}\Sigma_\varepsilon^{(\mathrm{M})}}\, n^{-1/2}$, where $\Sigma_\varepsilon^{(\mathrm{M})}$ is the Markovian noise covariance recorded in Section A. This is the usual parametric $n^{-1/2}$ benchmark for averaged LSA; this thesis does not prove a separate Hájek–Le Cam or minimax lower bound. Berry–Esseen type bounds and bootstrap inference procedures for the *standard* Polyak–Ruppert average $\overline{\theta}_n$ (without extrapolation) under Markovian noise have been obtained in Samsonov, Sheshukova, Moulines, and Naumov (2025).

These results do not directly give the distributional approximation needed for the PR-averaged Richardson–Romberg statistic under Markovian noise. Two distinctions are essential here. First, the stationary theorem is proved for an augmented-chain scalar statistic, where the martingale noise, Poisson boundary term, and Richardson–Romberg misadjustment are controlled separately. Second, the deterministic-start estimator requires an additional burn-in transfer, because the deterministic transient and startup remainders are absent from the stationary augmented-chain theorem.

The main goal of this work is therefore to build a non-asymptotic distributional approximation for scalar projections of the PR-averaged Richardson–Romberg statistic. The stationary result should be read as an $n_0 = 0$ augmented-chain theorem, not as a fixed-α central limit theorem centered exactly at θ^*. Its final deterministic-start consequence is obtained at the balanced triangular-array scale $\alpha_n = cn^{-1/2}$, where the residual RR terms are absorbed into explicit remainders and the covariance target is Σ_∞.

Concretely, we establish:

- A stationary full-window augmented-chain Berry–Esseen assembly for scalar RR statistics, including the martingale approximation, predictable-variance comparison, Poisson remainder, and stationary RR misadjustment. The controlled comparison statistic is

$$S^{\mathrm{RR}}_{n,\mathrm{stat}}(u) = -\frac{u^\top M^{\mathrm{RR}}_n}{\sqrt{n}} + u^\top \mathcal{R}^{\mathrm{RR}}_{n,\mathrm{stat}}, \qquad \mathcal{R}^{\mathrm{RR}}_{n,\mathrm{stat}} = D^{\mathrm{RR}}_{2,n} + R^{\mathrm{mis,RR}}_n.$$

  Its finite-window variance proxy is

$$\sigma^{2,\mathrm{RR}}_n(u) = u^\top \left(\frac{1}{n} \sum_{l=2}^{n-1} \mathcal{Q}^{\mathrm{RR}}_l \Sigma^{(\mathrm{M})}_\varepsilon (\mathcal{Q}^{\mathrm{RR}}_l)^\top \right) u,$$

  and the theorem has the schematic form

$$d_{\mathrm{K}}\left(\frac{S^{\mathrm{RR}}_{n,\mathrm{stat}}(u)}{\sigma^{\mathrm{RR}}_n(u)}, \mathcal{N}(0,1) \right) \le \frac{\mathrm{polylog}(n)}{n^{1/4}} + \frac{\|u^\top \mathcal{R}^{\mathrm{RR}}_{n,\mathrm{stat}}\|_{L^p}}{\sigma^{\mathrm{RR}}_n(u)}.$$

  In the balanced triangular-array specialization $\alpha_n = cn^{-1/2}$, the same stationary theorem identifies

$$\Sigma_\infty = \overline{A}^{-1} \Sigma^{(\mathrm{M})}_\varepsilon \overline{A}^{-\top}, \qquad \sigma^2(u) = u^\top \Sigma_\infty u,$$

  as the covariance target and gives, for every fixed non-degenerate direction u, the stationary $n_0 = 0$ CLT interpretation

$$d_{\mathrm{K}}\left(\frac{S^{\mathrm{RR}}_{n,\mathrm{stat}}(u)}{\sigma(u)}, \mathcal{N}(0,1) \right) \le C(u,c) \frac{\mathrm{polylog}(n)}{n^{1/4}}.$$

- A deterministic-start transfer theorem under mixing-scale burn-in conditions with logarithmic factors, yielding the corresponding balanced-scale bound for the main burned-in statistic. Writing $m = n - n_0$ and

$$\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} = 2\overline{\theta}^{(\alpha)}_{n,n_0} - \overline{\theta}^{(2\alpha)}_{n,n_0}, \qquad T^{\mathrm{RR}}_{n,n_0}(u) = \sqrt{m}\, u^\top \left(\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} - \theta^* \right),$$

  the transfer gives

$$d_{\mathrm{K}}\left(\frac{T^{\mathrm{RR}}_{n,n_0}(u)}{\sigma(u)}, \mathcal{N}(0,1) \right) \le C(u,c,\theta_0) \frac{\mathrm{polylog}(n)}{n^{1/4}}$$

  at $\alpha = cn^{-1/2}$ when

$$n_0 \asymp (\alpha a)^{-1} \log^2 n.$$

  Under the same burn-in window, the final $\sqrt{n}$ statistic satisfies the same polynomial rate up to logarithmic factors.

1.4 Setting and assumptions

We now formalize the setting and state the assumptions that will be used throughout this work.

Let $\{Z_k\}_{k\in\mathbb{N}}$ be a Markov chain on a complete separable metric space $(\mathsf{Z},\mathcal{Z})$ with transition kernel Q.

Assumption 1 (Uniform geometric ergodicity)**.** The kernel Q admits a unique invariant distribution π and is *uniformly geometrically ergodic*: there exists $t_{\mathrm{mix}} \in \mathbb{N}^*$ such that for all $k \in \mathbb{N}^*$,

$$\Delta(\mathsf{Q}^k) := \sup_{z,z'\in\mathsf{Z}} \frac{1}{2}\big\|\mathsf{Q}^k(z,\cdot) - \mathsf{Q}^k(z',\cdot)\big\|_{\mathrm{TV}} \le \left(\frac{1}{4}\right)^{\lfloor k/t_{\mathrm{mix}}\rfloor}.$$

Equivalently, there exist constants $\zeta > 0$ and $\rho \in (0,1)$ such that

$$\sup_z \big\|\mathsf{Q}^k(z,\cdot) - \pi\big\|_{\mathrm{TV}} \le \zeta\rho^k$$

for all $k \ge 1$.

Assumption 2 (Hurwitz condition and boundedness)**.** The matrix $-\overline{A}$ is Hurwitz; equivalently, all eigenvalues of $\overline{A}$ have strictly positive real parts. Moreover,

$$C_A := \max\left\{\sup_{z\in\mathsf{Z}}\|A(z)\|, \sup_{z\in\mathsf{Z}}\|\widetilde{A}(z)\|\right\} < \infty,$$

where $\widetilde{A}(z) := A(z) - \overline{A}$.

Assumption 3 (Noise regularity)**.** The noise function $\varepsilon(z) = \widetilde{A}(z)\theta^* - \widetilde{b}(z)$, where $\widetilde{b}(z) = b(z) - \bar{b}$, satisfies

$$\|\varepsilon\|_\infty := \sup_{z\in\mathsf{Z}}\|\varepsilon(z)\| < +\infty.$$

By construction, $\pi(\widetilde{A}) = 0$, $\pi(\widetilde{b}) = 0$, and hence $\pi(\varepsilon) = 0$.

Under Assumptions 1–3, the error $\theta_k^{(\alpha)} - \theta^*$ satisfies the recursion

$$\theta_k^{(\alpha)} - \theta^* = \big(I - \alpha A(Z_k)\big)\big(\theta_{k-1}^{(\alpha)} - \theta^*\big) - \alpha\varepsilon(Z_k). \tag{1.5}$$

1.5 Chapter Summary and Results

The chapter fixes the statistical problem, the Markovian LSA model, the constant-stepsize convention, and the Richardson–Romberg estimator used in the rest of the thesis. It also states the main assumptions and explains why the target is a non-asymptotic distributional approximation for scalar projections of the PR-averaged RR statistic.

Chapter 2

Zeroth-Order Richardson–Romberg Difference

2.1 LSA Error Decomposition

This chapter is a preliminary deterministic-product calculation for the zero-order RR difference. The final Berry–Esseen assembly uses the full PR-weight notation and hypotheses introduced in Chapter 4; the present chapter mainly fixes notation and explains the cancellation mechanism in a simpler terminal-iterate setting.

We consider the recursion $\theta_k = \theta_{k-1} - \alpha_k(A(Z_k)\theta_{k-1} - b(Z_k))$ with $\alpha_k = \alpha$ constant.

Define the transition products

$$\Gamma_{m:k} = \prod_{l=m}^{k} \big(I - \alpha A(Z_l)\big).$$

Write $B_\alpha := I - \alpha\overline{A}$ and introduce the deterministic-product linearized term

$$J_k^{(0,\alpha)} = -\alpha \sum_{l=1}^{k} B_\alpha^{k-l}\varepsilon(Z_l), \qquad J_0^{(0,\alpha)} = 0.$$

The difference between the exact random-product expansion and this deterministic-product term is denoted by $R_k^{(\alpha)}$:

$$\theta_k^{(\alpha)} - \theta^* = J_k^{(0,\alpha)} + B_\alpha^k(\theta_0 - \theta^*) + R_k^{(\alpha)}.$$

This convention matches the weight decomposition in Section 4.1.

A standard PR-averaged decomposition (cf. Chapter 4 for the full derivation) yields $\sqrt{n}(\bar\theta_n^{(\alpha)} - \theta^*) = W + D_1$, where the leading martingale-like term is

$$W = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l \varepsilon(Z_l), \qquad Q_l = \alpha \sum_{k=l}^{n-1} B_\alpha^{k-l}.$$

The residual term is

$$D_1 = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} B_\alpha^k(\theta_0 - \theta^*) + \frac{1}{\sqrt{n}} \sum_{k=1}^{n-1} R_k^{(\alpha)}.$$

2.2 Terminal RR Combination and the $\Delta J_n^{(0,\alpha)}$ Term

Applying the deterministic-product decomposition of the previous subsection separately to step sizes α and 2α, and writing $B_{2\alpha} := I - 2\alpha\overline{A}$, define the chapter-local terminal Richardson–Romberg object

$$\theta_n^{(\mathrm{tRR},\alpha)} := 2\theta_n^{(\alpha)} - \theta_n^{(2\alpha)}.$$

It is not the PR-averaged RR estimator $\overline{\theta}_n^{(\mathrm{RR},\alpha)}$ studied in Chapters 4–5. Its deterministic-product decomposition has the form

$$\begin{aligned}\theta_n^{(\mathrm{tRR},\alpha)} - \theta^* = \big[2B_\alpha^n - B_{2\alpha}^n\big](\theta_0 - \theta^*) \\ + \big[2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)}\big] + \big[2R_n^{(\alpha)} - R_n^{(2\alpha)}\big].\end{aligned}$$

The first bracket is the deterministic transient in this convention. The second bracket is the linearized stochastic RR difference, and the final one is the higher-order random-product remainder. In this subsection we focus on the zeroth-order RR difference, which we denote

$$\Delta J_n^{(0,\alpha)} := 2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)},$$

where, by definition,

$$J_n^{(0,\alpha)} = -\alpha \sum_{j=1}^{n} (I - \alpha\overline{A})^{n-j} \varepsilon(Z_j).$$

Substituting and using the elementary identity $X^m - Y^m = (X - Y)\sum_{i=1}^{m} X^{i-1}Y^{m-i}$ with $X = B_\alpha$, $Y = B_{2\alpha}$, and $X - Y = \alpha\overline{A}$, define the deterministic kernel

$$H_j^{(n)} := \sum_{i=1}^{n-j} B_\alpha^{i-1} B_{2\alpha}^{n-j-i}, \qquad 1 \le j \le n-1,$$

and set $H_n^{(n)} := 0$. Then

$$\begin{aligned}\Delta J_n^{(0,\alpha)} &= -2\alpha \sum_{j=1}^{n} \big(B_\alpha^{n-j} - B_{2\alpha}^{n-j}\big)\varepsilon(Z_j) \\ &= -2\alpha^2 \overline{A} \sum_{j=1}^{n-1} H_j^{(n)} \varepsilon(Z_j).\end{aligned}$$

The left-factored matrix-power identity is legitimate here because $I - \alpha\overline{A}$, $I - 2\alpha\overline{A}$, and $\overline{A}$ are polynomials in the same matrix $\overline{A}$ and therefore commute. The extra factor $\alpha\overline{A}$ pulled out front is the source of the additional α-decay of the RR difference compared to a single LSA trajectory.

2.3 Norm Estimate for $H_j^{(n)}$

To bound $\|H_j^{(n)}\|$ we use the following standard result.

Lemma 2.1 (Lyapunov contraction). *Let $-\overline{A}$ be a Hurwitz matrix. Then for any $P = P^\top \succ 0$ there exists a unique $Q = Q^\top \succ 0$ satisfying*

$$\overline{A}^\top Q + Q\overline{A} = P.$$

Let

$$a := \frac{\lambda_{\min}(P)}{2\|Q\|}, \qquad \kappa_Q := \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)},$$

and

$$\alpha_\infty := \min\left\{ \frac{\lambda_{\min}(P)}{2\kappa_Q \|\overline{A}\|_Q^2}, \frac{\|Q\|}{\lambda_{\min}(P)} \right\}.$$

Then for all $\alpha \in [0, \alpha_\infty]$,

$$\alpha a \le \frac{1}{2}, \qquad \|I - \alpha\overline{A}\|_Q^2 \le 1 - \alpha a.$$

In the RR estimates below we assume explicitly that $0 < \alpha$ and $2\alpha \le \alpha_\infty$. Hence the Lyapunov contraction applies at both step sizes α and 2α.

For $1 \le j \le n-1$, combine the triangle inequality, submultiplicativity, the norm equivalence, and the Lyapunov contraction at step sizes α and 2α:

$$\begin{aligned}\|H_j^{(n)}\| &\le \kappa_Q \sum_{i=1}^{n-j} (1-\alpha a)^{(i-1)/2} (1-2\alpha a)^{(n-j-i)/2} \\ &\le \kappa_Q (1-\alpha a)^{(n-j-1)/2} \frac{1}{1-\sqrt{(1-2\alpha a)/(1-\alpha a)}}.\end{aligned}$$

For $\alpha a \le 1/2$ the geometric rate is bounded below by an *elementary* inequality: combining

$$\sqrt{\frac{1-2\alpha a}{1-\alpha a}} \le \sqrt{1-\alpha a}$$

(since $1-2\alpha a \le (1-\alpha a)^2$) with $1-\sqrt{1-x} \ge x/2$ on $[0,1]$ gives

$$1-\sqrt{\frac{1-2\alpha a}{1-\alpha a}} \ge \frac{\alpha a}{2}, \qquad \frac{1}{1-\sqrt{(1-2\alpha a)/(1-\alpha a)}} \le \frac{2}{\alpha a}.$$

Defining the chapter-local norm-equivalence constant $K_Q := \kappa_Q$ (distinct from the Assumption 2 sup-norm constant C_A, which enters separately via $\|\overline{A}\|$ in the next subsection), we arrive at the final estimate

$$\|H_j^{(n)}\| \le K_Q (1-\alpha a)^{(n-j-1)/2} \frac{2}{\alpha a}.$$

The kernel decays geometrically in $n-j$ at rate $\sqrt{1-\alpha a}$ but its summed weight is of order $1/(\alpha a)$. The next subsection keeps the remaining powers of a explicit when this kernel is multiplied by the prefactor α^2 in $\Delta J_n^{(0,\alpha)}$.

2.4 Scalar L^p Bound for the Zeroth-Order Term

The expression

$$\Delta J_n^{(0,\alpha)} = -\sum_{j=1}^{n-1} 2\alpha^2 \overline{A} H_j^{(n)} \varepsilon(Z_j)$$

is a weighted sum of values of the centered noise function ε along the Markov chain. In the Berry–Esseen argument below only fixed scalar projections are used, so we state the concentration input in scalar form.

Lemma 2.2 (Weighted Markov concentration; Levin et al. (2025, Lemma 11))**.** *Assume **UGE 1**. Let $\{g_i\}_{i=1}^n$ be a family of measurable functions $g_i\colon \mathsf{Z} \to \mathbb{R}$ such that $c_i = \|g_i\|_\infty < \infty$ and $\pi(g_i) = 0$ for $i \in \{1,\dots,n\}$. Then, for any initial distribution ξ on $(\mathsf{Z}, \mathcal{Z})$, any $n \in \mathbb{N}$, and any $p \ge 2$,*

$$\left\| \sum_{i=1}^n g_i(Z_i) \right\|_{L^p(\xi)} \le C_{\mathrm{MC}} \sqrt{p\, t_{\mathrm{mix}} \sum_{i=1}^n c_i^2}. \tag{2.1}$$

Fix a deterministic direction $u \in \mathbb{R}^d$. We apply the lemma with

$$g_j^u(z) = -2\alpha^2 u^\top \overline{A} H_j^{(n)} \varepsilon(z), \qquad 1 \le j \le n-1,$$

and $g_n^u = 0$. Each g_j^u is centered under π since $\pi(\varepsilon) = 0$, so the centering hypothesis holds. Combining the bound $\|\overline{A}\| \le C_A$ (Assumption 2) with the previous subsection, define the constant $K_{A,Q} := C_A K_Q = C_A \kappa_Q$. Then the per-summand bound is

$$\|g_j^u\|_\infty \le \frac{4\alpha \|u\| K_{A,Q} \|\varepsilon\|_\infty}{a} (1-\alpha a)^{(n-j-1)/2}, \qquad 1 \le j \le n-1.$$

Note the prefactor α^2 collapsing to α/a: the $1/(\alpha a)$ blow-up in $H_j^{(n)}$ is absorbed only by one factor of α. Squaring and summing over j:

$$\begin{aligned}\sum_{j=1}^{n}\|g_j^u\|_\infty^2 &= \sum_{j=1}^{n-1}\|g_j^u\|_\infty^2 \\ &\le \frac{16\alpha^2\|u\|^2K_{A,Q}^2\|\varepsilon\|_\infty^2}{a^2}\sum_{j=1}^{n-1}(1-\alpha a)^{n-j-1} \\ &\le \frac{16\alpha\|u\|^2K_{A,Q}^2\|\varepsilon\|_\infty^2}{a^3}.\end{aligned}$$

Plugging this into the Markov concentration input (2.1) and defining

$$K_{\mathrm{RR},0} := 32K_{A,Q}\|\varepsilon\|_\infty\frac{\sqrt{t_{\mathrm{mix}}}}{a^{3/2}},$$

gives

$$\sqrt{p\,t_{\mathrm{mix}}\sum_{j=1}^{n}\|g_j^u\|_\infty^2} \le \sqrt{p}\,\|u\|K_{\mathrm{RR},0}\sqrt{\alpha}.$$

Applying (2.1) with $X = u^\top\Delta J_n^{(0,\alpha)}$ gives, for any $p \ge 2$,

$$\mathbb{E}^{1/p}\left[|u^\top\Delta J_n^{(0,\alpha)}|^p\right] \le C_{\mathrm{MC}}\sqrt{p}\,\|u\|K_{\mathrm{RR},0}\sqrt{\alpha},$$

or equivalently

$$\frac{1}{\sqrt{\alpha}}\left\|u^\top\Delta J_n^{(0,\alpha)}\right\|_{L^p} \le C_{\mathrm{MC}}\sqrt{p}\,\|u\|K_{\mathrm{RR},0}.$$

Thus every fixed scalar projection of the terminal-iterate zeroth-order RR difference is $O(\sqrt{\alpha})$ in L^p, uniformly in n. A Euclidean-norm bound can be recovered in fixed dimension by applying the scalar estimate to a coordinate basis, but no dimension-free vector concentration is claimed here.

2.5 Chapter Summary and Results

The chapter isolates the deterministic-product zeroth-order RR cancellation mechanism. The main output is a terminal-iterate decomposition, a geometric kernel bound for $H_j^{(n)}$, and a scalar L^p estimate showing the extra $\sqrt{\alpha}$-scale decay of the leading RR difference.

Chapter 3

Last Iterate Analysis

3.1 Centered Bound for the Shifted First-Order Perturbation

Let

$$B = I - \alpha \overline{A}. \tag{3.1}$$

Assume throughout that $0 < \alpha \le \alpha_\infty$. Then

$$\|B^m\|_Q \le (1 - \alpha a)^{m/2}, \quad m \ge 0. \tag{3.2}$$

Set also

$$\alpha_{\mathrm{inv}} := \frac{1}{2\|\overline{A}\|}. \tag{3.3}$$

If $w \le \alpha_{\mathrm{inv}}$, the Neumann series gives $\|(I - w\overline{A})^{-1}\| \le 2$. In Richardson–Romberg applications the single-stepsize estimates are applied at $w \in \{\alpha, 2\alpha\}$, so the local assumptions are $2\alpha \le \alpha_\infty$ and $2\alpha \le \alpha_{\mathrm{inv}}$. Constants denoted by C may depend on fixed problem parameters, but not on α, n, p, or q.

The deterministic-product components are defined by

$$J_n^{(0,\alpha)} = B J_{n-1}^{(0,\alpha)} - \alpha \varepsilon(Z_n), \quad J_0^{(0,\alpha)} = 0, \tag{3.4}$$

and

$$J_n^{(1,\alpha)} = B J_{n-1}^{(1,\alpha)} - \alpha \widetilde{A}(Z_n) J_{n-1}^{(0,\alpha)}, \quad J_0^{(1,\alpha)} = 0. \tag{3.5}$$

Hence

$$J_n^{(0,\alpha)} = -\alpha \sum_{k=1}^{n} B^{n-k} \varepsilon(Z_k), \tag{3.6}$$

and

$$\begin{aligned} J_n^{(1,\alpha)} &= -\alpha \sum_{j=1}^{n} B^{n-j} \widetilde{A}(Z_j) J_{j-1}^{(0,\alpha)} \\ &= \alpha^2 \sum_{k=1}^{n-1} \sum_{r=1}^{n-k} B^{n-k-r} \widetilde{A}(Z_{k+r}) B^{r-1} \varepsilon(Z_k). \end{aligned} \tag{3.7}$$

Use the shifted version

$$S_n = \sum_{t=0}^{n-1} B^{n-t} \widetilde{A}(Z_{t+1}) J_t^{(0,\alpha)}. \tag{3.8}$$

Then

$$S_n = \sum_{j=1}^{n} B^{n-j+1} \widetilde{A}(Z_j) J_{j-1}^{(0,\alpha)} = -\frac{1}{\alpha} B\, J_n^{(1,\alpha)}. \tag{3.9}$$

Thus the corresponding shifted first-order contribution is

$$T_n^{(1,\alpha)} = -\alpha S_n = B\, J_n^{(1,\alpha)}, \tag{3.10}$$

so passing from $T_n^{(1,\alpha)}$ back to $J_n^{(1,\alpha)}$ uses the local bound on B^{-1}.

Lemma 3.1. ***(Samsonov et al., 2025, Appendix D.2, Proposition 9)*** *Assume UGE 1 and stationarity. Let $\mathcal{F}_k = \sigma(Z_0, \dots, Z_k)$. For $1 \le k \le n-1$ and $1 \le l \le n-k$, let $g_{k,l} : \mathcal{Z} \to \mathbb{R}^d$ be deterministic functions satisfying $\pi(g_{k,l}) = 0$ and $\|g_{k,l}\|_\infty \le \beta_{k,l}$. Let ξ_k be $\mathcal{F}_k$-measurable vectors with $\sup_k \|\xi_k\|_\infty \le M_\xi$. Then, for every $p \ge 2$,*

$$\|\sum_{k=1}^{n-1} \left\{ \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l})^\top \xi_k - \mathbb{E}\left[\sum_{l=1}^{n-k} g_{k,l}(Z_{k+l})^\top \xi_k \,|\, \mathcal{F}_k \right] \right\} \|_{L_p} \qquad \le C p^{3/2} t_{mix}^{1/2} M_\xi \left(\sum_{k=1}^{n-1} \sum_{l=1}^{n-k} \beta_{k,l}^2 \right)^{1/2}. \tag{3.11}$$

Lemma 3.2. ***(Centered shifted first-order bound.)*** *Assume the Markov chain is started from stationarity, that is, the law of Z_1 is π, and $\pi(\widetilde{A}) = 0$. For every deterministic direction $u \in \mathbb{R}^d$ and every $p \ge 2$,*

$$\|u^\top (S_n - \mathbb{E} S_n)\|_{L_p} \le C \|u\| \, \|\varepsilon\|_\infty (p^{3/2} t_{mix}^{1/2} \frac{1}{a} + p^{1/2} t_{mix}^{3/2} \sqrt{\frac{\alpha}{a}}). \tag{3.12}$$

Proof. Since $J_0^{(0,\alpha)} = 0$, the term $t = 0$ in S_n vanishes. Substituting the explicit formula for $J_t^{(0,\alpha)}$ gives

$$S_n = -\alpha \sum_{t=1}^{n-1} \sum_{k=1}^{t} B^{n-t} \widetilde{A}(Z_{t+1}) B^{t-k} \varepsilon(Z_k) \quad = -\alpha \sum_{k=1}^{n-1} H_{k+1}^{(w)} \varepsilon(Z_k), \tag{3.13}$$

where, after the change of summation index $l = t - k + 1$,

$$H_{k+1}^{(w)} = \sum_{l=1}^{n-k} B^{n-k-l+1} \widetilde{A}(Z_{k+l}) B^{l-1}. \tag{3.14}$$

Define

$$\mu_k^{(w)} = \mathbb{E}_\pi \left[H_{k+1}^{(w)} \varepsilon(Z_k) \right]. \tag{3.15}$$

Then

$$S_n - \mathbb{E} S_n = -\alpha \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \varepsilon(Z_k) - \mu_k^{(w)} \right\}. \tag{3.16}$$

Let $\mathcal{F}_k = \sigma(Z_0, \dots, Z_k)$. The Markov property gives

$$\mathbb{E}[H_{k+1}^{(w)} \varepsilon(Z_k) | \mathcal{F}_k] = v_k^{(w,\epsilon)}(Z_k), \tag{3.17}$$

where

$$v_k^{(w)}(z) = \sum_{l=1}^{n-k} B^{n-k-l+1} (\mathsf{Q}^l \widetilde{A})(z) B^{l-1}, \quad v_k^{(w,\epsilon)}(z) = v_k^{(w)}(z) \varepsilon(z). \tag{3.18}$$

Here Q is the Markov transition kernel, so

$$(\mathsf{Q}^l \widetilde{A})(z) = \int \widetilde{A}(u)\, \mathsf{Q}^l(z, du) = \mathbb{E}\left[\widetilde{A}(Z_{k+l}) | Z_k = z \right]. \tag{3.19}$$

Under stationarity, $\mu_k^{(w)} = \pi(v_k^{(w,\epsilon)})$. Therefore

$$S_n - \mathbb{E} S_n = -\alpha (U_M + U_R), \tag{3.20}$$

with

$$U_M = \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \varepsilon(Z_k) - v_k^{(w,\epsilon)}(Z_k) \right\} \tag{3.21}$$

and

$$U_R = \sum_{k=1}^{n-1} \left\{ v_k^{(w,\epsilon)}(Z_k) - \pi(v_k^{(w,\epsilon)}) \right\}. \tag{3.22}$$

Fix a deterministic direction u, $\|u\| = 1$; the general statement follows by multiplying by $\|u\|$. For U_R, since $\pi(\widetilde{A}) = 0$, UGE gives

$$\|(\mathsf{Q}^l \widetilde{A})(z)\| = \left(\| \int \widetilde{A}(y) \left\{ \mathsf{Q}^l(z, dy) - \pi(dy) \right\} \| \right) \le 2C_A \Delta(\mathsf{Q}^l) \le 2C_A (1/4)^{\lfloor l/t_{\mathrm{mix}} \rfloor}. \tag{3.23}$$

Hence

$$\|v_k^{(w)}\|_\infty \le C \sum_{l=1}^{n-k} (1 - \alpha a)^{(n-k-l+1)/2} \Delta(\mathsf{Q}^l)(1 - \alpha a)^{(l-1)/2} \quad \le C t_{\mathrm{mix}} (1 - \alpha a)^{(n-k)/2}. \tag{3.24}$$

Thus

$$\|v_k^{(w,\epsilon)}\|_\infty \le C t_{\mathrm{mix}} \|\varepsilon\|_\infty (1 - \alpha a)^{(n-k)/2}. \tag{3.25}$$

Applying Markov concentration to the centered functions $v_k^{(w,\epsilon)} - \pi(v_k^{(w,\epsilon)})$ gives

$$\|u^\top U_R\|_{L_p} \le C p^{1/2} t_{\mathrm{mix}}^{1/2} \left(\sum_{k=1}^{n-1} \|v_k^{(w,\epsilon)}\|_\infty^2 \right)^{1/2} \quad \le C p^{1/2} t_{\mathrm{mix}}^{3/2} \|\varepsilon\|_\infty \frac{1}{\sqrt{\alpha a}}. \tag{3.26}$$

For U_M, unfold the projected matrix kernel through $(H_{k+1}^{(w)})^\top u$:

$$(H_{k+1}^{(w)})^\top u = \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l}), \quad g_{k,l}(z) = (B^{l-1})^\top \, \widetilde{A}(z)^\top \, (B^{n-k-l+1})^\top u. \tag{3.27}$$

Each $g_{k,l}$ is centered under π, and

$$\|g_{k,l}\|_\infty \le C(1 - \alpha a)^{(n-k)/2}, \tag{3.28}$$

$$\sum_{l=1}^{n-k} \|g_{k,l}\|_\infty^2 \le C(n-k)(1 - \alpha a)^{n-k}. \tag{3.29}$$

The conditional expectation in the future-centered estimate is exactly

$$\mathbb{E}\left[\sum_{l=1}^{n-k} g_{k,l}(Z_{k+l})^\top \varepsilon(Z_k) \,|\, \mathcal{F}_k \right] = u^\top v_k^{(w,\epsilon)}(Z_k). \tag{3.30}$$

Apply that estimate with $\xi_k = \varepsilon(Z_k)$ and $\beta_{k,l} = C(1 - \alpha a)^{(n-k)/2}$:

$$\|u^\top U_M\|_{L_p} \le C p^{3/2} t_{\mathrm{mix}}^{1/2} \|\varepsilon\|_\infty \left(\sum_{k=1}^{n-1} \sum_{l=1}^{n-k} (1 - \alpha a)^{n-k} \right)^{1/2} \quad \le C p^{3/2} t_{\mathrm{mix}}^{1/2} \|\varepsilon\|_\infty \frac{1}{\alpha a}. \tag{3.31}$$

$$\|u^\top (S_n - \mathbb{E} S_n)\|_{L_p} \le \alpha \left(\|u^\top U_M\|_{L_p} + \|u^\top U_R\|_{L_p} \right) \quad \le C \|\varepsilon\|_\infty \left(p^{3/2} t_{\mathrm{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\mathrm{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right). \tag{3.32}$$

3.2 Chapter Summary and Results

The chapter proves a centered bound for the shifted first-order perturbation and records how it applies to PR-averaged RR misadjustment terms. It also identifies the limitation of a single-α estimate, motivating the separate RR weight analysis developed in the following chapter.

Chapter 4

Richardson–Romberg PR Weight Bounds

Throughout this chapter, unnamed constants C may change from line to line but do not hide dependence on n, α, p, or q.

4.1 Stationary Small-Step Convention

We collect the stationary small-step restrictions used in this chapter. The individual ceilings are stated in Section B.

For stationary depth-two arguments define

$$\begin{aligned}\alpha_*(q, t_{\mathrm{mix}}) := \min\bigl\{&\alpha_{\mathrm{L,P2}}(q, t_{\mathrm{mix}}), \alpha_{\mathrm{L,C6}}(q, t_{\mathrm{mix}}), \alpha_{\mathrm{L,P5}}(q, t_{\mathrm{mix}}),\\ &\alpha_{\mathrm{L,P8}}(q, t_{\mathrm{mix}}), \alpha_{\mathrm{L,P9}}(q, t_{\mathrm{mix}}), \alpha_{\mathrm{L,inv}}(q, t_{\mathrm{mix}})\bigr\}.\end{aligned} \tag{4.1}$$

For the shifted-to-unshifted first-order transfer we also use the local inverse ceiling

$$\alpha_{\mathrm{inv}} := \frac{1}{2\|\overline{A}\|}. \tag{4.2}$$

If $w \le \alpha_{\mathrm{inv}}$, the Neumann series yields $\|(I - w\overline{A})^{-1}\| \le 2$.

Define the stationary admissibility threshold

$$\alpha_{\mathrm{stat}}(q) := \min(\alpha_\infty, \alpha_{\mathrm{inv}}, \alpha_*(q, t_{\mathrm{mix}})). \tag{4.3}$$

Throughout this chapter, the small-step condition is $2\alpha \le \alpha_{\mathrm{stat}}(q)$.

4.2 PR-Averaged Error Decomposition and the RR Weight

We start from the full-window PR weight representation of the RR average.

Depth-one expansion. Unfolding the error recursion of Chapter 1 gives, for every $k \ge 1$,

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^{k} \Gamma_{l+1:k}^{(\alpha)} \varepsilon(Z_l) + \Gamma_{1:k}^{(\alpha)}(\theta_0 - \theta^*), \quad \Gamma_{l+1:k}^{(\alpha)} := \prod_{j=l+1}^{k} (I - \alpha A(Z_j)). \tag{4.4}$$

Replacing the random products in the noise sum by their deterministic counterparts $B_\alpha^{k-l} := (I - \alpha\overline{A})^{k-l}$ and keeping the initial-product discrepancy separate yields the **depth-one decomposition**:

$$R_{k,\mathrm{init}}^{(\alpha)} := \left(\Gamma_{1:k}^{(\alpha)} - B_\alpha^k\right)(\theta_0 - \theta^*), \tag{4.5}$$

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^{k} B_\alpha^{k-l} \varepsilon(Z_l) + B_\alpha^k(\theta_0 - \theta^*) + R_{k,\mathrm{init}}^{(\alpha)} + R_k^{(\alpha)}, \tag{4.6}$$

where $R_k^{(\alpha)} := J_k^{(1,\alpha)} + H_k^{(1,\alpha)}$ is the first noise-driven misadjustment remainder.

PR averaging produces $Q_l^{(\alpha)}$. Recall the PR average $\overline{\theta}_n^{(\alpha)} = (n-n_0)^{-1}\sum_{k=n_0}^{n-1}\theta_k^{(\alpha)}$. For the stationary result in this chapter, set $n_0 = 0$. A burned-in average uses the different deterministic weight

$$Q_{l,n_0}^{(\alpha)} = \frac{n}{n-n_0}\alpha \sum_{k=\max(n_0,l)}^{n-1} B_\alpha^{k-l}, \tag{4.7}$$

when the statistic is normalized as $-n^{-1/2}\sum_l Q_{l,n_0}^{(\alpha)}\varepsilon(Z_l)$. Subtracting θ^*, substituting the depth-one decomposition above, and exchanging the order of summation in the depth-zero piece,

$$\sum_{k=0}^{n-1}\sum_{l=1}^{k} B_\alpha^{k-l}\varepsilon(Z_l) = \sum_{l=1}^{n-1}\left(\sum_{k=l}^{n-1} B_\alpha^{k-l}\right)\varepsilon(Z_l), \tag{4.8}$$

gives the deterministic PR weight

$$Q_l^{(\alpha)} := \alpha\sum_{k=l}^{n-1} B_\alpha^{k-l} = \alpha\sum_{j=0}^{n-l-1} B_\alpha^{j}. \tag{4.9}$$

Multiplying by $\sqrt{n}$ gives

$$\sqrt{n}\,(\overline{\theta}_n^{(\alpha)} - \theta^*) = W^{(\alpha)} + D^{(\alpha)}, \tag{4.10}$$

where

$$W^{(\alpha)} := -\frac{1}{\sqrt{n}}\sum_{l=1}^{n-1} Q_l^{(\alpha)}\,\varepsilon(Z_l), \tag{4.11}$$

and

$$\begin{gathered} D^{(\alpha)} := D_{\mathrm{tr}}^{(\alpha)} + D_{\mathrm{init}}^{(\alpha)} + D_R^{(\alpha)}, \\ D_{\mathrm{tr}}^{(\alpha)} := \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1} B_\alpha^k(\theta_0-\theta^*), \quad D_{\mathrm{init}}^{(\alpha)} := \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1} R_{k,\mathrm{init}}^{(\alpha)}, \quad D_R^{(\alpha)} := \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1} R_k^{(\alpha)}. \end{gathered} \tag{4.12}$$

The RR combination of the noise-driven part is the misadjustment controlled below by the Levin depth-two transfer.

RR combination produces $\mathcal{Q}_l^{\mathrm{RR}}$. By linearity,

$$\sqrt{n}\,(\overline{\theta}_n^{(\mathrm{RR},\alpha)} - \theta^*) = W^{\mathrm{RR}} + D^{\mathrm{RR}}, \tag{4.13}$$

$$W^{\mathrm{RR}} := 2W^{(\alpha)} - W^{(2\alpha)} = -\frac{1}{\sqrt{n}}\sum_{l=1}^{n-1}\mathcal{Q}_l^{\mathrm{RR}}\,\varepsilon(Z_l), \tag{4.14}$$

where the **RR weight** is

$$\mathcal{Q}_l^{\mathrm{RR}} := 2Q_l^{(\alpha)} - Q_l^{(2\alpha)}. \tag{4.15}$$

4.3 Closed-Form Identities

Throughout this chapter write

$$B_\alpha := I - \alpha\overline{A}, \quad B_{2\alpha} := I - 2\alpha\overline{A}, \quad k := n-l. \tag{4.16}$$

Assume $0 < \alpha$ and $2\alpha \le \alpha_\infty$.

The geometric-series identity gives

$$Q_l^{(\alpha)} = \alpha\sum_{j=0}^{n-l-1} B_\alpha^j = \alpha\,(\alpha\overline{A})^{-1}\,(I - B_\alpha^{n-l}) = \overline{A}^{-1}(I - B_\alpha^{n-l}). \tag{4.17}$$

Hence

$$Q_l^{(\alpha)} - \overline{A}^{-1} = -\overline{A}^{-1} B_\alpha^{n-l}, \quad Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{n-l-1}. \tag{4.18}$$

Applying the identities at α and 2α gives

$$\mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1} = -\overline{A}^{-1}\,(2B_\alpha^k - B_{2\alpha}^k), \tag{4.19}$$

$$\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}} = -2\alpha\,(B_\alpha^{k-1} - B_{2\alpha}^{k-1}). \tag{4.20}$$

Only the discrete derivative gains an additional local factor of α.

4.4 Pointwise Bounds for the RR Weights

Lemma 4.1. ***(Pointwise RR weight bounds.)*** *Let $0 < \alpha$ and $2\alpha \le \alpha_\infty$, set $C_Q := \kappa_Q^{1/2}\|\overline{A}^{-1}\|$ and $\widetilde{C}_A := \kappa_Q\|\overline{A}\|$, and write $k = n - l$.*

(i) *For every $1 \le l \le n-1$,*

$$\|\mathcal{Q}_l^{RR} - \overline{A}^{-1}\| \le 3C_Q(1-\alpha a)^{k/2}. \tag{4.21}$$

(ii) *For every $1 \le l \le n-2$,*

$$\|\mathcal{Q}_{l+1}^{RR} - \mathcal{Q}_l^{RR}\| \le 2\widetilde{C}_A\,\alpha^2\,(k-1)\,(1-\alpha a)^{(k-2)/2}. \tag{4.22}$$

Proof of (i). From the RR identity, norm equivalence, and Lyapunov contraction,

$$\|2B_\alpha^k - B_{2\alpha}^k\|_Q \le 2\|B_\alpha^k\|_Q + \|B_{2\alpha}^k\|_Q \le 2(1-\alpha a)^{k/2} + (1-2\alpha a)^{k/2} \le 3(1-\alpha a)^{k/2}. \tag{4.23}$$

Thus

$$\|\mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1}\| \le \kappa_Q^{1/2}\|\overline{A}^{-1}\| \cdot \|2B_\alpha^k - B_{2\alpha}^k\|_Q \le 3C_Q(1-\alpha a)^{k/2}. \tag{4.24}$$

Proof of (ii). Use

$$X^m - Y^m = (X-Y)\sum_{i=1}^{m} X^{i-1}Y^{m-i} \tag{4.25}$$

with $X = B_\alpha$, $Y = B_{2\alpha}$, and $m = k-1$:

$$B_\alpha^{k-1} - B_{2\alpha}^{k-1} = \alpha\overline{A}\sum_{i=1}^{k-1} B_\alpha^{i-1}\,B_{2\alpha}^{k-1-i}. \tag{4.26}$$

Each summand satisfies

$$\|B_\alpha^{i-1}\,B_{2\alpha}^{k-1-i}\| \le \kappa_Q^{1/2}\,(1-\alpha a)^{(i-1)/2}(1-2\alpha a)^{(k-1-i)/2} \quad \le \kappa_Q^{1/2}\,(1-\alpha a)^{(k-2)/2}, \tag{4.27}$$

Summing over i gives

$$\|B_\alpha^{k-1} - B_{2\alpha}^{k-1}\| \le \alpha\kappa_Q\|\overline{A}\|(k-1)(1-\alpha a)^{(k-2)/2}. \tag{4.28}$$

By the discrete-difference identity from the previous section,

$$\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}} = -2\alpha\,(B_\alpha^{k-1} - B_{2\alpha}^{k-1}). \tag{4.29}$$

$$\|\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}}\| \le 2\alpha\,\|B_\alpha^{k-1} - B_{2\alpha}^{k-1}\| \quad \le 2\widetilde{C}_A\,\alpha^2\,(k-1)\,(1-\alpha a)^{(k-2)/2}, \tag{4.30}$$

which is the claimed bound. □

4.5 Summed Bounds and Comparison with the Single-Step Case

Corollary 4.1. ***(Summed RR weight bounds.)*** *Under the assumptions of the previous lemma, uniformly in $n \ge 2$,*

$$\sum_{l=1}^{n-1}\|\mathcal{Q}_l^{RR} - \overline{A}^{-1}\|^2 \le \frac{C_1}{\alpha a}, \quad \sum_{l=1}^{n-2}\|\mathcal{Q}_{l+1}^{RR} - \mathcal{Q}_l^{RR}\| \le \frac{C_2}{a^2}, \tag{4.31}$$

with $C_1 = 9C_Q^2$ and $C_2 = 32\widetilde{C}_A$.

4.6 Variance Comparison

Use the deterministic variance proxy

$$\Sigma_n^{\mathrm{RR}} := \frac{1}{n}\sum_{l=2}^{n-1} \mathcal{Q}_l^{\mathrm{RR}}\,\Sigma_\varepsilon^{(\mathrm{M})}\,(\mathcal{Q}_l^{\mathrm{RR}})^\top, \quad \sigma_n^{2,\mathrm{RR}}(u) := u^\top \Sigma_n^{\mathrm{RR}} u, \tag{4.32}$$

where

$$\Sigma_\varepsilon^{(\mathrm{M})} = \mathbb{E}_\pi[\varepsilon(Z_0)\varepsilon(Z_0)^\top] + \sum_{j\geq 1}\Big(\mathbb{E}_\pi[\varepsilon(Z_0)\varepsilon(Z_j)^\top] + \mathbb{E}_\pi[\varepsilon(Z_j)\varepsilon(Z_0)^\top]\Big). \tag{4.33}$$

The asymptotic covariance is

$$\Sigma_\infty = \overline{A}^{-1}\Sigma_\varepsilon^{(\mathrm{M})}\overline{A}^{-\top}, \quad \sigma^2(u) = u^\top\Sigma_\infty u. \tag{4.34}$$

Lemma 4.2. ***(Stationary finite-window variance comparison.)*** *Let $0 < \alpha$ and $2\alpha \leq \alpha_\infty$, set $\Sigma := \Sigma_\varepsilon^{(M)}$, and assume $\|\Sigma\| < \infty$. Then*

$$\|\Sigma_n^{RR} - \Sigma_\infty\| \leq \frac{C_3}{n\,\alpha a}, \tag{4.35}$$

with

$$C_3 = 12\,C_Q\,\|\overline{A}^{-1}\|\,\|\Sigma\| + 9\,C_Q^2\,\|\Sigma\| + 2\,\|\Sigma_\infty\|. \tag{4.36}$$

Consequently, for every $u \in \mathbb{R}^d$,

$$|\sigma_n^{2,RR}(u) - \sigma^2(u)| \leq \frac{C_3\,\|u\|^2}{n\,\alpha a}. \tag{4.37}$$

Proof. Write $\Delta_l := \mathcal{Q}_l^{\mathrm{RR}} - \overline{A}^{-1}$ and expand

$$\mathcal{Q}_l^{\mathrm{RR}}\,\Sigma\,(\mathcal{Q}_l^{\mathrm{RR}})^\top - \Sigma_\infty = R_{1,l} + R_{2,l}, \tag{4.38}$$

where

$$R_{1,l} := \Delta_l\Sigma\overline{A}^{-\top} + \overline{A}^{-1}\Sigma\Delta_l^\top, \quad R_{2,l} := \Delta_l\Sigma\Delta_l^\top. \tag{4.39}$$

The pointwise bounds are

$$\|R_{1,l}\| \leq 2\,\|\overline{A}^{-1}\|\,\|\Sigma\|\,\|\Delta_l\|, \quad \|R_{2,l}\| \leq \|\Sigma\|\,\|\Delta_l\|^2. \tag{4.40}$$

The linear part satisfies

$$\sum_{l=1}^{n-1}\|\Delta_l\| \leq 3C_Q\sum_{k=1}^{n-1}(1-\alpha a)^{k/2} \leq \frac{6C_Q}{\alpha a}. \tag{4.41}$$

The quadratic part satisfies

$$\sum_{l=1}^{n-1}\|\Delta_l\|^2 \leq \frac{9C_Q^2}{\alpha a}. \tag{4.42}$$

Combining, including the missing two boundary terms in the $l = 2$ to $n-1$ sum,

$$\begin{aligned}\|\Sigma_n^{\mathrm{RR}} - \Sigma_\infty\| &\leq \frac{2\|\Sigma_\infty\|}{n} + \frac{1}{n}\sum_{l=2}^{n-1}(\|R_{1,l}\| + \|R_{2,l}\|)\\ &\leq \frac{1}{n}(2\,\|\overline{A}^{-1}\|\,\|\Sigma\|\cdot\frac{6C_Q}{\alpha a} + \|\Sigma\|\cdot\frac{9C_Q^2}{\alpha a} + 2\,\|\Sigma_\infty\|/(\alpha a)) = \frac{C_3}{n\,\alpha a}\end{aligned} \tag{4.43}$$

$$\square \tag{4.44}$$

4.7 Poisson Martingale Approximation

We convert W^{RR} into a martingale plus an Abel remainder via the Poisson equation.

Poisson kernel. Let Q be the one-step transition kernel:

$$(\mathsf{Q}f)(z) = \mathbb{E}\left[f(Z_{k+1})|Z_k = z\right]. \tag{4.45}$$

Under UGE 1,

$$\|\mathsf{Q}^k\varepsilon\|_\infty \le 2\|\varepsilon\|_\infty(1/4)^{\lfloor k/t_{\mathrm{mix}}\rfloor} \tag{4.46}$$

(valid for centered ε, $\pi(\varepsilon) = 0$) makes the Poisson series

$$\widehat{\varepsilon} := \sum_{k=0}^{\infty} \mathsf{Q}^k\varepsilon \tag{4.47}$$

absolutely convergent in sup-norm:

$$\|\widehat{\varepsilon}\|_\infty \le 2\|\varepsilon\|_\infty \sum_{k=0}^{\infty}(1/4)^{\lfloor k/t_{\mathrm{mix}}\rfloor} \le 3\,t_{\mathrm{mix}}\,\|\varepsilon\|_\infty, \quad \|\mathsf{Q}\widehat{\varepsilon}\|_\infty \le \|\widehat{\varepsilon}\|_\infty, \tag{4.48}$$

and $\widehat{\varepsilon}$ solves

$$\widehat{\varepsilon} - \mathsf{Q}\widehat{\varepsilon} = \varepsilon. \tag{4.49}$$

Conditional centering. Let $\mathcal{F}_l := \sigma(Z_0, \dots, Z_l)$. For $l \ge 2$,

$$\varepsilon(Z_l) = m_l + t_l, \quad l \ge 2, \tag{4.50}$$

where

$$m_l := \widehat{\varepsilon}(Z_l) - \mathsf{Q}\widehat{\varepsilon}(Z_{l-1}), \quad t_l := \mathsf{Q}\widehat{\varepsilon}(Z_{l-1}) - \mathsf{Q}\widehat{\varepsilon}(Z_l), \quad l \ge 2, \tag{4.51}$$

and m_l is centered conditionally on $\mathcal{F}_{l-1}$.

Lemma 4.3. ***(Stationary Poisson martingale decomposition.)*** *Assume* ***UGE 1*** *and $\pi(\varepsilon) = 0$. Set*

$$\Delta M_l^{RR} := \mathcal{Q}_l^{RR}\left(\widehat{\varepsilon}(Z_l) - \mathsf{Q}\widehat{\varepsilon}(Z_{l-1})\right), \quad 2 \le l \le n-1, \tag{4.52}$$

and let $M_n^{RR} := \sum_{l=2}^{n-1} \Delta M_l^{RR}$. Then $\left\{\Delta M_l^{RR}\right\}_{l=2}^{n-1}$ is a sequence of $\mathcal{F}_l$-martingale differences, and

$$W^{RR} = -\frac{1}{\sqrt{n}} M_n^{RR} + D_{2,n}^{RR}, \tag{4.53}$$

with the ***Abel remainder***

$$D_{2,n}^{RR} := -\frac{1}{\sqrt{n}}\left[\mathcal{Q}_1^{RR}\,\widehat{\varepsilon}(Z_1) + \sum_{l=1}^{n-2}(\mathcal{Q}_{l+1}^{RR} - \mathcal{Q}_l^{RR})\,\mathsf{Q}\widehat{\varepsilon}(Z_l)\right]. \tag{4.54}$$

Moreover, with $C_{\mathcal{Q}} := \|\overline{A}^{-1}\| + 3C_Q$ a uniform bound on $\|\mathcal{Q}_l^{RR}\|$, and C_2 the constant from the Corollary of Section 4.4,

$$\|D_{2,n}^{RR}\|_\infty \le \frac{3\,t_{mix}\,\|\varepsilon\|_\infty}{\sqrt{n}}\left(C_{\mathcal{Q}} + \frac{C_2}{a^2}\right). \tag{4.55}$$

Consequently, for every $p \ge 1$,

$$\|D_{2,n}^{RR}\|_{L_p} \le \frac{C\,t_{mix}\,\|\varepsilon\|_\infty}{a^2\,\sqrt{n}}, \tag{4.56}$$

with a constant C depending only on $\|\overline{A}^{-1}\|$, κ_Q, and $\|\overline{A}\|$.

Proof. The increments are martingale differences by the Markov property.

Substitute (4.49) into (4.14):

$$W^{\mathrm{RR}} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} \mathcal{Q}_l^{\mathrm{RR}} \left(\widehat{\varepsilon}(Z_l) - \mathsf{Q}\widehat{\varepsilon}(Z_l)\right). \tag{4.57}$$

Peel off $l = 1$:

$$W^{\mathrm{RR}} = -\frac{1}{\sqrt{n}} \mathcal{Q}_1^{\mathrm{RR}}\, \widehat{\varepsilon}(Z_1) + \frac{1}{\sqrt{n}} \mathcal{Q}_1^{\mathrm{RR}}\, \mathsf{Q}\widehat{\varepsilon}(Z_1) - \frac{1}{\sqrt{n}} M_n^{\mathrm{RR}} - \frac{1}{\sqrt{n}} \sum_{l=2}^{n-1} \mathcal{Q}_l^{\mathrm{RR}} \left(\mathsf{Q}\widehat{\varepsilon}(Z_{l-1}) - \mathsf{Q}\widehat{\varepsilon}(Z_l)\right). \tag{4.58}$$

Set $g_l := \mathsf{Q}\widehat{\varepsilon}(Z_l)$. Abel summation gives

$$\sum_{l=2}^{n-1} \mathcal{Q}_l^{\mathrm{RR}}(g_{l-1} - g_l) = \mathcal{Q}_2^{\mathrm{RR}}\, g_1 - \mathcal{Q}_{n-1}^{\mathrm{RR}}\, g_{n-1} + \sum_{l=2}^{n-2} (\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}})\, g_l. \tag{4.59}$$

The boundary terms combine as

$$\begin{aligned} &- \mathcal{Q}_1^{\mathrm{RR}}\, \widehat{\varepsilon}(Z_1) + \mathcal{Q}_1^{\mathrm{RR}}\, g_1 - \mathcal{Q}_2^{\mathrm{RR}}\, g_1 + \mathcal{Q}_{n-1}^{\mathrm{RR}}\, g_{n-1} - \sum_{l=2}^{n-2} (\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}})\, g_l \\ &= -\mathcal{Q}_1^{\mathrm{RR}}\, \widehat{\varepsilon}(Z_1) - \sum_{l=1}^{n-2} (\mathcal{Q}_{l+1}^{\mathrm{RR}} - \mathcal{Q}_l^{\mathrm{RR}})\, g_l + \mathcal{Q}_{n-1}^{\mathrm{RR}}\, g_{n-1}. \end{aligned} \tag{4.60}$$

The rightmost boundary vanishes because $\mathcal{Q}_{n-1}^{\mathrm{RR}} = 2\alpha I - 2\alpha I = 0$. This gives the stated formula.

For (4.55), use the sup-norm Poisson bound, the uniform weight bound, and the summed-total-variation bound:

$$\|\sqrt{n}\, D_{2,n}^{\mathrm{RR}}\| \le 3t_{\mathrm{mix}} \|\varepsilon\|_\infty (C_{\mathcal{Q}} + C_2/a^2). \tag{4.61}$$

The L_p bound is immediate from the deterministic sup-norm bound. ☐

4.8 Predictable Quadratic Variation Concentration

We control the predictable quadratic variation of M_n^{RR} by applying (B.1) to a centered quadratic functional of the chain.

Predictable quadratic variation. The increments

$$\Delta M_l^{\mathrm{RR}} = \mathcal{Q}_l^{\mathrm{RR}} \left(\widehat{\varepsilon}(Z_l) - \mathsf{Q}\widehat{\varepsilon}(Z_{l-1})\right) \tag{4.62}$$

are $\mathcal{F}_l$-martingale differences. The Markov property gives

$$\mathbb{E}[\widehat{\varepsilon}(Z_l) | \mathcal{F}_{l-1}] = \mathsf{Q}\widehat{\varepsilon}(Z_{l-1}) \tag{4.63}$$

$$\mathbb{E}[\Delta M_l^{\mathrm{RR}}\, (\Delta M_l^{\mathrm{RR}})^\top \,|\, \mathcal{F}_{l-1}] = \mathcal{Q}_l^{\mathrm{RR}}\, \mathcal{V}_\varepsilon(Z_{l-1})\, (\mathcal{Q}_l^{\mathrm{RR}})^\top, \tag{4.64}$$

where

$$\mathcal{V}_\varepsilon(z) := \mathsf{Q}(\widehat{\varepsilon}\widehat{\varepsilon}^\top)(z) - (\mathsf{Q}\widehat{\varepsilon})(z)\, (\mathsf{Q}\widehat{\varepsilon})(z)^\top. \tag{4.65}$$

Lemma 4.4. ***(Poisson covariance identity and absolute convergence.)*** *Assume **UGE 1**, $\pi(\varepsilon) = 0$, and $\|\varepsilon\|_\infty < \infty$. Then $\Sigma_\varepsilon^{(M)}$ is absolutely convergent in operator norm, and*

$$\pi(\mathcal{V}_\varepsilon) = \Sigma_\varepsilon^{(M)} =: \Sigma. \tag{4.66}$$

Proof. Boundedness and UGE give absolute convergence. For $j \geq 1$,

$$\|\mathbb{E}_\pi[\varepsilon(Z_0)\varepsilon(Z_j)^\top]\| = \|\pi(\varepsilon\,(\mathsf{Q}^j\varepsilon)^\top)\| \leq \|\varepsilon\|_\infty \|\mathsf{Q}^j\varepsilon\|_\infty \leq 2\|\varepsilon\|_\infty^2 (1/4)^{\lfloor j/t_{\mathrm{mix}}\rfloor}, \tag{4.67}$$

and similarly for the transposed covariance. Set $h := \widehat{\varepsilon}$ and $r := \mathsf{Q}h = h - \varepsilon$. Since $\pi\mathsf{Q} = \pi$,

$$\pi(\mathcal{V}_\varepsilon) = \pi(hh^\top) - \pi(rr^\top) \quad = \pi(\varepsilon\varepsilon^\top) + \pi(\varepsilon r^\top) + \pi(r\varepsilon^\top). \tag{4.68}$$

The Poisson series gives $r = \sum_{j\geq 1} \mathsf{Q}^j\varepsilon$ in sup-norm:

$$\begin{aligned} \pi(\varepsilon r^\top) &= \sum_{j\geq 1} \mathbb{E}_\pi[\varepsilon(Z_0)\varepsilon(Z_j)^\top], \\ \pi(r\varepsilon^\top) &= \sum_{j\geq 1} \mathbb{E}_\pi[\varepsilon(Z_j)\varepsilon(Z_0)^\top]. \end{aligned} \tag{4.69}$$

Substitution proves (4.66). $\square$

Thus

$$\langle M^{\mathrm{RR}}\rangle_n := \sum_{l=2}^{n-1} \mathbb{E}[\Delta M_l^{\mathrm{RR}}\,(\Delta M_l^{\mathrm{RR}})^\top | \mathcal{F}_{l-1}] \quad = \sum_{l=2}^{n-1} \mathcal{Q}_l^{\mathrm{RR}}\,\mathcal{V}_\varepsilon(Z_{l-1})\,(\mathcal{Q}_l^{\mathrm{RR}})^\top. \tag{4.70}$$

By Lemma 4.4, $\pi(\mathcal{V}_\varepsilon) = \Sigma$.

Sup-norm bound on $\mathcal{V}_\varepsilon$. Since

$$\|\widehat{\varepsilon}\|_\infty \leq 3\,t_{\mathrm{mix}}\,\|\varepsilon\|_\infty \tag{4.71}$$

and Q is a Markov kernel,

$$\begin{aligned} \|\mathcal{V}_\varepsilon\|_\infty &\leq \|\mathsf{Q}(\widehat{\varepsilon}\widehat{\varepsilon}^\top)\|_\infty + \|(\mathsf{Q}\widehat{\varepsilon})(\mathsf{Q}\widehat{\varepsilon})^\top\|_\infty \\ &\leq 2\,\|\widehat{\varepsilon}\|_\infty^2 \leq 18\,t_{\mathrm{mix}}^2\,\|\varepsilon\|_\infty^2. \end{aligned} \tag{4.72}$$

Lemma 4.5. ***(Stationary predictable-variation concentration.)*** *Assume **UGE 1** and $\pi(\varepsilon) = 0$, with $\|\varepsilon\|_\infty < \infty$. Let $C_{\mathcal{Q}}$ be the uniform bound on $\|\mathcal{Q}_l^{RR}\|$ from the previous lemma. There exists a universal constant $C_4 > 0$ such that, for every $u \in \mathbb{R}^d$, every $p \geq 2$, every initial distribution ξ, and every $n \geq 2$,*

$$\mathbb{E}_\xi^{1/p}\left[|u^\top\langle M^{RR}\rangle_n u - n\,\sigma_n^{2,RR}(u)|^p\right] \leq C_4\,C_{\mathcal{Q}}^2\,\|u\|^2\,\|\varepsilon\|_\infty^2\,t_{mix}^{5/2}\sqrt{p\,n}. \tag{4.73}$$

Proof. Write $h_l(z) := u^\top\mathcal{Q}_l^{\mathrm{RR}}\,\mathcal{V}_\varepsilon(z)\,(\mathcal{Q}_l^{\mathrm{RR}})^\top u$ and $g_l(z) := h_l(z) - \pi(h_l)$. Then

$$|h_l(z)| \leq C_{\mathcal{Q}}^2\,\|u\|^2\,\|\mathcal{V}_\varepsilon\|_\infty \leq 18\,C_{\mathcal{Q}}^2\,\|u\|^2\,t_{\mathrm{mix}}^2\,\|\varepsilon\|_\infty^2, \quad \|g_l\|_\infty \leq 2\,|h_l(z)| \leq 36\,C_{\mathcal{Q}}^2\,\|u\|^2\,t_{\mathrm{mix}}^2\,\|\varepsilon\|_\infty^2. \tag{4.74}$$

By (4.70), $n\,\sigma_n^{2,\mathrm{RR}}(u) = \sum_{l=2}^{n-1}\pi(h_l)$,

$$u^\top\langle M^{\mathrm{RR}}\rangle_n u - n\,\sigma_n^{2,\mathrm{RR}}(u) = \sum_{l=2}^{n-1} g_l(Z_{l-1}). \tag{4.75}$$

Set $\widetilde{g}_i := g_{i+1}$. Then $\pi(\widetilde{g}_i) = 0$ and $\|\widetilde{g}_i\|_\infty \leq c := 36\,C_{\mathcal{Q}}^2\,\|u\|^2\,t_{\mathrm{mix}}^2\,\|\varepsilon\|_\infty^2$. Apply (B.1) with $c_i = c$:

$$\mathbb{E}_\xi^{1/p}\left[\left(|\sum_{l=2}^{n-1} g_l(Z_{l-1})|\right)^p\right] \leq C_{\mathrm{MC}}\sqrt{p\,t_{\mathrm{mix}}\,(n-2)c^2}. \tag{4.76}$$

Substitution gives

$$\mathbb{E}_\xi^{1/p}\left[\left(|\sum_{l=2}^{n-1} g_l(Z_{l-1})|\right)^p\right] \leq C_{\mathrm{MC}}\cdot 36\,C_{\mathcal{Q}}^2\,\|u\|^2\,\|\varepsilon\|_\infty^2\,t_{\mathrm{mix}}^{5/2}\sqrt{p\,n}. \tag{4.77}$$

Absorb the universal prefactor into C_4. $\square$

Corollary 4.2. ***(Stationary asymptotic-variance bracket comparison.)*** *Under the assumptions of the previous lemma, for every $u \in \mathbb{R}^d$, every $p \geq 2$, and every $n \geq 2$,*

$$\mathbb{E}_\xi^{1/p}\left[|u^\top\langle M^{RR}\rangle_n u - n\,\sigma^2(u)|^p\right] \leq C_4\,C_{\mathcal{Q}}^2\,\|u\|^2\,\|\varepsilon\|_\infty^2\,t_{mix}^{5/2}\sqrt{p\,n} + \frac{C_3\,\|u\|^2}{\alpha\,a}, \tag{4.78}$$

with C_3 the variance-comparison constant of Section 4.5.

4.9 Martingale Berry–Esseen Step

We apply the martingale Berry–Esseen theorem to the scalar martingale $u^\top M_n^{\mathrm{RR}}$.

Bounded increments. For $X_l := u^\top \Delta M_l^{\mathrm{RR}}$,

$$|X_l| \le \varkappa(u), \quad \varkappa(u) := 6\, t_{\mathrm{mix}}\, C_{\mathcal{Q}}\, \|\varepsilon\|_\infty\, \|u\|. \tag{4.79}$$

The conditional variances sum to

$$V_n^2 := \sum_{l=2}^{n-1} \sigma_l^2 = u^\top \langle M^{\mathrm{RR}} \rangle_n u. \tag{4.80}$$

Variance lower bound. Set $s_n^2 := n\, \sigma_n^{2,\mathrm{RR}}(u)$. If

$$n\, \alpha\, a \ge \frac{2\, C_3\, \|u\|^2}{\sigma^2(u)} \tag{4.81}$$

then $s_n^2 \ge n\sigma^2(u)/2$. The trivial upper bound

$$\sigma_n^{2,\mathrm{RR}}(u) \le C_{\mathcal{Q}}^2 \|\Sigma_\varepsilon^{(\mathrm{M})}\| \|u\|^2 \tag{4.82}$$

gives $s_n^2 \le K^2(u)\, n$ with $K(u) := C_{\mathcal{Q}}\, \|\Sigma_\varepsilon^{(\mathrm{M})}\|^{1/2}\, \|u\|$.

We apply Lemma B.2 to this scalar martingale; the number of increments is at most n, and the increment bound is $\varkappa(u)$.

Theorem 4.1. ***(Stationary martingale Berry–Esseen bound.)*** *Assume* ***UGE 1****, $\pi(\varepsilon) = 0$, $\|\varepsilon\|_\infty < \infty$, $\sigma^2(u) > 0$, $0 < \alpha$, and $2\alpha \le \alpha_\infty$. There exist constants $C_{K,1}(u), C_{K,2}(u) > 0$ depending only on $\|u\|$, $\sigma(u)$, $C_{\mathcal{Q}}$, t_{mix}, $\|\varepsilon\|_\infty$, $\|\Sigma_\varepsilon^{(M)}\|$, and the constants $L_B(\varkappa(u)), C_1, C_2$ of Lemma B.2, such that for every $n \ge 3$ satisfying the variance lower-bound condition* (4.81)*,*

$$d_K\left(\frac{u^\top M_n^{RR}}{\sqrt{n}\, \sigma_n^{RR}(u)}, \mathcal{N}(0,1)\right) \le \frac{C_{K,1}(u)\, \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u)\, \log n}{\sqrt{n}}. \tag{4.83}$$

Proof. Apply Lemma B.2 to $(X_l)_{l=2}^{n-1}$ with partial sum $u^\top M_n^{\mathrm{RR}}$, deterministic scale $s_n^2 = n\, \sigma_n^{2,\mathrm{RR}}(u)$, increment bound $\varkappa(u)$, and $p = \lceil \log n \rceil$. This martingale array has at most n increments, so the first term in Lemma B.2 is bounded using $(2n+1)\log(2n+1)$.

Term I.

$$L_B(\varkappa(u))\, \frac{(2n+1)\log(2n+1)}{s_n^3} \le \frac{6\sqrt{2}\, L_B(\varkappa(u))}{\sigma^3(u)}\, \frac{\log n}{\sqrt{n}} =: \frac{C^{(\mathrm{I})}(u)\, \log n}{\sqrt{n}}. \tag{4.84}$$

Term III. With $a_p := 2p/(2p+1)$,

$$s_n^{1/(2p+1)} \le \max(1, K(u))^{1/(2p+1)}\, n^{1/(2(2p+1))}. \tag{4.85}$$

For $p \ge \log n$,

$$C_2\, s_n^{-a_p}\, p\, \varkappa(u)^{a_p} \le \sqrt{2}\, e^{3/4}\, \frac{C_2\, \max(1, \varkappa(u))}{\sigma(u)}\, \frac{p}{\sqrt{n}} \le \frac{C^{(\mathrm{III})}(u)\, \log n}{\sqrt{n}}, \tag{4.86}$$

Term II. By bracket concentration,

$$(\mathbb{E}|V_n^2 - s_n^2|^p)^{1/p} \le B(u)\, \sqrt{p\, n}, \quad B(u) := C_4\, C_{\mathcal{Q}}^2\, \|u\|^2\, \|\varepsilon\|_\infty^2\, t_{\mathrm{mix}}^{5/2}, \tag{4.87}$$

The conditional-variance term is bounded by

$$\begin{aligned} &C_1\, \sqrt{p}\, s_n^{-a_p}\, (\mathbb{E}|V_n^2 - s_n^2|^p)^{1/(2p+1)} \\ &\quad \le C\, C_1\, \sigma^{-1}(u)\, \max(1, K(u))^{1/(2p+1)}\, \max(1, B(u))^{p/(2p+1)} \\ &\qquad \times p^{(3p+1)/(2(2p+1))} n^{-p/(2(2p+1))}. \end{aligned} \tag{4.88}$$

For $p = \lceil \log n \rceil$, $p^{(3p+1)/(2(2p+1))} \le C \log^{3/4} n$ and $n^{-p/(2(2p+1))} \le C n^{-1/4}$. Thus

$$\text{Term II} \le \frac{C^{(\mathrm{II})}(u)\, \log^{3/4} n}{n^{1/4}}. \tag{4.89}$$

Adding the three bounds and setting $C_{K,1}(u) := C^{(\mathrm{II})}(u)$, $C_{K,2}(u) := C^{(\mathrm{I})}(u) + C^{(\mathrm{III})}(u)$ proves the martingale Berry–Esseen bound. □

Corollary 4.3. ***(Stationary martingale asymptotic-normalization bound.)*** *Under the hypotheses of the previous theorem,*

$$d_K\left(\frac{u^\top M_n^{RR}}{\sqrt{n}\,\sigma(u)}, \mathcal{N}(0,1)\right) \le \frac{C_{K,1}(u)\, \log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u)\, \log n}{\sqrt{n}} + \frac{C_3\, \|u\|^2}{n\,\alpha\, a\, \sigma^2(u)}. \tag{4.90}$$

Proof. Set $r := \sigma_n^{\mathrm{RR}}(u)/\sigma(u)$ and $W := u^\top M_n^{\mathrm{RR}}/(\sqrt{n}\sigma_n^{\mathrm{RR}}(u))$. Then the statistic normalised by $\sigma(u)$ is Wr. Under the variance lower-bound condition (4.81), $r \ge 1/\sqrt{2}$, while the trivial upper bound on $\sigma_n^{2,\mathrm{RR}}(u)$ gives $r \le r_{\max}(u) < \infty$. On this compact interval the standard normal cdf satisfies

$$\sup_x |\Phi(x/r) - \Phi(x)| \le C_\Phi\, |r-1|, \quad C_\Phi := \sqrt{2}/\sqrt{\pi e}. \tag{4.91}$$

Therefore

$$d_K\left(Wr, \mathcal{N}(0,1)\right) \le d_K\left(W, \mathcal{N}(0,1)\right) + C_\Phi\, |r-1|. \tag{4.92}$$

The variance comparison of Section 4.5 gives

$$|r-1| = \frac{|\sigma_n^{2,\mathrm{RR}}(u) - \sigma^2(u)|}{\sigma(u)\,(\sigma_n^{\mathrm{RR}}(u) + \sigma(u))} \le \frac{C_3\, \|u\|^2}{n\,\alpha\, a\, \sigma^2(u)}. \tag{4.93}$$

Adding this perturbation to the martingale Berry–Esseen bound proves the stated claim after absorbing C_Φ into the constants. □

4.10 Depth-Two Misadjustment Bound

The remaining stationary non-martingale contribution is

$$R_n^{\text{mis, RR}} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} (2R_k^{(\alpha)} - R_k^{(2\alpha)}), \tag{4.94}$$

where $R_k^{(\alpha)} := J_k^{(1,\alpha)} + H_k^{(1,\alpha)}$.

Depth-two refinement. For $\ell \ge 1$ define

$$J_n^{(\ell,\alpha)} := (I - \alpha\overline{A})\, J_{n-1}^{(\ell,\alpha)} - \alpha\widetilde{A}(Z_n)\, J_{n-1}^{(\ell-1,\alpha)}, \tag{4.95}$$

$$H_n^{(\ell,\alpha)} := (I - \alpha A(Z_n))\, H_{n-1}^{(\ell,\alpha)} - \alpha\widetilde{A}(Z_n)\, J_{n-1}^{(\ell,\alpha)}, \tag{4.96}$$

with $J_0^{(\ell,\alpha)} = H_0^{(\ell,\alpha)} = 0$ and the $\ell = 0$ processes as in Section 3. Then

$$H_n^{(\ell,\alpha)} = J_n^{(\ell+1,\alpha)} + H_n^{(\ell+1,\alpha)}, \quad \ell \ge 0. \tag{4.97}$$

Applying (4.97) with $\ell = 1$ refines $R_k^{(\alpha)}$ to

$$R_k^{(\alpha)} = J_k^{(1,\alpha)} + J_k^{(2,\alpha)} + H_k^{(2,\alpha)}, \tag{4.98}$$

so

$$R_n^{\text{mis, RR}} = T_n^{(1)} + T_n^{(2)} + T_n^{(H)}, \tag{4.99}$$

$$T_n^{(j)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \left(2J_k^{(j,\alpha)} - J_k^{(j,2\alpha)} \right), \quad j \in \{1, 2\}, \tag{4.100}$$

with $T_n^{(H)}$ defined identically with $H^{(2)}$ in place of $J^{(j)}$. Each piece is bounded separately.

Finite-past truncation. Fix $q \geq 2$, $2 \leq p \leq q/2$, and $0 < w \leq \alpha_{\mathrm{stat}}(q)$. Let $(Z_t)_{t \in \mathbb{Z}}$ be a two-sided stationary version of the driving Markov chain, and write $B_w := I - w\overline{A}$. For $m \geq 1$ start at time $-m$ from zero,

$$J_{-m,m}^{(0,w)} = J_{-m,m}^{(1,w)} = J_{-m,m}^{(2,w)} = H_{-m,m}^{(2,w)} = 0, \tag{4.101}$$

and evolve, for $r = -m+1, \dots, 0$,

$$J_{r,m}^{(0,w)} = B_w J_{r-1,m}^{(0,w)} - w\varepsilon(Z_r), \tag{4.102}$$

$$J_{r,m}^{(\ell,w)} = B_w J_{r-1,m}^{(\ell,w)} - w\widetilde{A}(Z_r) J_{r-1,m}^{(\ell-1,w)}, \quad \ell \in \{1, 2\}, \tag{4.103}$$

$$H_{r,m}^{(2,w)} = (I - wA(Z_r)) H_{r-1,m}^{(2,w)} - w\widetilde{A}(Z_r) J_{r-1,m}^{(2,w)}. \tag{4.104}$$

Lemma 4.6. ***(Stationary full depth-two augmented state.)*** *There exist constants $C_{fp}, c_{fp} > 0$, depending only on the fixed problem parameters allowed by the chapter convention, on C_W, D_J, D_H from Lemma B.8, Lemma B.5, Lemma B.7, and on C_{prod}, c_{prod} from Lemma 5.3, such that, with*

$$A_{fp}(p, q, w) := C_{fp}(1 + d^{1/q}) p^8 t_{mix}^5 \frac{1}{a} \sqrt{w/a} \log^3(1/(wa)), \tag{4.105}$$

the terminal vector of the finite-past truncation above,

$$(J_{0,m}^{(0,w)}, J_{0,m}^{(1,w)}, J_{0,m}^{(2,w)}, H_{0,m}^{(2,w)}) \tag{4.106}$$

is Cauchy in L_p. More precisely, for $m' \geq m \geq 1$,

$$\|H_{0,m'}^{(2,w)} - H_{0,m}^{(2,w)}\|_{L_p} \leq A_{fp}(p, q, w) \exp(-c_{fp} wam/p), \tag{4.107}$$

The J coordinates converge to the invariant law $\Pi_{J,2,w}$ of Lemma B.9, and the L_p limit

$$H_0^{(2,w)} := \lim_{m \to \infty} H_{0,m}^{(2,w)} \tag{4.108}$$

satisfies the one-trajectory bound of Lemma B.7.

Proof. First control the J-coordinates. Compare the truncation started at $-m'$ with the one started at $-m$ by applying the component contraction (B.6) after time $-m$. The contraction constant is C_W and its exponential rate is $1/12$. The initial depth-two cost at time $-m$ is bounded by the elementary moment bounds for $J^{(0,w)}$ and $J^{(1,w)}$, together with Lemma B.5 for $J^{(2,w)}$. Write

$$Y_{r,m}^{(w)} := (Z_{r+1}, J_{r,m}^{(0,w)}, J_{r,m}^{(1,w)}, J_{r,m}^{(2,w)}). \tag{4.109}$$

Thus there is a constant C_{cost}, depending only on the fixed problem parameters and on D_J, such that

$$\|c_{J,2}^{(w)}(Y_{-m,m'}^{(w)}, Y_{-m,m}^{(w)})\|_{L_p} \leq C_{\mathrm{cost}}(1 + d^{1/q}) p^{7/2} t_{\mathrm{mix}}^{5/2} \sqrt{w/a} \log^{3/2}(1/(wa)). \tag{4.110}$$

Consequently, for $r \geq -m$ and $\ell \in \{0, 1, 2\}$,

$$\|J_{r,m'}^{(\ell,w)} - J_{r,m}^{(\ell,w)}\|_{L_p} \leq A_J(p, q, w) \exp(-wa(r+m)/(12p)), \tag{4.111}$$

where

$$A_J(p, q, w) := C_W C_{\mathrm{cost}}(1 + d^{1/q}) p^7 t_{\mathrm{mix}}^5 \sqrt{w/a} \log^3(1/(wa)). \tag{4.112}$$

Now subtract the two $H^{(2,w)}$ finite-past recursions. With $\Delta H_r := H^{(2,w)}_{r,m'} - H^{(2,w)}_{r,m}$ and similarly for $\Delta J^{(2,w)}_r$,

$$\Delta H_0 = \Gamma^{(w)}_{-m+1:0} H^{(2,w)}_{-m,m'} - w \sum_{l=-m+1}^{0} \Gamma^{(w)}_{l+1:0} \widetilde{A}(Z_l) \Delta J^{(2,w)}_{l-1}. \tag{4.113}$$

For the initial term, use the conditional product-stability estimate (5.24) and Lemma B.7:

$$\begin{aligned}\|\Gamma^{(w)}_{-m+1:0} H^{(2,w)}_{-m,m'}\|_{L_p} &\le C_{\mathrm{prod}} \exp(-c_{\mathrm{prod}} wam/p) \|H^{(2,w)}_{-m,m'}\|_{L_p} \\ &\le C_{\mathrm{prod}} D_H d^{1/q} t_{\mathrm{mix}}^{5/2} p^{7/2} w^{3/2} \log^{3/2}(1/(wa)) \exp(-c_{\mathrm{prod}} wam/p).\end{aligned} \tag{4.114}$$

For the convolution term, combine (5.24), $\|\widetilde{A}(z)\| \le C_A$, and (4.111):

$$\begin{aligned}& w \sum_{l=-m+1}^{0} \|\Gamma^{(w)}_{l+1:0} \widetilde{A}(Z_l) \Delta J^{(2,w)}_{l-1}\|_{L_p} \\ &\quad \le w C_{\mathrm{prod}} C_A A_J(p,q,w) \sum_{l=-m+1}^{0} \exp(-c_{\mathrm{prod}} wa(-l)/p) \exp(-wa(l-1+m)/(12p)).\end{aligned} \tag{4.115}$$

The elementary convolution bound

$$\sum_{l=-m+1}^{0} \exp(-c_{\mathrm{prod}} wa(-l)/p) \exp(-wa(l-1+m)/(12p)) \le C_{\mathrm{conv}} \frac{p}{wa} \exp(-c_{\mathrm{conv}} wam/p) \tag{4.116}$$

holds with $C_{\mathrm{conv}}, c_{\mathrm{conv}} > 0$ depending only on c_{prod} and the numerical rate $1/12$. Therefore the convolution term is bounded by

$$C_{\mathrm{prod}} C_A C_{\mathrm{conv}} \frac{p}{a} A_J(p,q,w) \exp(-c_{\mathrm{conv}} wam/p). \tag{4.117}$$

Choose

$$c_{\mathrm{fp}} := \frac{1}{2} \min(c_{\mathrm{prod}}, c_{\mathrm{conv}}) \tag{4.118}$$

and choose C_{fp} large enough to dominate the constants in (4.114) and (4.117) after substituting (4.112). Since $w \le \alpha_\infty$ implies $wa \le 1/2$, the initial-term factor $w^{3/2}$ is absorbed by $a^{-1}\sqrt{w/a}$, and the powers $p^{7/2} t_{\mathrm{mix}}^{5/2} \log^{3/2}(1/(wa))$ are absorbed by the displayed $p^8 t_{\mathrm{mix}}^5 \log^3(1/(wa))$ in $A_{\mathrm{fp}}(p,q,w)$. This gives (4.107).

The J-bounds and (4.107) show that the full terminal vector is Cauchy in L_p. Passing to the L_p limit gives the stationary full augmented-chain law.

$$\square \tag{4.119}$$

Shifting the finite-past construction before taking the limit defines a two-sided stationary process

$$(Z_{k+1}, J^{(0,w)}_k, J^{(1,w)}_k, J^{(2,w)}_k, H^{(2,w)}_k)_{k\in\mathbb{Z}}. \tag{4.120}$$

It solves the full depth-two recursions, and its law is the stationary full augmented-chain law used below.

Joint RR stationary construction. Apply Lemma 4.6 at $w = \alpha$ and $w = 2\alpha$ on the same two-sided stationary base chain. The joint limit

$$\left(Z_{k+1}, J^{(0,\alpha)}_k, J^{(1,\alpha)}_k, J^{(2,\alpha)}_k, H^{(2,\alpha)}_k, J^{(0,2\alpha)}_k, J^{(1,2\alpha)}_k, J^{(2,2\alpha)}_k, H^{(2,2\alpha)}_k\right)_{k\in\mathbb{Z}} \tag{4.121}$$

is the stationary RR augmented state used below.

Stationary augmented-chain convention. The estimates below use the stationary versions constructed above. A zero-start full average is not obtained by a single terminal contraction: summing startup contractions yields

$$\frac{1}{\sqrt{n}}\sum_{k\geq 0}\rho_\alpha^k \asymp \frac{1}{\sqrt{n}\,\alpha a} \tag{4.122}$$

when $\rho_\alpha \approx \exp(-c\alpha a)$. This term is deferred to the burn-in chapter.

For the shifted-to-unshifted first-order transfer we use the local inverse ceiling $\alpha_{\rm inv}$ defined in (4.2).

Lemma 4.7. ***(Stationary-limit transfer for the centered first-order iterate.)*** *Fix $w \in (0, \alpha_\infty]$ and set $B_w := I - w\overline{A}$. Assume the stationary augmented-chain convention above, **UGE 1**, $\pi(\widetilde{A}) = 0$, and $\|\varepsilon\|_\infty < \infty$. Let $(J_t^{(0,w)}, J_t^{(1,w)})_{t\in\mathbb{Z}}$ be the stationary two-sided solution of the first two perturbation recursions, and set $T_t^{(1,w)} := B_w\, J_t^{(1,w)}$. Then, with*

$$\Phi_+(p, w) := 1 + p^{3/2}\, t_{mix}^{1/2}/a + p^{1/2}\, t_{mix}^{3/2}\sqrt{w/a}, \tag{4.123}$$

there exists a constant $C_{stat,1}$, depending only on the constants in the zero-start centered shifted first-order bound in Lemma 3.2, such that for every $p \geq 2$ and every $t \in \mathbb{Z}$,

$$\|T_t^{(1,w)} - \mathbb{E}_\pi T_t^{(1,w)}\|_{L_p} \leq C_{stat,1}\, w\, \Phi_+(p, w). \tag{4.124}$$

Consequently, whenever $w \leq \alpha_{inv}$,

$$\|J_t^{(1,w)} - \mathbb{E}_\pi J_t^{(1,w)}\|_{L_p} \leq 2C_{stat,1}\, w\, \Phi_+(p, w). \tag{4.125}$$

Proof. Use finite-past zero-start versions $J_{t,m}^{(0,w)}$ and $J_{t,m}^{(1,w)}$. By stationarity and Lemma 3.2,

$$\|B_w J_{t,m}^{(1,w)} - \mathbb{E} B_w J_{t,m}^{(1,w)}\|_{L_p} \leq C_{\rm stat,1}\, w\, \Phi_+(p, w). \tag{4.126}$$

The deterministic-product expansions give

$$J_{t,m}^{(0,w)} = -w\sum_{r=0}^{m-1} B_w^r \varepsilon(Z_{t-r}), \tag{4.127}$$

and the corresponding $J_{t,m}^{(1,w)}$ series is Cauchy for fixed admissible w. Passing to the limit gives the bound for $T_t^{(1,w)}$. The bound for $J_t^{(1,w)}$ follows from

$$J_t^{(1,w)} - \mathbb{E}_\pi J_t^{(1,w)} = B_w^{-1}\big(T_t^{(1,w)} - \mathbb{E}_\pi T_t^{(1,w)}\big). \tag{4.128}$$

□

Telescoping identity for $J^{(1)}$. Summing

$$J_k^{(1,\alpha)} = (I - \alpha\overline{A})J_{k-1}^{(1,\alpha)} - \alpha\widetilde{A}(Z_k)J_{k-1}^{(0,\alpha)} \tag{4.129}$$

from $k = 1$ to n and rearranging gives, for an arbitrary initial value,

$$\overline{A}\sum_{k=0}^{n-1} J_k^{(1,\alpha)} = -\sum_{k=1}^{n}\widetilde{A}(Z_k)\, J_{k-1}^{(0,\alpha)} + \frac{1}{\alpha}\left(J_0^{(1,\alpha)} - J_n^{(1,\alpha)}\right). \tag{4.130}$$

In stationarity,

$$\mathbb{E}_\pi\Big[\widetilde{A}(Z_1)J_0^{(0,\alpha)}\Big] = -\overline{A}\,\mathbb{E}_\pi\Big[J_\infty^{(1,\alpha)}\Big]. \tag{4.131}$$

Subtracting $n\,\mathbb{E}_\pi[J_\infty^{(1,\alpha)}]$ from both sides and applying $\overline{A}^{-1}$ gives the **centered telescoping identity**

$$\sum_{k=0}^{n-1}\left(J_k^{(1,\alpha)} - \mathbb{E}_\pi[J_\infty^{(1,\alpha)}]\right) = -\overline{A}^{-1}\sum_{k=1}^{n}\overline{\psi}_\alpha(J_{k-1}^{(0,\alpha)}, Z_k) + \frac{1}{\alpha}\,\overline{A}^{-1}\left(J_0^{(1,\alpha)} - J_n^{(1,\alpha)}\right). \tag{4.132}$$

Lemma 4.8. ***(Stationary first-order misadjustment bound.)*** *Assume the stationary augmented-chain convention above,* ***UGE 1****,* $\pi(\widetilde{A}) = 0$*,* $\pi(\varepsilon) = 0$*,* $\|\varepsilon\|_\infty < \infty$*, and* $0 < \alpha$*,* $2\alpha \le \alpha_\infty$*, and* $2\alpha \le \alpha_{inv}$*. Set*

$$\Phi_+(p, \alpha) := 1 + p^{3/2}\, t_{mix}^{1/2}/a + p^{1/2}\, t_{mix}^{3/2}\sqrt{\alpha/a}. \tag{4.133}$$

There exists a constant $C_{mis,1}$ *depending on* $\|\overline{A}\|$*,* $\|\overline{A}^{-1}\|$*,* κ_Q*,* C_A*,* $\|\varepsilon\|_\infty$*,* t_{mix}*, and the universals* $c_{W,1}$*,* $c_{W,2}$ *of Lemma B.4, and the constant of the centered bound in Lemma 3.2, such that for every* $p \ge 2$ *and every* $n \ge 2$*,*

$$\begin{aligned} \|T_n^{(1)}\|_{L_p} &\le C_{mis,1}\sqrt{n}\,\alpha^2 + C_{mis,1}\, p^{3/2}\sqrt{\alpha} \\ &\quad + C_{mis,1}\, p^3\,(\alpha n)^{-1/2}\,\log^{1/2}(1/(\alpha a)) + C_{mis,1}\,\Phi_+(p,\alpha)\, n^{-1/2}. \end{aligned} \tag{4.134}$$

Proof. Decompose $T_n^{(1)} = T_n^{(1,\text{ b})} + T_n^{(1,\text{ c})}$ via the bias-fluctuation split

$$T_n^{(1,\text{ b})} := \sqrt{n}\,\left(2\,\mathbb{E}_\pi[J_\infty^{(1,\alpha)}] - \mathbb{E}_\pi[J_\infty^{(1,2\alpha)}]\right), \tag{4.135}$$

$$T_n^{(1,\text{ c})} := \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}\sum_{w\in\{\alpha,2\alpha\}} c_w\left(J_k^{(1,w)} - \mathbb{E}_\pi[J_\infty^{(1,w)}]\right), \quad c_\alpha = 2,\, c_{2\alpha} = -1. \tag{4.136}$$

Bias. By Lemma B.3, $\mathbb{E}_\pi[J_\infty^{(1,w)}] = w\,\Delta + R(w)$ with $\|R(w)\| \le 12\,\|\overline{A}^{-1}\|\,C_A^2\,t_{\text{mix}}^2\,w^2\,\|\varepsilon\|_\infty$. The leading $w\,\Delta$ cancels, leaving

$$\|2\,\mathbb{E}_\pi[J_\infty^{(1,\alpha)}] - \mathbb{E}_\pi[J_\infty^{(1,2\alpha)}]\| \le 2\,\|R(\alpha)\| + \|R(2\alpha)\| \le 72\,\|\overline{A}^{-1}\|\,C_A^2\,t_{\text{mix}}^2\,\alpha^2\,\|\varepsilon\|_\infty. \tag{4.137}$$

Multiplying by $\sqrt{n}$ gives the first term.

Centered. Apply (4.132) at $w \in \{\alpha, 2\alpha\}$:

$$\begin{aligned} \|T_n^{(1,\text{ c})}\|_{L_p} \le \frac{1}{\sqrt{n}}\sum_{w\in\{\alpha,2\alpha\}} |c_w|\Bigg(&\|\overline{A}^{-1}\|\cdot\|\sum_{k=1}^{n}\overline{\psi}_w(J_{k-1}^{(0,w)}, Z_k)\|_{L_p} \\ &+ \frac{1}{w}\,\|\overline{A}^{-1}\|\cdot\|J_0^{(1,w)} - J_n^{(1,w)}\|_{L_p}\Bigg). \end{aligned} \tag{4.138}$$

For the bilinear sum,

$$\|\sum_{k=1}^{n}\overline{\psi}_w\|_{L_p} \le c_{W,1}\,p^{3/2}\sqrt{wn} + c_{W,2}\,p^3\,w^{-1/2}\,\log^{1/2}(1/(wa)), \tag{4.139}$$

so

$$\frac{1}{\sqrt{n}}\|\sum\overline{\psi}_w\|_{L_p} \le \sqrt{2}c_{W,1}\,p^{3/2}\sqrt{\alpha} + \sqrt{2}c_{W,2}\,p^3\,(\alpha n)^{-1/2}\,\log^{1/2}(1/(\alpha a)). \tag{4.140}$$

For the boundary term, Lemma 4.7 gives

$$\|J_n^{(1,w)} - \mathbb{E}J_n^{(1,w)}\|_{L_p} \le Cw\,\Phi_+(p,w), \tag{4.141}$$

and Lemma B.3 gives $\|\mathbb{E}J_n^{(1,w)}\| \le Cw$, hence

$$\|J_n^{(1,w)}\|_{L_p} \le Cw\,\Phi_+(p,w). \tag{4.142}$$

The same bound holds for $J_0^{(1,w)}$, so

$$w^{-1}\|J_0^{(1,w)} - J_n^{(1,w)}\|_{L_p} \le C\Phi_+(p,w) \le \sqrt{2}\,C\Phi_+(p,\alpha) \tag{4.143}$$

after absorbing the factor 2 into C. Dividing by $\sqrt{n}$ gives the last term. □

Lemma 4.9. ***(Stationary second-order and depth-two remainder bound.)*** *Under the assumptions of the previous lemma, for every $p \geq 2$ and every $q \geq 2$ satisfying $p \leq q/2$ and $2\alpha \leq \alpha_{stat}(q)$,*

$$\|T_n^{(2)}\|_{L_p} + \|T_n^{(H)}\|_{L_p} \leq C_{mis,2}\,(1 + d^{1/q})\, p^{7/2}\, t_{mix}^{5/2} \sqrt{n}\, \alpha^{3/2}\, \log^{3/2}(1/(\alpha a)), \tag{4.144}$$

with $C_{mis,2} := 6\,\sqrt{2}^3\,(D_J + D_H)$.

Proof. By the triangle inequality and Lemma B.5,

$$\begin{aligned} \|T_n^{(2)}\|_{L_p} &\leq \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \sum_{w \in \{\alpha, 2\alpha\}} |c_w|\, \|J_k^{(2,w)}\|_{L_p} \\ &\leq 3\sqrt{n} \sup_{w \in \{\alpha, 2\alpha\}} \|J_k^{(2,w)}\|_{L_p} \\ &\leq 3\, D_J\, (2\alpha)^{3/2}\, t_{\mathrm{mix}}^{5/2}\, p^{7/2} \sqrt{n}\, \log^{3/2}(1/(\alpha a)), \end{aligned} \tag{4.145}$$

The same argument with Lemma B.7 controls $T_n^{(H)}$. ☐

Theorem 4.2. ***(Stationary PR-averaged RR misadjustment bound.)*** *Assume **UGE 1**, $\pi(\widetilde{A}) = 0$, $\pi(\varepsilon) = 0$, $\|\varepsilon\|_\infty < \infty$, $0 < \alpha$, and the stationary augmented-chain convention above. There exists a constant C depending on the universal and problem constants of the previous two lemmas such that for every $p \geq 2$, every $q \geq 2$ satisfying $p \leq q/2$ and $2\alpha \leq \alpha_{stat}(q)$, and every $n \geq 2$,*

$$\begin{aligned} \|R_n^{mis,\, RR}\|_{L_p} \leq{} & C\sqrt{n}\,\alpha^2 + C\,(1 + d^{1/q})\, p^{7/2}\, t_{mix}^{5/2} \sqrt{n}\, \alpha^{3/2}\, \log^{3/2}(1/(\alpha a)) \\ & + C\, p^{3/2}\sqrt{\alpha} + C\, p^3\, (\alpha n)^{-1/2}\, \log^{1/2}(1/(\alpha a)) + C\, \Phi_+(p, \alpha)\, n^{-1/2}. \end{aligned} \tag{4.146}$$

Proof. From (4.99), $\|R_n^{\mathrm{mis,\, RR}}\|_{L_p} \leq \|T_n^{(1)}\|_{L_p} + \|T_n^{(2)}\|_{L_p} + \|T_n^{(H)}\|_{L_p}$. The result follows from Lemma 4.8 and Lemma 4.9. ☐

Corollary 4.4. ***(Stationary balanced-scale misadjustment rate.)*** *At the working scale $\alpha = c\,n^{-1/2}$, with $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e\,d) \rceil, 2)$, and for n large enough that $2\alpha \leq \alpha_{stat}(q)$,*

$$\|R_n^{mis,\, RR}\|_{L_p} \leq C\, polylog(n)\, n^{-1/4}, \tag{4.147}$$

of the same polynomial order as the leading martingale Berry–Esseen rate, up to logarithmic factors.

Proof. At $\alpha = c\,n^{-1/2}$: $\sqrt{n}\alpha^2 = c^2\, n^{-1/2}$, $\sqrt{n}\alpha^{3/2} = c^{3/2}\, n^{-1/4}$, $\sqrt{\alpha} = c^{1/2}\, n^{-1/4}$, $(\alpha n)^{-1/2} = c^{-1/2}\, n^{-1/4}$, and $\Phi_+(p, \alpha)\, n^{-1/2} = O(p^{3/2}\, n^{-1/2})$. With $p \asymp \log n$ and $q \geq \log(ed)$, the displayed bound is $\mathrm{polylog}(n)\, n^{-1/4}$. ☐

This is a stationary $n_0 = 0$ augmented-chain bound; the deterministic-start version is handled in the burn-in transfer chapter.

4.11 Smoothing Assembly

We now assemble the stationary martingale bound and the composite remainder via the Bobkov–Götze smoothing inequality.

Stationary assembled statistic. Define

$$S_{n,\mathrm{stat}}^{\mathrm{RR}}(u) := -\frac{u^\top M_n^{\mathrm{RR}}}{\sqrt{n}} + u^\top \mathcal{R}_{n,\mathrm{stat}}^{\mathrm{RR}}, \tag{4.148}$$

$$\mathcal{R}_{n,\mathrm{stat}}^{\mathrm{RR}} := D_{2,n}^{\mathrm{RR}} + R_n^{\mathrm{mis,\, RR}}, \tag{4.149}$$

where both terms are read under the stationary augmented-chain convention. Dividing by $\sigma_n^{\mathrm{RR}}(u)$,

$$\frac{S_{n,\mathrm{stat}}^{\mathrm{RR}}(u)}{\sigma_n^{\mathrm{RR}}(u)} = X_n + Y_n, \tag{4.150}$$

$$X_n := -\frac{u^\top M_n^{\mathrm{RR}}}{\sqrt{n}\,\sigma_n^{\mathrm{RR}}(u)}, \quad Y_n := \frac{u^\top \mathcal{R}_{n,\mathrm{stat}}^{\mathrm{RR}}}{\sigma_n^{\mathrm{RR}}(u)}. \tag{4.151}$$

The martingale bound applies to X_n with direction $-u$.

Smoothing inequality. For any $t > 0$,

$$d_K(X+Y, \mathcal{N}(0,1)) \le d_K(X, \mathcal{N}(0,1)) + \mathbb{P}(|Y| > t) + \frac{t}{\sqrt{2\pi}}, \tag{4.152}$$

Markov's inequality with $t = e\,\|Y\|_{L_p}$ gives

$$d_K(X+Y, \mathcal{N}(0,1)) \le d_K(X, \mathcal{N}(0,1)) + \frac{e\,\|Y\|_{L_p}}{\sqrt{2\pi}} + e^{-p}, \tag{4.153}$$

L_p-bound on the composite remainder. The two stationary remainder pieces contribute additively.

Lemma 4.10. ***(Stationary composite remainder bound.)*** *Assume the hypotheses and step-size restrictions of the stationary PR-averaged RR misadjustment bound above. Then for every $u \in \mathbb{R}^d$, every $p \ge 2$, and every $q \ge 2$ satisfying $p \le q/2$ and $2\alpha \le \alpha_{stat}(q)$,*

$$\begin{aligned}\|u^\top \mathcal{R}_{n,stat}^{RR}\|_{L_p} \le \frac{C_{D2}\,\|u\|}{\sqrt{n}} + \|u\|\,C_{mis}\Bigg(&\sqrt{n}\,\alpha^2 + (1+d^{1/q})\,p^{7/2}\,t_{mix}^{5/2}\sqrt{n}\,\alpha^{3/2}\,\log^{3/2}(1/(\alpha a)) \\ &+ p^{3/2}\sqrt{\alpha} + p^3\,(\alpha n)^{-1/2}\,\log^{1/2}(1/(\alpha a)) + \Phi_+(p,\alpha)\,n^{-1/2}\Bigg),\end{aligned} \tag{4.154}$$

with constants

$$C_{D2} := 3\,t_{mix}\,\|\varepsilon\|_\infty\,(C_{\mathcal{Q}} + C_2/a^2), \tag{4.155}$$

and C_{mis} the constant in the misadjustment theorem.

Proof. Combine the sup-norm bound for $D_{2,n}^{\mathrm{RR}}$ with Theorem 4.2. □

Theorem 4.3. ***(Stationary augmented-chain Berry–Esseen assembly for the RR comparison statistic.)*** *Assume **UGE 1**, $\pi(\widetilde{A}) = 0$, $\pi(\varepsilon) = 0$, $\|\varepsilon\|_\infty < \infty$, $\sigma^2(u) > 0$, $0 < \alpha$, and the stationary augmented-chain convention above. Use the external inputs summarized in Section B. Set $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e\,d) \rceil, 2)$. Then $p \ge 2$, $q \ge 2$, and $p \le q/2$. If $n \ge 3$ satisfies the small-step and variance lower-bound conditions*

$$2\alpha \le \alpha_{stat}(q), \quad n\,\alpha\,a \ge \frac{2\,C_3\,\|u\|^2}{\sigma^2(u)}, \tag{4.156}$$

where the second inequality is (4.81)*, then*

$$\begin{aligned}d_K\left(\frac{S_{n,stat}^{RR}(u)}{\sigma_n^{RR}(u)}, \mathcal{N}(0,1)\right) \le \frac{C_{K,1}(u)\,\log^{3/4} n}{n^{1/4}} + \frac{C_{K,2}(u)\,\log n}{\sqrt{n}} \\ + \frac{e\,\|u^\top \mathcal{R}_{n,stat}^{RR}\|_{L_p}}{\sqrt{2\pi}\,\sigma_n^{RR}(u)} + \frac{e}{n},\end{aligned} \tag{4.157}$$

with $\|u^\top \mathcal{R}_{n,stat}^{RR}\|_{L_p}$ bounded by the preceding lemma, and $C_{K,1}(u), C_{K,2}(u)$ the constants of the martingale Berry–Esseen theorem.

Proof. Apply (4.153) with $X = X_n$ and $Y = Y_n$. The martingale terms come from Theorem 4.1, and the remainder term is

$$\frac{e\|Y_n\|_{L_p}}{\sqrt{2\pi}} = \frac{e\|u^\top \mathcal{R}_{n,\mathrm{stat}}^{\mathrm{RR}}\|_{L_p}}{\sqrt{2\pi}\,\sigma_n^{\mathrm{RR}}(u)}.$$

Since $p \ge \log n$, $e^{-p} \le n^{-1}$. □

Corollary 4.5. ***(Stationary balanced-scale augmented-chain Berry–Esseen bound.)*** *At the balanced scale $\alpha = c\,n^{-1/2}$ with $c > 0$, put $p = \max(2, \lceil \log n \rceil)$ and $q = \max(2p, \lceil \log(e\,d) \rceil, 2)$. Then $p \le q/2$. Assume the stationary augmented-chain hypotheses of Theorem 4.3, including $\sigma^2(u) > 0$. If $n \ge 3$ satisfies*

$$2\alpha \le \alpha_{stat}(q), \quad n\,\alpha\,a \ge \frac{2\,C_3\,\|u\|^2}{\sigma^2(u)}, \tag{4.158}$$

the bound (4.157) *reduces to*

$$d_K\left(\frac{S^{RR}_{n,stat}(u)}{\sigma^{RR}_n(u)}, \mathcal{N}(0,1)\right) \le \frac{C(u)\,polylog(n)}{n^{1/4}}, \tag{4.159}$$

where $C(u)$ depends on $\|u\|$, $\sigma(u)$, $C_{\mathcal{Q}}$, $\|\overline{A}\|$, $\|\overline{A}^{-1}\|$, κ_Q, C_A, $\|\varepsilon\|_\infty$, $\|\Sigma^{(M)}_\varepsilon\|$, t_{mix}, a, α_∞, c, and the universal constants of the smoothing inequality, Lemma B.2, and (B.1).

4.12 Chapter Summary and Results

The chapter derives the closed forms, pointwise estimates, energy bounds, and total-variation controls for the PR-averaged RR weights. These bounds feed into the Poisson martingale approximation, predictable-variation comparison, depth-two misadjustment control, and the stationary Berry–Esseen assembly.

Chapter 5

Burn-in Transfer for Deterministic Starts

This chapter transfers the stationary $n_0 = 0$ Berry–Esseen bound to the deterministic-start Richardson–Romberg average after burn-in.

For deterministic starts we strengthen the stationary condition by adding the startup/product-stability restrictions:

$$\alpha_{\mathrm{st}}(p) := \min(\alpha_{\mathrm{prod}}(p), \alpha_{\mathrm{prod}}(2p), \frac{1}{2a}), \quad \alpha_{\mathrm{burn}}(p,q) := \min(\alpha_{\mathrm{stat}}(q), \alpha_{\mathrm{st}}(p)). \tag{5.1}$$

Throughout this chapter, the burn-in small-step condition is $2\alpha \leq \alpha_{\mathrm{burn}}(p,q)$.

5.1 Target Statistic and Normalization

Let $\xi = \mathcal{L}(Z_0)$ be the initial law of the base Markov chain, let $0 \leq n_0 < n$, and set $m := n - n_0$. The burned-in PR average is

$$\overline{\theta}^{(\alpha)}_{n,n_0} := \frac{1}{m} \sum_{k=n_0}^{n-1} \theta^{(\alpha)}_k,$$

and the two-level RR average is

$$\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} := 2\overline{\theta}^{(\alpha)}_{n,n_0} - \overline{\theta}^{(2\alpha)}_{n,n_0}. \tag{5.2}$$

Both stepsizes are run from the same deterministic initial point θ_0 and on the same Markov trajectory.

The finite-start burned-in vector statistic is

$$\mathcal{T}^{\mathrm{RR}}_{n,n_0} := \sqrt{m}\left(\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} - \theta^*\right). \tag{5.3}$$

Its scalar projection in direction $u \in \mathbb{R}^d$ is

$$T^{\mathrm{RR}}_{n,n_0}(u) := u^\top \mathcal{T}^{\mathrm{RR}}_{n,n_0} = \sqrt{m}\, u^\top \left(\overline{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} - \theta^*\right). \tag{5.4}$$

There are two normalizations. The finite-window normalization is

$$\sigma^{\mathrm{bRR}}_{n,n_0}(u) := \sqrt{\sigma^{2,\mathrm{bRR}}_{n,n_0}(u)}, \quad \Xi^{\mathrm{bRR}}_{n,n_0}(u) := \frac{T^{\mathrm{RR}}_{n,n_0}(u)}{\sigma^{\mathrm{bRR}}_{n,n_0}(u)}, \tag{5.5}$$

where $\sigma^{2,\mathrm{bRR}}_{n,n_0}(u)$ is the deterministic variance proxy defined in (5.30). The asymptotic normalization is

$$\sigma(u) := \sqrt{u^\top \Sigma_\infty u}, \quad \Xi^{\mathrm{asy},\mathrm{RR}}_{n,n_0}(u) := \frac{T^{\mathrm{RR}}_{n,n_0}(u)}{\sigma(u)}. \tag{5.6}$$

We first control $\Xi^{\mathrm{bRR}}_{n,n_0}(u)$ and then pass to $\Xi^{\mathrm{asy},\mathrm{RR}}_{n,n_0}(u)$.

5.2 Burned-in Deterministic Weights

With the $\sqrt{m}$ normalization, the depth-zero linearized term has the weights

$$Q_{l;n_0,n}^{(\alpha)} := \alpha \sum_{k=\max\{n_0,l\}}^{n-1} B_\alpha^{k-l}, \quad 1 \le l \le n-1, \tag{5.7}$$

and $Q_{l;n_0,n}^{(\alpha)} = 0$ for $l \ge n$. The RR weight is

$$Q_{l;n_0,n}^{\mathrm{RR}} := 2Q_{l;n_0,n}^{(\alpha)} - Q_{l;n_0,n}^{(2\alpha)}. \tag{5.8}$$

The leading burned-in sum is

$$W_{n,n_0}^{\mathrm{RR}} := -\frac{1}{\sqrt{m}} \sum_{l=1}^{n-1} Q_{l;n_0,n}^{\mathrm{RR}} \varepsilon(Z_l). \tag{5.9}$$

5.3 Closed Forms and Burned-in Weight Bounds

Write $B_w := I - w\overline{A}$ for $w \in \{\alpha, 2\alpha\}$ and recall $m = n - n_0$. The burned-in weights have two closed forms. If $l \ge n_0$, then

$$Q_{l;n_0,n}^{(w)} = w \sum_{k=l}^{n-1} B_w^{k-l} = \overline{A}^{-1} \left(I - B_w^{n-l}\right). \tag{5.10}$$

If $l < n_0$, then

$$Q_{l;n_0,n}^{(w)} = w \sum_{k=n_0}^{n-1} B_w^{k-l} = \overline{A}^{-1} \left(B_w^{n_0-l} - B_w^{n-l}\right). \tag{5.11}$$

Only the post-burn-in weights are compared with $\overline{A}^{-1}$.

Lemma 5.1. ***(Burned-in RR weight bounds.)*** *Assume $0 < \alpha$, $2\alpha \le \alpha_\infty$, and the Lyapunov contraction* (A.3)*. Let*

$$\rho_\alpha := \sqrt{1-\alpha a}, \quad C_Q := \kappa_Q^{1/2} \|\overline{A}^{-1}\|, \quad \tilde{C}_A := \kappa_Q \|\overline{A}\|.$$

(i) Post-burn-in comparison. *If $l \ge n_0$ and $k := n - l$, then*

$$\|Q_{l;n_0,n}^{\mathrm{RR}} - \overline{A}^{-1}\| \le 3C_Q \rho_\alpha^k. \tag{5.12}$$

(ii) Pre-burn-in energy. *If $1 \le l < n_0$ and $r := n_0 - l$, then*

$$\|Q_{l;n_0,n}^{\mathrm{RR}}\| \le 6C_Q \rho_\alpha^r. \tag{5.13}$$

Consequently, there are constants $C_{\mathrm{burn,E}}$ and $C_{\mathrm{burn,V}}$, depending only on κ_Q, $\|\overline{A}^{-1}\|$, and $\|\overline{A}\|$, such that, uniformly in n and n_0,

$$\sum_{l=1}^{n_0-1} \|Q_{l;n_0,n}^{\mathrm{RR}}\|^2 + \sum_{l=\max\{1,n_0\}}^{n-1} \|Q_{l;n_0,n}^{\mathrm{RR}} - \overline{A}^{-1}\|^2 \le \frac{C_{\mathrm{burn,E}}}{\alpha a}, \tag{5.14}$$

and

$$\sum_{l=1}^{n-2} \|Q_{l+1;n_0,n}^{\mathrm{RR}} - Q_{l;n_0,n}^{\mathrm{RR}}\| \le \frac{C_{\mathrm{burn,V}}}{a^2}. \tag{5.15}$$

Proof. For $l \geq n_0$, (5.10) gives

$$Q^{\mathrm{RR}}_{l;n_0,n} - \overline{A}^{-1} = -\overline{A}^{-1}\left(2B^k_\alpha - B^k_{2\alpha}\right).$$

The same contraction argument as in the full-window estimate gives (5.12).

For $l < n_0$, set $r := n_0 - l$. By (5.11),

$$Q^{\mathrm{RR}}_{l;n_0,n} = \overline{A}^{-1}\left(2\left(B^r_\alpha - B^{r+m}_\alpha\right) - \left(B^r_{2\alpha} - B^{r+m}_{2\alpha}\right)\right).$$

Taking norms and using $\|B^j_{2\alpha}\|_Q \leq \rho^j_\alpha$ gives (5.13). Squaring (5.12) and (5.13) and summing the resulting geometric series gives (5.14).

It remains to bound the total variation. In the post-burn-in region $l, l+1 \geq n_0$, the full-window identity gives

$$Q^{\mathrm{RR}}_{l+1;n_0,n} - Q^{\mathrm{RR}}_{l;n_0,n} = -2\alpha\left(B^s_\alpha - B^s_{2\alpha}\right), \quad s := n - l - 1.$$

The elementary identity $X^s - Y^s = (X - Y)\sum_{i=1}^s X^{i-1}Y^{s-i}$ with $X = B_\alpha$ and $Y = B_{2\alpha}$ yields, for $s \geq 1$,

$$\|B^s_\alpha - B^s_{2\alpha}\| \leq \alpha\tilde{C}_A\, s\, \rho^{s-1}_\alpha. \tag{5.16}$$

The post-burn-in contribution is $O(a^{-2})$, as in the full-window proof.

For $l, l+1 < n_0$, put $s := n_0 - l - 1$. The pre-burn-in closed form gives

$$Q^{\mathrm{RR}}_{l+1;n_0,n} - Q^{\mathrm{RR}}_{l;n_0,n} = 2\alpha\left(B^s_\alpha - B^s_{2\alpha} - B^{s+m}_\alpha + B^{s+m}_{2\alpha}\right).$$

Applying (5.16) to the powers s and $s+m$ and summing over $s \geq 1$ gives $O(a^{-2})$. If the boundary $l = n_0 - 1$ exists, then

$$Q^{\mathrm{RR}}_{n_0;n_0,n} - Q^{\mathrm{RR}}_{n_0-1;n_0,n} = 2\alpha\left(B^m_{2\alpha} - B^m_\alpha\right),$$

and (5.16) with $s = m$ bounds this term by the same a^{-2} scale. Combining the pre-burn-in, boundary, and post-burn-in contributions proves (5.15). ☐

5.4 Deterministic Transient After Burn-in

For a generic step size $w \in \{\alpha, 2\alpha\}$ define the deterministic transient

$$D^{(w)}_{\mathrm{tr},n,n_0} := \frac{1}{\sqrt{m}}\sum_{k=n_0}^{n-1} B^k_w(\theta_0 - \theta^*), \quad B_w := I - w\overline{A}. \tag{5.17}$$

The Richardson–Romberg transient entering (5.4) is

$$D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u) := u^\top\left(2D^{(\alpha)}_{\mathrm{tr},n,n_0} - D^{(2\alpha)}_{\mathrm{tr},n,n_0}\right). \tag{5.18}$$

Lemma 5.2. ***(Burned-in deterministic transient.)*** *Assume* $0 < \alpha$, $2\alpha \leq \alpha_\infty$*, and the Lyapunov contraction* (A.3)*. Then, for each* $w \in \{\alpha, 2\alpha\}$,

$$\|D^{(w)}_{\mathrm{tr},n,n_0}\| \leq \frac{2\sqrt{\kappa_Q}}{wa\sqrt{m}}(1 - wa)^{n_0/2}\|\theta_0 - \theta^*\|. \tag{5.19}$$

Consequently,

$$|D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u)| \leq \frac{5\sqrt{\kappa_Q}\|u\|\|\theta_0 - \theta^*\|}{\alpha a\sqrt{m}}(1 - \alpha a)^{n_0/2}. \tag{5.20}$$

Proof. Let $r_w := \sqrt{1 - wa}$. By (A.3) and equivalence of the Euclidean and Q-norms,

$$\|B^k_w(\theta_0 - \theta^*)\| \leq \sqrt{\kappa_Q}r^k_w\|\theta_0 - \theta^*\|.$$

Therefore

$$\|D^{(w)}_{\mathrm{tr},n,n_0}\| \le \frac{\sqrt{\kappa_Q}\|\theta_0-\theta^*\|}{\sqrt{m}}\sum_{k=n_0}^{n-1} r_w^k \le \frac{\sqrt{\kappa_Q}\|\theta_0-\theta^*\|}{\sqrt{m}}\frac{r_w^{n_0}}{1-r_w}.$$

Since $1-\sqrt{1-x}\ge x/2$ for $0\le x\le 1$, the last display gives (5.19). Applying this estimate at $w=\alpha$ and $w=2\alpha$, using $1-2\alpha a\le 1-\alpha a$, and taking the triangle inequality in (5.18) gives

$$|D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u)| \le \|u\|\left(2\|D^{(\alpha)}_{\mathrm{tr},n,n_0}\| + \|D^{(2\alpha)}_{\mathrm{tr},n,n_0}\|\right) \le \frac{5\sqrt{\kappa_Q}\|u\|\|\theta_0-\theta^*\|}{\alpha a\sqrt{m}}(1-\alpha a)^{n_0/2}.$$

□

Corollary 5.1. ***(Mixing-scale burn-in removes the deterministic transient.)*** *If, for some $\beta>0$,*

$$n_0 \ge \frac{2\beta}{\alpha a}\log n, \tag{5.21}$$

then

$$|D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u)| \le \frac{5\sqrt{\kappa_Q}\|u\|\|\theta_0-\theta^*\|}{\alpha a\sqrt{m}}n^{-\beta}. \tag{5.22}$$

In particular, at the balanced scale $\alpha=cn^{-1/2}$ and $m\ge n/2$, this is $O(n^{-\beta})$.

The exact recursion also contains the random product $\Gamma^{(w)}_{1:k} := \prod_{j=1}^k (I-wA(Z_j))$.

Lemma 5.3. ***(Product stability and its conditional form.)*** *Assume lemma B.6. If $p\ge 2$, $2\alpha\le\alpha_{\mathrm{st}}(p)$, $w\in\{\alpha,2\alpha\}$, and $0\le s\le k$, then for every deterministic vector v,*

$$\mathbb{E}^{\frac{1}{p}}\left[\|\Gamma^{(w)}_{s+1:k}v\|^p|Z_s\right] \le C_{\mathrm{prod}}\exp\left(-\frac{c_{\mathrm{prod}}wa(k-s)}{p}\right)\|v\|. \tag{5.23}$$

The empty product is $\Gamma^{(w)}_{k+1:k}=I$. Moreover, on any coupling extension with a joint filtration $(\mathcal{G}_r)$ such that the future product, conditionally on $\mathcal{G}_s$, has the base-chain law started from Z_s, every $\mathcal{G}_s$-measurable vector W_s satisfies

$$\mathbb{E}^{\frac{1}{p}}\left[\|\Gamma^{(w)}_{s+1:k}W_s\|^p|\mathcal{G}_s\right] \le C_{\mathrm{prod}}\exp\left(-\frac{c_{\mathrm{prod}}wa(k-s)}{p}\right)\|W_s\|. \tag{5.24}$$

Consequently,

$$\|\Gamma^{(w)}_{s+1:k}V_s\|_{L_p} \le C_{\mathrm{prod}}\exp\left(-\frac{c_{\mathrm{prod}}wa(k-s)}{p}\right)\|V_s\|_{L_p} \le C_{\mathrm{prod}}\exp\left(-\frac{c_{\mathrm{prod}}wa(k-s)}{p}\right)\|V_s\|_{L_{2p}} \tag{5.25}$$

Proof. Fix $s\le k$. The adapted-product assumption gives

$$\mathcal{L}\left(\Gamma^{(w)}_{s+1:k}|\mathcal{G}_s\right) = \mathcal{L}\left(\Gamma^{(w)}_{s+1:k}|Z_s\right).$$

Set

$$\Phi_s(v) := \mathbb{E}^{\frac{1}{p}}\left[\|\Gamma^{(w)}_{s+1:k}v\|^p|Z_s\right].$$

Hence, for $\mathcal{G}_s$-measurable W_s,

$$\mathbb{E}^{\frac{1}{p}}\left[\|\Gamma^{(w)}_{s+1:k}W_s\|^p|\mathcal{G}_s\right] \quad = \Phi_s(W_s) \quad \le C_{\mathrm{prod}}\exp\left(-\frac{c_{\mathrm{prod}}wa(k-s)}{p}\right)\|W_s\|,$$

where the last step is (5.23). This is (5.24); the case $s=k$ follows after taking $C_{\mathrm{prod}}\ge 1$.

For $V_s\in L_{2p}$,

$$\|\Gamma^{(w)}_{s+1:k}V_s\|^p_{L_p} = \mathbb{E}\mathbb{E}\left[\|\Gamma^{(w)}_{s+1:k}V_s\|^p|\mathcal{G}_s\right] \quad \le C^p_{\mathrm{prod}}\exp\left(-c_{\mathrm{prod}}wa(k-s)\right)\mathbb{E}\|V_s\|^p.$$

Taking p-th roots and using $\|V_s\|_{L_p}\le\|V_s\|_{L_{2p}}$ gives (5.25). The natural-filtration case corresponds to $\mathcal{G}_s=\sigma(\{Z_r : r\le s\})$. □

Define the accumulated RR random initial-product discrepancy by

$$\mathcal{I}^{\mathrm{init,RR}}_{n,n_0}(u) := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} u^\top \left[2\left(\Gamma^{(\alpha)}_{1:k} - B^k_\alpha\right) - \left(\Gamma^{(2\alpha)}_{1:k} - B^k_{2\alpha}\right)\right] (\theta_0 - \theta^*), \tag{5.26}$$

where the empty product at $k = 0$ is the identity.

Lemma 5.4. ***(Burned-in random initial-product transient.)*** *Let $p \geq 2$. Assume $0 < \alpha$, $2\alpha \leq \alpha_{\mathrm{st}}(p)$, $2\alpha \leq \alpha_\infty$, $\alpha a \leq 1/4$, and the Lyapunov contraction* (A.3). *Then*

$$\|\mathcal{I}^{\mathrm{init,RR}}_{n,n_0}(u)\|_{L_p} \leq \frac{C_{\mathrm{init,RR}}\|u\|\|\theta_0 - \theta^*\|p}{\alpha a\sqrt{m}} \exp\left(-c_{\mathrm{init}}\alpha a n_0/p\right), \tag{5.27}$$

where $C_{\mathrm{init,RR}} < \infty$ and $c_{\mathrm{init}} > 0$ depend only on C_{prod}, c_{prod}, and the Lyapunov norm-equivalence constant.

Proof. Let $e_0 := \theta_0 - \theta^*$. By lemma 5.3 with $s = 0$ and $V_0 = e_0$,

$$\|\Gamma^{(w)}_{1:k} e_0\|_{L_p} \leq C_{\mathrm{prod}} \exp\left(-\frac{c_{\mathrm{prod}} wak}{p}\right) \|e_0\|.$$

The deterministic Lyapunov contraction gives

$$\|B^k_w e_0\| \leq \sqrt{\kappa_Q}(1 - wa)^{k/2}\|e_0\| \leq \sqrt{\kappa_Q} \exp\left(-wak/(2p)\right) \|e_0\|,$$

since $p \geq 1$. Hence, after decreasing the exponential constant,

$$\|\left(\Gamma^{(w)}_{1:k} - B^k_w\right) e_0\|_{L_p} \leq C \exp\left(-c_{\mathrm{init}} wak/p\right) \|e_0\|.$$

Apply this estimate at $w = \alpha$ and $w = 2\alpha$, use the triangle inequality in (5.26), and extend the geometric sum to infinity:

$$\|\mathcal{I}^{\mathrm{init,RR}}_{n,n_0}(u)\|_{L_p} \leq \frac{C\|u\|\|e_0\|}{\sqrt{m}} \sum_{k=n_0}^{\infty} \exp\left(-c_{\mathrm{init}}\alpha ak/p\right).$$

Because $\alpha a \leq 1/4$ and $p \geq 2$, $1 - \exp\left(-c_{\mathrm{init}}\alpha a/p\right) \geq C^{-1}\alpha a/p$, which gives (5.27). ☐

Corollary 5.2. ***(Mixing-scale burn-in removes the random initial-product discrepancy.)*** *If, for some $\beta > 0$,*

$$n_0 \geq \frac{\beta p}{c_{\mathrm{init}}\alpha a} \log n, \tag{5.28}$$

then

$$\|\mathcal{I}^{\mathrm{init,RR}}_{n,n_0}(u)\|_{L_p} \leq \frac{C_{\mathrm{init,RR}}\|u\|\|\theta_0 - \theta^*\|p}{\alpha a\sqrt{m}} n^{-\beta}. \tag{5.29}$$

At the balanced scale $\alpha = cn^{-1/2}$, with $m \geq n/2$ and p logarithmic in n, this is $\mathrm{polylog}(n)\, n^{-\beta}$.

5.5 Burned-in Variance Proxy

For the stochastic depth-zero term, write $Q^{\mathrm{bRR}}_l := Q^{\mathrm{RR}}_{l;n_0,n}$ and $\Sigma := \Sigma^{(\mathrm{M})}_\varepsilon$. We take $n \geq 2$ in this subsection. The martingale part produced by the Poisson decomposition starts at $l = 2$, so the deterministic variance proxy is

$$\Sigma^{\mathrm{bRR}}_{n,n_0} := \frac{1}{m} \sum_{l=2}^{n-1} Q^{\mathrm{bRR}}_l \, \Sigma \, (Q^{\mathrm{bRR}}_l)^\top, \quad \sigma^{2,\mathrm{bRR}}_{n,n_0}(u) := u^\top \Sigma^{\mathrm{bRR}}_{n,n_0} u. \tag{5.30}$$

Lemma 5.5. ***(Burned-in variance comparison.)*** *Assume the conditions of lemma 5.1 and $\|\Sigma\| < \infty$. There exists a constant $C_{\mathrm{burn},3}$, depending only on κ_Q, $\|\overline{A}^{-1}\|$, $\|\overline{A}\|$, and $\|\Sigma\|$, such that*

$$\|\Sigma^{\mathrm{bRR}}_{n,n_0} - \Sigma_\infty\| \le \frac{C_{\mathrm{burn},3}}{m\,\alpha a}. \tag{5.31}$$

Consequently,

$$|\sigma^{2,\mathrm{bRR}}_{n,n_0}(u) - \sigma^2(u)| \le \frac{C_{\mathrm{burn},3}\|u\|^2}{m\,\alpha a}. \tag{5.32}$$

Proof. Put

$$I_{\mathrm{post}} := \{l : 2 \le l \le n-1, l \ge n_0\}, \quad I_{\mathrm{pre}} := \{l : 2 \le l < n_0\}, \quad r_m := m - |I_{\mathrm{post}}| \in \{0,1,2\}.$$

For $l \in I_{\mathrm{post}}$, set $\Delta_l := Q^{\mathrm{bRR}}_l - \overline{A}^{-1}$. Then

$$Q^{\mathrm{bRR}}_l \Sigma (Q^{\mathrm{bRR}}_l)^\mathsf{T} - \Sigma_\infty = \Delta_l \Sigma \overline{A}^{-\mathsf{T}} + \overline{A}^{-1}\Sigma\Delta_l^\mathsf{T} + \Delta_l\Sigma\Delta_l^\mathsf{T}.$$

Therefore

$$\Sigma^{\mathrm{bRR}}_{n,n_0} - \Sigma_\infty = \frac{1}{m}\sum_{l\in I_{\mathrm{post}}}\Big(Q^{\mathrm{bRR}}_l \Sigma (Q^{\mathrm{bRR}}_l)^\mathsf{T} - \Sigma_\infty\Big) \qquad + \frac{1}{m}\sum_{l\in I_{\mathrm{pre}}} Q^{\mathrm{bRR}}_l \Sigma (Q^{\mathrm{bRR}}_l)^\mathsf{T} - \frac{r_m}{m}\Sigma_\infty.$$

The post-burn-in contribution is bounded by

$$\sum_{l\in I_{\mathrm{post}}} \|Q^{\mathrm{bRR}}_l \Sigma (Q^{\mathrm{bRR}}_l)^\mathsf{T} - \Sigma_\infty\| \qquad \le 2\|\overline{A}^{-1}\|\|\Sigma\| \sum_{l\in I_{\mathrm{post}}}\|\Delta_l\| + \|\Sigma\|\sum_{l\in I_{\mathrm{post}}}\|\Delta_l\|^2 \quad \le \frac{C}{\alpha a},$$

where (5.12) gives the first geometric sum and (5.14) gives the squared sum. For the pre-burn-in part, (5.14) gives

$$\sum_{l\in I_{\mathrm{pre}}} \|Q^{\mathrm{bRR}}_l \Sigma (Q^{\mathrm{bRR}}_l)^\mathsf{T}\| \le \|\Sigma\|\sum_{l\in I_{\mathrm{pre}}}\|Q^{\mathrm{bRR}}_l\|^2 \le \frac{C}{\alpha a}.$$

Finally,

$$\frac{r_m}{m}\|\Sigma_\infty\| \le \frac{2\|\overline{A}^{-1}\|^2\|\Sigma\|}{m} \le \frac{C}{m\alpha a},$$

using $\alpha a \le 1$. Combining the last three displays and dividing by m gives (5.31). Also,

$$|\sigma^{2,\mathrm{bRR}}_{n,n_0}(u) - \sigma^2(u)| = |u^\mathsf{T}(\Sigma^{\mathrm{bRR}}_{n,n_0} - \Sigma_\infty)u| \le \|\Sigma^{\mathrm{bRR}}_{n,n_0} - \Sigma_\infty\|\|u\|^2,$$

which gives (5.32). □

Assume $\sigma^2(u) > 0$ and impose the burned-in variance lower-bound condition

$$m\,\alpha\,a \ge \frac{2C_{\mathrm{burn},3}\|u\|^2}{\sigma^2(u)}. \tag{5.33}$$

Then (5.32) gives $\sigma^{2,\mathrm{bRR}}_{n,n_0}(u) \ge \sigma^2(u)/2$, hence $\sigma^{\mathrm{bRR}}_{n,n_0}(u) \ge \sigma(u)/\sqrt{2}$.

We say that $(n, n_0, p, q, \alpha, u)$ is in the *admissible burn-in regime* if $m := n - n_0$ and

$$n \ge 3, \quad 0 \le n_0 < n, \quad p \ge 2, \quad q \ge 2, \quad p \le q/4, \quad \sigma^2(u) > 0,$$

$$m \ge n/2, \quad 0 < \alpha, \quad 2\alpha \le \alpha_{\mathrm{burn}}(p,q), \quad m\,\alpha\,a \ge \frac{2C_{\mathrm{burn},3}\|u\|^2}{\sigma^2(u)}. \tag{5.34}$$

The last inequality is (5.33).

5.6 Burned-in Poisson Martingale Approximation

Let $\hat{\varepsilon} := \sum_{j=0}^{\infty} \mathsf{Q}^j \varepsilon$ be the Poisson solution, so

$$\hat{\varepsilon} - \mathsf{Q}\hat{\varepsilon} = \varepsilon, \quad \|\hat{\varepsilon}\|_\infty \le 3\, t_{\mathrm{mix}}\, \|\varepsilon\|_\infty.$$

Lemma 5.6. ***(Burned-in Poisson martingale decomposition.)*** *Assume* ***UGE 1*** *and* $\pi(\varepsilon) = 0$. *Define, for* $2 \le l \le n-1$,

$$\Delta M_l^{\mathrm{bRR}} := Q_l^{\mathrm{bRR}} \left(\hat{\varepsilon}(Z_l) - \mathsf{Q}\hat{\varepsilon}(Z_{l-1})\right),$$

and set $M_{n,n_0}^{\mathrm{bRR}} := \sum_{l=2}^{n-1} \Delta M_l^{\mathrm{bRR}}$. *Then* $\{\Delta M_l^{\mathrm{bRR}}\}_{l=2}^{n-1}$ *is a sequence of* $\mathcal{F}_l$*-martingale differences and*

$$W_{n,n_0}^{\mathrm{RR}} = -\frac{1}{\sqrt{m}} M_{n,n_0}^{\mathrm{bRR}} + D_{2,n,n_0}^{\mathrm{bRR}}, \tag{5.35}$$

where

$$D_{2,n,n_0}^{\mathrm{bRR}} := -\frac{1}{\sqrt{m}} \left[Q_1^{\mathrm{bRR}}\, \hat{\varepsilon}(Z_1) + \sum_{l=1}^{n-2} \left(Q_{l+1}^{\mathrm{bRR}} - Q_l^{\mathrm{bRR}}\right) \mathsf{Q}\hat{\varepsilon}(Z_l) \right].$$

Moreover, with $C_{\mathrm{burn,Q}} := \|\overline{A}^{-1}\| + 6C_Q$,

$$\|D_{2,n,n_0}^{\mathrm{bRR}}\|_\infty \le \frac{3\, t_{\mathrm{mix}}\, \|\varepsilon\|_\infty}{\sqrt{m}} \left(C_{\mathrm{burn,Q}} + \frac{C_{\mathrm{burn,V}}}{a^2} \right). \tag{5.36}$$

5.7 Burned-in Predictable Variance Comparison

For the martingale in lemma 5.6 define

$$\mathcal{V}_\varepsilon(z) := \mathsf{Q}(\hat{\varepsilon}\hat{\varepsilon}^\top)(z) - (\mathsf{Q}\hat{\varepsilon})(z)\, (\mathsf{Q}\hat{\varepsilon})(z)^\top. \tag{5.37}$$

By the Poisson covariance identity lemma 4.4, $\pi(\mathcal{V}_\varepsilon) = \Sigma$ and

$$\|\mathcal{V}_\varepsilon\|_\infty \le 18\, t_{\mathrm{mix}}^2\, \|\varepsilon\|_\infty^2. \tag{5.38}$$

The predictable quadratic variation is

$$\langle M^{\mathrm{bRR}} \rangle_{n,n_0} := \sum_{l=2}^{n-1} \mathbb{E}[\Delta M_l^{\mathrm{bRR}}\, (\Delta M_l^{\mathrm{bRR}})^\top | \mathcal{F}_{l-1}] = \sum_{l=2}^{n-1} Q_l^{\mathrm{bRR}}\, \mathcal{V}_\varepsilon(Z_{l-1})\, (Q_l^{\mathrm{bRR}})^\top. \tag{5.39}$$

Lemma 5.7. ***(Burned-in predictable-variation concentration.)*** *Assume* ***UGE 1***, $\pi(\varepsilon) = 0$, *and* $\|\varepsilon\|_\infty < \infty$. *There exists a universal constant* $C_4 > 0$ *such that, for every* $u \in \mathbb{R}^d$, *every* $p \ge 2$, *every initial distribution* ξ, *every* $n \ge 2$, *and every* $0 \le n_0 < n$,

$$\mathbb{E}_\xi^{1/p} \left[|u^\top \langle M^{\mathrm{bRR}} \rangle_{n,n_0} u - m\, \sigma_{n,n_0}^{2,\mathrm{bRR}}(u)|^p \right] \le C_4\, C_{\mathrm{burn,Q}}^2\, \|u\|^2\, \|\varepsilon\|_\infty^2\, t_{\mathrm{mix}}^{5/2} \sqrt{p\, n}. \tag{5.40}$$

Proof. Set

$$h_l(z) := u^\top Q_l^{\mathrm{bRR}}\, \mathcal{V}_\varepsilon(z)\, (Q_l^{\mathrm{bRR}})^\top u, \quad g_l(z) := h_l(z) - \pi(h_l).$$

By (5.38) and $\|Q_l^{\mathrm{bRR}}\| \le C_{\mathrm{burn,Q}}$,

$$\|g_l\|_\infty \le 36\, C_{\mathrm{burn,Q}}^2\, \|u\|^2\, t_{\mathrm{mix}}^2\, \|\varepsilon\|_\infty^2.$$

Moreover, (5.39) and (5.30) imply

$$u^\top \langle M^{\mathrm{bRR}} \rangle_{n,n_0} u - m\, \sigma_{n,n_0}^{2,\mathrm{bRR}}(u) = \sum_{l=2}^{n-1} g_l(Z_{l-1}).$$

Apply (B.1) to the time-inhomogeneous centered functions g_l with

$$c_l := 36\, C_{\mathrm{burn,Q}}^2\, \|u\|^2\, t_{\mathrm{mix}}^2\, \|\varepsilon\|_\infty^2,$$

it gives

$$\|\sum_{l=2}^{n-1} g_l(Z_{l-1})\|_{L_p(\xi)} \le C_{\mathrm{MC}} \sqrt{p\, t_{\mathrm{mix}} \sum_{l=2}^{n-1} c_l^2} \le C\, C_{\mathrm{burn,Q}}^2\, \|u\|^2\, \|\varepsilon\|_\infty^2\, t_{\mathrm{mix}}^{5/2} \sqrt{p\, n},$$

which is (5.40) after increasing C_4. ☐

Corollary 5.3. ***(Burned-in asymptotic-variance bracket comparison.)*** *Under the assumptions of lemma 5.7,*

$$\mathbb{E}_\xi^{1/p}\left[|u^\mathsf{T} \langle M^{\mathrm{bRR}} \rangle_{n,n_0} u - m\, \sigma^2(u)|^p\right] \le C_4\, C_{\mathrm{burn,Q}}^2\, \|u\|^2\, \|\varepsilon\|_\infty^2\, t_{\mathrm{mix}}^{5/2} \sqrt{p\, n} + \frac{C_{\mathrm{burn,3}} \|u\|^2}{\alpha a}. \tag{5.41}$$

Proof. Use the triangle inequality and (5.32):

$$m\, |\sigma_{n,n_0}^{2,\mathrm{bRR}}(u) - \sigma^2(u)| \le \frac{C_{\mathrm{burn,3}} \|u\|^2}{\alpha a}.$$

☐

5.8 Startup Transfer for Augmented-Chain Remainders

The remaining startup discrepancy is the difference between the finite-start and stationary augmented-chain remainders.

For $w \in \{\alpha, 2\alpha\}$ write

$$Y_k^{(w)} := (Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)}).$$

For every admissible $w \in \{\alpha, 2\alpha\}$, let

$$(Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)}, H_k^{(2,w)})_{k \in \mathbb{Z}}$$

be the stationary copy from lemma 4.6.

Lemma 5.8. ***(Conditional product stability at a coupling time.)*** *On a coupling space with joint filtration $(\mathcal{G}_s)_{s \ge 0}$, let T be the integer-valued exact-coupling stopping time of two base chains. Assume, uniformly over the initial states used below,*

$$\mathbb{P}(T > r) \le C_T \exp\left(-c_T r\right), \quad r \ge 0. \tag{5.42}$$

Products are taken along the post-coupling chain, with $\Gamma_{k+1:k}^{(w)} = I$. Then, for every $p \ge 2$, $k \ge 0$, and $w \in \{\alpha, 2\alpha\}$ for which lemma 5.3 applies and $wa \le 1$:

(i) Random restart. *If $(V_s)_{s \ge 0}$ is $\mathcal{G}_s$-adapted and $\sup_s \|V_s\|_{L_{2p}} \le B$, then*

$$\|\Gamma_{T+1:k}^{(w)} V_T 1_{T \le k}\|_{L_p} \le C \frac{p}{wa} B \exp\left(-\frac{cwak}{p}\right). \tag{5.43}$$

(ii) Stable convolution. *If $c_0 > 0$, $(U_l)_{l \ge 0}$ is $\mathcal{G}_l$-adapted, and, for all l,*

$$\|U_l\|_{L_{2p}} \le B \exp\left(-\frac{c_0 wal}{p}\right),$$

then

$$w \sum_{l=1}^{k} \|\Gamma_{l+1:k}^{(w)} U_l\|_{L_p} \le C \frac{p}{a} B \exp\left(-\frac{cwak}{p}\right). \tag{5.44}$$

Proof. For $s \le k$ and $\mathcal{G}_s$-measurable W_s,

$$\mathbb{E}\left[\|\Gamma^{(w)}_{s+1:k} W_s\|^p | \mathcal{G}_s\right]^{\frac{1}{p}} \le C_{\text{prod}} \exp\left(-\frac{c_{\text{prod}} wa(k-s)}{p}\right) \|W_s\|. \tag{5.45}$$

and therefore

$$\|\Gamma^{(w)}_{s+1:k} W_s\|_{L_p} \le C_{\text{prod}} \exp\left(-\frac{c_{\text{prod}} wa(k-s)}{p}\right) \|W_s\|_{L_p}. \tag{5.46}$$

For $V_s 1_{T=s}$, which is $\mathcal{G}_s$-measurable, $\Gamma^{(w)}_{T+1:k} V_T 1_{T\le k} = \sum_{s=0}^k \Gamma^{(w)}_{s+1:k} V_s 1_{T=s}$,

$$\|\Gamma^{(w)}_{T+1:k} V_T 1_{T\le k}\|_{L_p} \le C \sum_{s=0}^{k} \exp\left(-\frac{cwa(k-s)}{p}\right) \|V_s 1_{T=s}\|_{L_p}.$$

$$\|V_s 1_{T=s}\|_{L_p} \le \|V_s\|_{L_{2p}} \mathbb{P}(T=s)^{1/(2p)} \le B\, \mathbb{P}(T=s)^{1/(2p)}.$$

$$\mathbb{P}(T=s)^{1/(2p)} \le C \exp\left(-\frac{c_T s}{4p}\right).$$

$$\sum_{s=0}^{k} \exp\left(-\frac{cwa(k-s)}{p}\right) \exp\left(-\frac{c_T was}{4p}\right) \le C \frac{p}{wa} \exp\left(-\frac{c' wak}{p}\right).$$

This proves (5.43).

For (ii), (5.46) gives

$$\|\Gamma^{(w)}_{l+1:k} U_l\|_{L_p} \le C \exp\left(-\frac{cwa(k-l)}{p}\right) \|U_l\|_{L_p} \le C \exp\left(-\frac{cwa(k-l)}{p}\right) B \exp\left(-\frac{c_0 wal}{p}\right).$$

$$w \sum_{l=1}^{k} \|\Gamma^{(w)}_{l+1:k} U_l\|_{L_p} \le CwB \sum_{l=1}^{k} \exp\left(-\frac{cwa(k-l)}{p}\right) \exp\left(-\frac{c_0 wal}{p}\right).$$

The last sum is bounded by

$$C \frac{p}{wa} \exp\left(-\frac{c' wak}{p}\right),$$

hence (5.44). ∎

For $w \in \{\alpha, 2\alpha\}$ write

$$R^{(w)}_{k,\text{fin}} := J^{(1,w)}_{k,\text{fin}} + J^{(2,w)}_{k,\text{fin}} + H^{(2,w)}_{k,\text{fin}}$$

and

$$R^{(w)}_{k,\text{aug}} := J^{(1,w)}_{k,\text{aug}} + J^{(2,w)}_{k,\text{aug}} + H^{(2,w)}_{k,\text{aug}}.$$

Here the augmented copy is initialized from the invariant law of

$$(Z_{k+1}, J^{(0,w)}_k, J^{(1,w)}_k, J^{(2,w)}_k, H^{(2,w)}_k).$$

Lemma 5.9. ***(Full-state startup contraction for the depth-two augmented remainder.)*** *Assume **UGE 1**, $\pi(\varepsilon) = 0$, $\|\varepsilon\|_\infty < \infty$, and $0 < \alpha$. There exist constants $c_{\text{st}} > 0$ and $C_{\text{st}} < \infty$, depending only on the problem constants and on the constants in* (B.6) *and lemma B.7, such that for every $p \ge 2$, every $q \ge 2$ satisfying $p \le q/4$ and $2\alpha \le \alpha_{\text{burn}}(p, q)$, every initial distribution ξ of the base chain with finite-start perturbation coordinates initialized at zero, every $w \in \{\alpha, 2\alpha\}$, and every $k \ge 0$, the finite-start and stationary augmented remainders can be coupled so that*

$$\|R^{(w)}_{k,\text{fin}} - R^{(w)}_{k,\text{aug}}\|_{L_p} \le A_{\text{st}}(p, q, w) \exp\left(-c_{\text{st}} wak/p\right), \tag{5.47}$$

where

$$A_{\text{st}}(p, q, w) := C_{\text{st}}(1 + d^{1/q}) \left(p^7 + \frac{p^8}{a}\right) t^5_{\text{mix}} \sqrt{w/a} \log^3(1/(wa)).$$

Proof. Couple the finite-start copy to the invariant full augmented chain of lemma 4.6. Let

$$\Delta J_k^{(\ell,w)} := J_{k,\mathrm{fin}}^{(\ell,w)} - J_{k,\mathrm{aug}}^{(\ell,w)}, \quad \Delta H_k^{(2,w)} := H_{k,\mathrm{fin}}^{(2,w)} - H_{k,\mathrm{aug}}^{(2,w)}.$$

Let T be the exact-coupling time from lemma B.8. UGE gives $\mathbb{P}(T > r) \le C_T \exp(-c_T r)$, uniformly in ξ.

The $J^{(1)} + J^{(2)}$ part. By lemma B.8 and (B.5), for $\ell \in \{0, 1, 2\}$,

$$\|J_{k,\mathrm{fin}}^{(\ell,w)} - J_{k,\mathrm{aug}}^{(\ell,w)}\|_{L_p} \le C p^{7/2} t_{\mathrm{mix}}^{5/2} \log^{3/2}(1/(wa)) e^{-cwak/p} \mathbb{E}^{1/p}[c_{J,2}^{(w)}(Y_0^{\mathrm{fin}}, Y_0^{\mathrm{aug}})^p]. \tag{5.48}$$

The finite-start J coordinates are zero, and the stationary copy satisfies

$$\mathbb{E}^{1/p}[c_{J,2}^{(w)}(Y_0^{\mathrm{fin}}, Y_0^{\mathrm{aug}})^p] \le C(1 + d^{1/q}) p^{7/2} t_{\mathrm{mix}}^{5/2} \sqrt{w/a} \log^{3/2}(1/(wa)). \tag{5.49}$$

Put

$$A_{\mathrm{J}}(p, q, w) := C(1 + d^{1/q}) p^7 t_{\mathrm{mix}}^5 \sqrt{w/a} \log^3(1/(wa)). \tag{5.50}$$

$$\|\Delta J_k^{(1,w)}\|_{L_p} + \|\Delta J_k^{(2,w)}\|_{L_p} \le A_{\mathrm{J}}(p, q, w) \exp(-cwak/p). \tag{5.51}$$

The $H^{(2)}$ part. By lemma B.7 and Fatou,

$$\|H_s^{(2,w)}\|_{L_{2p}} \le B_H(p, q, w) := C(1 + d^{1/q}) p^{7/2} t_{\mathrm{mix}}^{5/2} w^{3/2} \log^{3/2}(1/(wa)), \tag{5.52}$$

$$\begin{aligned} &B_H(p, q, w) \le C A_{\mathrm{J}}(p, q, w), \\ &\frac{p}{wa} B_H(p, q, w) \le C \frac{p}{a} A_{\mathrm{J}}(2p, q, w), \\ &\frac{p}{a} A_{\mathrm{J}}(2p, q, w) \le C(1 + d^{1/q}) \frac{p^8}{a} t_{\mathrm{mix}}^5 \sqrt{w/a} \log^3(1/(wa)). \end{aligned} \tag{5.53}$$

On $\{T \le k\}$, (B.4) gives

$$H_k^{(2,w)} = \Gamma_{T+1:k}^{(w)} H_T^{(2,w)} - w \sum_{l=T+1}^{k} \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) J_{l-1}^{(2,w)},$$

$$\Delta H_k^{(2,w)} = \Gamma_{T+1:k}^{(w)} \Delta H_T^{(2,w)} - w \sum_{l=T+1}^{k} \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)}, \tag{5.54}$$

$$\|\Delta H_k^{(2,w)} 1_{T>k}\|_{L_p} \le \left(\|H_{k,\mathrm{fin}}^{(2,w)}\|_{L_{2p}} + \|H_{k,\mathrm{aug}}^{(2,w)}\|_{L_{2p}} \right) \mathbb{P}(T > k)^{1/(2p)}.$$

$$\|\Delta H_k^{(2,w)} 1_{T>k}\|_{L_p} \le C B_H(p, q, w) e^{-ck/p}. \tag{5.55}$$

Set $V_s := \Delta H_s^{(2,w)}$. Then

$$\sup_s \|V_s\|_{L_{2p}} \le \|H_{s,\mathrm{fin}}^{(2,w)}\|_{L_{2p}} + \|H_{s,\mathrm{aug}}^{(2,w)}\|_{L_{2p}} \le C B_H(p, q, w).$$

By (5.43),

$$\|\Gamma_{T+1:k}^{(w)} \Delta H_T^{(2,w)} 1_{T \le k}\|_{L_p} \le C \frac{p}{wa} B_H(p, q, w) e^{-cwak/p} \le C A_{\mathrm{st}}(p, q, w) e^{-cwak/p}. \tag{5.56}$$

For $U_l := \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)}$,

$$\|U_l\|_{L_{2p}} \le C A_{\mathrm{J}}(2p, q, w) \exp(-cwal/p),$$

$$w \sum_{l=1}^{k} \|\Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) \Delta J_{l-1}^{(2,w)}\|_{L_p} \qquad \le C A_{\mathrm{st}}(p, q, w) e^{-cwak/p}. \tag{5.57}$$

$$\|\Delta H_k^{(2,w)}\|_{L_p} \le C A_{\mathrm{st}}(p, q, w) e^{-cwak/p}. \tag{5.58}$$

$$R_{k,\mathrm{fin}}^{(w)} - R_{k,\mathrm{aug}}^{(w)} = \Delta J_k^{(1,w)} + \Delta J_k^{(2,w)} + \Delta H_k^{(2,w)}.$$

Together with (5.51) this proves (5.47). □

Define the RR startup discrepancy accumulated over the burned-in window by

$$\mathcal{U}_{n,n_0}^{\mathrm{start,RR}} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \left[2\left(R_{k,\mathrm{fin}}^{(\alpha)} - R_{k,\mathrm{aug}}^{(\alpha)} \right) - \left(R_{k,\mathrm{fin}}^{(2\alpha)} - R_{k,\mathrm{aug}}^{(2\alpha)} \right) \right]. \tag{5.59}$$

Lemma 5.10. ***(Accumulated startup transfer.)*** *Under lemma 5.9, if $0 < \alpha$, then for every $p \geq 2$ and every $q \geq 2$ satisfying $p \leq q/4$ and $2\alpha \leq \alpha_{\mathrm{burn}}(p,q)$,*

$$\|\mathcal{U}_{n,n_0}^{\mathrm{start,RR}}\|_{L_p} \leq \frac{C_{\mathrm{start,RR}} p\, A_{\mathrm{st}}(p,q,\alpha)}{\alpha a \sqrt{m}} \exp\left(-c_{\mathrm{st}} \alpha a n_0 / p\right). \tag{5.60}$$

Proof. By (5.47) at $w = \alpha, 2\alpha$,

$$\|\mathcal{U}_{n,n_0}^{\mathrm{start,RR}}\|_{L_p} \leq \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \left(2A_{\mathrm{st}}(p,q,\alpha) e^{-c_{\mathrm{st}}\alpha a k/p} + A_{\mathrm{st}}(p,q,2\alpha) e^{-2c_{\mathrm{st}}\alpha a k/p} \right).$$

$$A_{\mathrm{st}}(p,q,2\alpha) \leq C A_{\mathrm{st}}(p,q,\alpha), \quad 1 - \exp\left(-c_{\mathrm{st}}\alpha a/p\right) \geq C^{-1} \alpha a/p.$$

The geometric sum gives (5.60). □

Corollary 5.4. ***(Mixing-scale burn-in removes the startup discrepancy.)*** *If, for some $\beta > 0$,*

$$n_0 \geq \frac{\beta p}{c_{\mathrm{st}} \alpha a} \log n, \tag{5.61}$$

then

$$\|\mathcal{U}_{n,n_0}^{\mathrm{start,RR}}\|_{L_p} \leq \frac{C_{\mathrm{start,RR}} p\, A_{\mathrm{st}}(p,q,\alpha)}{\alpha a \sqrt{m}} n^{-\beta}. \tag{5.62}$$

At the balanced scale $\alpha = cn^{-1/2}$, with $m \geq n/2$ and p, q logarithmic in n, this is $\mathrm{polylog}(n)\, n^{-1/4-\beta}$.

5.9 Burned-in Depth-Two Misadjustment Bound

Define the finite-start burned-in RR misadjustment by

$$R_{n,n_0,\mathrm{fin}}^{\mathrm{mis,RR}} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \left(2R_{k,\mathrm{fin}}^{(\alpha)} - R_{k,\mathrm{fin}}^{(2\alpha)} \right). \tag{5.63}$$

For comparison, let

$$R_{m,\mathrm{aug}}^{\mathrm{mis,RR}} := \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} \left(2R_{j,\mathrm{aug}}^{(\alpha)} - R_{j,\mathrm{aug}}^{(2\alpha)} \right) \tag{5.64}$$

be the stationary augmented-chain depth-two misadjustment over a window of length m.

Theorem 5.1. ***(Burned-in PR-averaged RR misadjustment bound.)*** *Assume **UGE 1**, $\pi(\varepsilon) = 0$, $\|\varepsilon\|_\infty < \infty$, and $0 < \alpha$. Set*

$$\Phi_+(p,\alpha) := 1 + p^{3/2}\, t_{\mathrm{mix}}^{1/2}/a + p^{1/2}\, t_{\mathrm{mix}}^{3/2} \sqrt{\alpha/a}.$$

There exists a constant $C_{\mathrm{burn,mis}}$ depending only on the stationary misadjustment constants, the startup-contraction constants, and the problem constants such that, for every $p \geq 2$, every $q \geq 2$ satisfying $p \leq q/4$ and $2\alpha \leq \alpha_{\mathrm{burn}}(p,q)$, and every $m \geq 2$,

$$\begin{aligned} \|R_{n,n_0,\mathrm{fin}}^{\mathrm{mis,RR}}\|_{L_p} \leq C_{\mathrm{burn,mis}} \Big(&\sqrt{m}\, \alpha^2 + (1 + d^{1/q})\, p^{7/2}\, t_{\mathrm{mix}}^{5/2} \sqrt{m}\, \alpha^{3/2} \log^{3/2}(1/(\alpha a)) \\ &+ p^{3/2} \sqrt{\alpha} + p^3 (\alpha m)^{-1/2} \log^{1/2}(1/(\alpha a)) \\ &+ \Phi_+(p,\alpha)\, m^{-1/2} + \frac{p\, A_{\mathrm{st}}(p,q,\alpha)}{\alpha a \sqrt{m}} \exp\left(-c_{\mathrm{st}} \alpha a n_0/p\right) \Big). \end{aligned} \tag{5.65}$$

Proof. Couple the finite-start and stationary augmented-chain remainders as in lemma 5.9, and define the stationary window on the same indices by

$$\tilde{R}^{\mathrm{mis,RR}}_{n,n_0,\mathrm{aug}} := \frac{1}{\sqrt{m}} \sum_{k=n_0}^{n-1} \left(2R^{(\alpha)}_{k,\mathrm{aug}} - R^{(2\alpha)}_{k,\mathrm{aug}}\right).$$

Then, with the notation of (5.59),

$$R^{\mathrm{mis,RR}}_{n,n_0,\mathrm{fin}} = \tilde{R}^{\mathrm{mis,RR}}_{n,n_0,\mathrm{aug}} + \mathcal{U}^{\mathrm{start,RR}}_{n,n_0}$$

under this coupling. Therefore

$$\|R^{\mathrm{mis,RR}}_{n,n_0,\mathrm{fin}}\|_{L_p} \le \|\tilde{R}^{\mathrm{mis,RR}}_{n,n_0,\mathrm{aug}}\|_{L_p} + \|\mathcal{U}^{\mathrm{start,RR}}_{n,n_0}\|_{L_p}.$$

By stationarity, $\|\tilde{R}^{\mathrm{mis,RR}}_{n,n_0,\mathrm{aug}}\|_{L_p} = \|R^{\mathrm{mis,RR}}_{m,\mathrm{aug}}\|_{L_p}$. Apply theorem 4.2 with n replaced by m and lemma 5.10. □

Corollary 5.5. ***(Balanced-scale burned-in misadjustment rate.)*** *Assume the hypotheses of theorem 5.1. Let $\alpha = c\,n^{-1/2}$, $m \ge n/2$, $p = \max\{2, \lceil \log n \rceil\}$, and $q = \max\{4p, \lceil \log(e\,d) \rceil, 2\}$. If, for some fixed $\beta > 0$,*

$$n_0 \ge \frac{\beta p}{c_{\mathrm{st}}\alpha a} \log n,$$

and n is large enough that $2\alpha \le \alpha_{\mathrm{burn}}(p, q)$, then

$$\|R^{\mathrm{mis,RR}}_{n,n_0,\mathrm{fin}}\|_{L_p} \le C_{\mathrm{burn,mis}}\, \mathrm{polylog}(n)\, n^{-1/4}. \tag{5.66}$$

5.10 Burned-in Martingale Berry–Esseen

The depth-zero martingale in lemma 5.6 is normalized at the effective sample size $m = n - n_0$.

Fix $u \in \mathbb{R}^d$ and put $X^{\mathrm{bRR}}_l := u^\top \Delta M^{\mathrm{bRR}}_l$ for $2 \le l \le n-1$. Then

$$|X^{\mathrm{bRR}}_l| \le \kappa_{\mathrm{burn}}(u), \quad \kappa_{\mathrm{burn}}(u) := 6\, t_{\mathrm{mix}}\, C_{\mathrm{burn,Q}}\, \|\varepsilon\|_\infty\, \|u\|. \tag{5.67}$$

Set

$$s^2_{n,n_0}(u) := m\, \sigma^{2,\mathrm{bRR}}_{n,n_0}(u).$$

Under (5.33), $s^2_{n,n_0}(u) \ge m\sigma^2(u)/2$. Also, $m \ge n/2$ and the uniform bound $\|Q^{\mathrm{bRR}}_l\| \le C_{\mathrm{burn,Q}}$ imply the deterministic upper bound

$$s^2_{n,n_0}(u) \le 2m\, C^2_{\mathrm{burn,Q}}\, \|\Sigma^{(\mathrm{M})}_\varepsilon\|\, \|u\|^2. \tag{5.68}$$

Theorem 5.2. ***(Burned-in martingale Berry–Esseen bound.)*** *Assume **UGE 1**, $\pi(\varepsilon) = 0$, $\|\varepsilon\|_\infty < \infty$, $\sigma^2(u) > 0$, $0 < \alpha$, $2\alpha \le \alpha_\infty$, $m \ge n/2$, and the burned-in variance lower-bound condition (5.33). There exist constants $C_{\mathrm{bK},1}(u), C_{\mathrm{bK},2}(u) > 0$, depending only on $\|u\|$, $\sigma(u)$, $C_{\mathrm{burn,Q}}$, t_{mix}, $\|\varepsilon\|_\infty$, $\|\Sigma^{(\mathrm{M})}_\varepsilon\|$, and the universal constants in lemma B.2, such that for every $n \ge 3$,*

$$d_K\left(\frac{u^\top M^{\mathrm{bRR}}_{n,n_0}}{\sqrt{m}\,\sigma^{\mathrm{bRR}}_{n,n_0}(u)}, \mathcal{N}(0,1)\right) \le \frac{C_{\mathrm{bK},1}(u)\, \log^{3/4} n}{m^{1/4}} + \frac{C_{\mathrm{bK},2}(u)\, \log n}{\sqrt{m}}. \tag{5.69}$$

Proof. Apply lemma B.2 to the martingale differences X^{bRR}_l. The bounded-increment input is (5.67), and the target variance is $s^2_{n,n_0}(u)$. The first and third terms of lemma B.2 are bounded exactly as before, with n replaced by m in the denominator and the harmless factor $n/m \le 2$ absorbed into the constants. Explicitly, from $s^2_{n,n_0}(u) \ge m\sigma^2(u)/2$ and $m \ge n/2$,

$$\frac{(2n+1)\log(2n+1)}{s^3_{n,n_0}(u)} \le C(u) \frac{\log n}{\sqrt{m}}. \tag{5.70}$$

The bounded-increment Lindeberg term is controlled in the same way, using (5.68) and $s^2_{n,n_0}(u) \geq m\sigma^2(u)/2$. For the conditional-variance term use lemma 5.7:

$$\mathbb{E}^{1/p}\left[|u^\mathsf{T}\langle M^{\mathrm{bRR}}\rangle_{n,n_0}u - s^2_{n,n_0}(u)|^p\right] \leq C\sqrt{p\,n}.$$

Together with $m \geq n/2$, (5.33), and (5.68), this gives

$$\mathrm{TermII} \leq C(u)\, p^{(3p+1)/(2(2p+1))}\, m^{-p/(2(2p+1))}.$$

Taking $p = \lceil \log n \rceil$ gives the displayed $\log^{3/4}(n)m^{-1/4}$ bound. The bounded-increment terms give the second displayed term. □

5.11 Finite-Window Smoothing Assembly

The finite-start scalar statistic decomposes as

$$T^{\mathrm{RR}}_{n,n_0}(u) = -\frac{u^\mathsf{T} M^{\mathrm{bRR}}_{n,n_0}}{\sqrt{m}} + \mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u), \tag{5.71}$$

where the composite non-Gaussian remainder is

$$\mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u) := D^{\mathrm{RR}}_{\mathrm{tr},n,n_0}(u) + \mathcal{I}^{\mathrm{init,RR}}_{n,n_0}(u) + u^\mathsf{T} D^{\mathrm{bRR}}_{2,n,n_0} + u^\mathsf{T} R^{\mathrm{mis,RR}}_{n,n_0,\mathrm{fin}}. \tag{5.72}$$

Let $\mathcal{B}_{\mathrm{mis}}(m, n_0, p, q, \alpha)$ denote the right-hand side of (5.65).

Lemma 5.11. ***(Burned-in composite remainder bound.)*** *Assume the hypotheses of lemma 5.2, lemma 5.4, lemma 5.6, and theorem 5.1. Then, for every $p \geq 2$ and every $q \geq 2$ satisfying $p \leq q/4$ and $2\alpha \leq \alpha_{\mathrm{burn}}(p,q)$,*

$$\|\mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u)\|_{L_p} \leq \|u\| \left(\mathcal{B}_{\mathrm{start}}(m,n_0,p,\alpha) + \frac{C_{\mathrm{burn,D2}}}{\sqrt{m}} + \mathcal{B}_{\mathrm{mis}}(m,n_0,p,q,\alpha)\right), \tag{5.73}$$

where

$$\begin{aligned}\mathcal{B}_{\mathrm{start}}(m,n_0,p,\alpha) &:= \frac{\|\theta_0 - \theta^*\|}{\alpha a\sqrt{m}}\left[5\sqrt{\kappa_Q}\,(1-\alpha a)^{n_0/2} + C_{\mathrm{init,RR}}\, p\, \exp\left(-c_{\mathrm{init}}\alpha a n_0/p\right)\right],\\ C_{\mathrm{burn,D2}} &:= 3\, t_{\mathrm{mix}}\, \|\varepsilon\|_\infty \left(C_{\mathrm{burn,Q}} + \frac{C_{\mathrm{burn,V}}}{a^2}\right).\end{aligned}$$

Proof. Apply the triangle inequality in (5.72). The deterministic part is (5.20). The random initial-product term is (5.27). The Poisson boundary term is bounded by

$$\|u^\mathsf{T} D^{\mathrm{bRR}}_{2,n,n_0}\|_{L_p} \leq \|u\|\,\|D^{\mathrm{bRR}}_{2,n,n_0}\|_\infty \leq \frac{C_{\mathrm{burn,D2}}\|u\|}{\sqrt{m}}$$

using (5.36). The last term is exactly (5.65). □

Dividing (5.71) by the finite-window normalization gives

$$\Xi^{\mathrm{bRR}}_{n,n_0}(u) = X^{\mathrm{bRR}}_{n,n_0} + Y^{\mathrm{bRR}}_{n,n_0}, \tag{5.74}$$

where

$$X^{\mathrm{bRR}}_{n,n_0} := -\frac{u^\mathsf{T} M^{\mathrm{bRR}}_{n,n_0}}{\sqrt{m}\,\sigma^{\mathrm{bRR}}_{n,n_0}(u)}, \quad Y^{\mathrm{bRR}}_{n,n_0} := \frac{\mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u)}{\sigma^{\mathrm{bRR}}_{n,n_0}(u)}.$$

Theorem 5.3. ***(Finite-window deterministic-start burned-in PR-averaged RR Berry–Esseen bound.)*** *Assume the hypotheses of theorem 5.2, lemma 5.4, and theorem 5.1. Let* $m := n - n_0$ *and* $p = \max\{2, \lceil \log n \rceil\}$ *and* $q = \max\{4p, \lceil \log(e\, d) \rceil, 2\}$*. Then* $p \geq 2$*,* $q \geq 2$*, and* $p \leq q/4$*. If* $(n, n_0, p, q, \alpha, u)$ *is in the admissible burn-in regime* (5.34)*, then*

$$d_K\left(\Xi^{\mathrm{bRR}}_{n,n_0}(u), \mathcal{N}(0,1)\right) \leq \frac{C_{\mathrm{bK},1}(u)\, \log^{3/4} n}{m^{1/4}} + \frac{C_{\mathrm{bK},2}(u)\, \log n}{\sqrt{m}} + \frac{e\, \|\mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u)\|_{L_p}}{\sqrt{2\pi}\, \sigma^{\mathrm{bRR}}_{n,n_0}(u)} + \frac{e}{n}, \tag{5.75}$$

with the composite remainder bounded by lemma 5.11. The bound is uniform over $\xi = \mathcal{L}(Z_0)$*.*

Proof. Apply the smoothing inequality (4.153) to the split (5.74). The martingale Berry–Esseen term is theorem 5.2; the minus sign in X^{bRR}_{n,n_0} is handled by applying that theorem to the signed increments $-u^\mathsf{T} \Delta M^{\mathrm{bRR}}_l$. The bounded-increment constant and predictable quadratic variation are unchanged. Since $p = \max\{2, \lceil \log n \rceil\}$, the smoothing tail e^{-p} is at most e/n. The L_p norm of the perturbation is

$$\|Y^{\mathrm{bRR}}_{n,n_0}\|_{L_p} = \frac{\|\mathcal{R}^{\mathrm{bRR}}_{n,n_0,\mathrm{fin}}(u)\|_{L_p}}{\sigma^{\mathrm{bRR}}_{n,n_0}(u)},$$

which gives (5.75). □

5.12 Balanced Burn-in Berry–Esseen Bound

The final inference statement uses $\sigma(u)$ instead of $\sigma^{\mathrm{bRR}}_{n,n_0}(u)$.

Lemma 5.12. ***(Burned-in normalization transfer.)*** *Assume* $\sigma^2(u) > 0$ *and the burned-in variance lower-bound condition* (5.33)*. Put*

$$r_{n,n_0}(u) := \frac{\sigma^{\mathrm{bRR}}_{n,n_0}(u)}{\sigma(u)}.$$

Then

$$\frac{1}{\sqrt{2}} \leq r_{n,n_0}(u) \leq \sqrt{3/2},$$

and, for any real random variable W*,*

$$d_K\left(r_{n,n_0}(u) W, \mathcal{N}(0,1)\right) \leq d_K\left(W, \mathcal{N}(0,1)\right) + \frac{C_{\mathrm{norm}} C_{\mathrm{burn},3} \|u\|^2}{m\, \alpha\, a\, \sigma^2(u)}, \tag{5.76}$$

where C_{norm} *is a universal constant.*

Proof. The variance lower-bound condition and (5.32) give

$$|\sigma^{2,\mathrm{bRR}}_{n,n_0}(u) - \sigma^2(u)| \leq \sigma^2(u)/2.$$

Hence $r_{n,n_0}(u) \in [1/\sqrt{2}, \sqrt{3/2}]$. For r in this compact interval,

$$\sup_x |\Phi(x/r) - \Phi(x)| \leq C_{\mathrm{norm}}\, |r - 1|,$$

because $\sup_x |x\phi(x)| < \infty$. Therefore

$$d_K(rW, \mathcal{N}(0,1)) \leq d_K(W, \mathcal{N}(0,1)) + C_{\mathrm{norm}}\, |r - 1|.$$

Finally,

$$|r_{n,n_0}(u) - 1| = \frac{|\sigma^{2,\mathrm{bRR}}_{n,n_0}(u) - \sigma^2(u)|}{\sigma(u)\, (\sigma^{\mathrm{bRR}}_{n,n_0}(u) + \sigma(u))} \leq \frac{C_{\mathrm{burn},3} \|u\|^2}{m\, \alpha\, a\, \sigma^2(u)},$$

using (5.32) and $\sigma^{\mathrm{bRR}}_{n,n_0}(u) + \sigma(u) \geq \sigma(u)$. □

Theorem 5.4. ***(Balanced-scale deterministic-start burned-in PR-averaged RR Berry–Esseen bound.)*** *Assume Assumptions 1–3 from section 1.4. Assume also the Lyapunov contraction* (A.3) *for the two step sizes used below and the external inputs and local extensions summarized in chapter B. Fix $c > 0$ and set*

$$\alpha := c\,n^{-1/2}, \quad m := n - n_0, \quad p := \max\{2, \lceil \log n \rceil\}, \quad q := \max\{4p, \lceil \log(e\,d) \rceil, 2\}.$$

Then $p \geq 2$, $q \geq 2$, and $p \leq q/4$. Use the deterministic-start admissibility threshold $\alpha_{\mathrm{burn}}(p, q)$ defined in (5.1)*. Suppose $(n, n_0, p, q, \alpha, u)$ is in the admissible burn-in regime* (5.34)*. Assume also that the burn-in satisfies the explicit mixing-scale conditions with logarithmic factors*

$$n_0 \geq \frac{2}{\alpha a} \log n, \quad n_0 \geq \frac{p}{c_{\mathrm{init}} \alpha a} \log n, \quad n_0 \geq \frac{p}{c_{\mathrm{st}} \alpha a} \log n. \tag{5.77}$$

Then there exists a finite constant $C_{\mathrm{burn,final}}(u, c, \theta_0)$, depending only on u, c, $\|\theta_0 - \theta^*\|$, and the problem and universal constants in the preceding bounds, but not on ξ, such that

$$d_K\left(\Xi^{\mathrm{bRR}}_{n,n_0}(u), \mathcal{N}(0,1)\right) \leq \frac{C_{\mathrm{burn,final}}(u, c, \theta_0)\,\mathrm{polylog}(n)}{n^{1/4}}, \tag{5.78}$$

and

$$d_K\left(\Xi^{\mathrm{asy,RR}}_{n,n_0}(u), \mathcal{N}(0,1)\right) \leq \frac{C_{\mathrm{burn,final}}(u, c, \theta_0)\,\mathrm{polylog}(n)}{n^{1/4}}. \tag{5.79}$$

Proof. Apply theorem 5.3. Since $m \geq n/2$, the martingale terms satisfy

$$\frac{\log^{3/4} n}{m^{1/4}} + \frac{\log n}{\sqrt{m}} \leq C\,\frac{\mathrm{polylog}(n)}{n^{1/4}}.$$

The composite remainder in lemma 5.11 is controlled by corollary 5.1, corollary 5.2, (5.36), and corollary 5.5. In particular,

$$\|R^{\mathrm{mis,RR}}_{n,n_0,\mathrm{fin}}\|_{L_p} \leq C\,\mathrm{polylog}(n)\,n^{-1/4}.$$

Together with $\sigma^{\mathrm{bRR}}_{n,n_0}(u) \geq \sigma(u)/\sqrt{2}$, the smoothing remainder is $O(\mathrm{polylog}(n) n^{-1/4})$, and (5.78) follows.

For the asymptotic normalization, write

$$\Xi^{\mathrm{asy,RR}}_{n,n_0}(u) = r_{n,n_0}(u)\,\Xi^{\mathrm{bRR}}_{n,n_0}(u).$$

By lemma 5.12, the additional cost is at most

$$\frac{C\,\|u\|^2}{m\,\alpha\,a\,\sigma^2(u)} = O(n^{-1/2}),$$

which is absorbed into the balanced finite-window rate. □

The remaining non-elementary large-n requirement is

$$2cn^{-1/2} \leq \alpha_{\mathrm{burn}}(p, q), \tag{5.80}$$

with p, q as in the theorem.

Corollary 5.6. ***($\sqrt{n}$-normalization for the burned-in RR statistic.)*** *Under the assumptions of theorem 5.4 and the admissible burn-in regime* (5.34)*, in particular $m = n - n_0 \geq n/2$, $p \leq q/4$, $\sigma^2(u) > 0$, $2\alpha \leq \alpha_{\mathrm{burn}}(p, q)$, and the burned-in variance lower bound* (5.33)*, define the final scalar statistic*

$$T^{\mathrm{RR,n}}_{n,n_0}(u) := \sqrt{n}\,u^\top\left(\bar{\theta}^{(\mathrm{RR},\alpha)}_{n,n_0} - \theta^*\right) = \sqrt{n/m}\,T^{\mathrm{RR}}_{n,n_0}(u),$$

and its asymptotic normalization

$$\Xi^{\mathrm{n,RR}}_{n,n_0}(u) := \frac{T^{\mathrm{RR,n}}_{n,n_0}(u)}{\sigma(u)}.$$

For the Berry–Esseen moment choice $p = \max\{2, \lceil \log n \rceil\}$ *in theorem 5.4, the lower burn-in conditions* (5.77) *are implied by*

$$n_0 \geq C_- (\alpha a)^{-1} \log^2 n$$

with any fixed C_- *large enough. If, in addition, the burn-in window stays in the same mixing-scale window with logarithmic-square factor,*

$$C_- (\alpha a)^{-1} \log^2 n \leq n_0 \leq C_0 (\alpha a)^{-1} \log^2 n \tag{5.81}$$

for some finite C_0 *and such a fixed* C_-*, then there exists a finite constant*

$$C_{\mathrm{burn,n}}(u, c, \theta_0, C_0, C_-),$$

independent of ξ*, such that*

$$d_K \left(\Xi^{\mathrm{n,RR}}_{n,n_0}(u), \mathcal{N}(0,1) \right) \leq \frac{C_{\mathrm{burn,n}}(u, c, \theta_0, C_0, C_-) \operatorname{polylog}(n)}{n^{1/4}}. \tag{5.82}$$

Proof. Put $s_{n,n_0} := \sqrt{n/m}$. Since (5.34) gives $m \geq n/2$,

$$0 \leq s_{n,n_0} - 1 = \frac{n_0}{m\,(s_{n,n_0} + 1)} \leq \frac{2n_0}{n}.$$

For every real random variable W and every $s \in [1, \sqrt{2}]$,

$$d_K (sW, \mathcal{N}(0,1)) \leq d_K (W, \mathcal{N}(0,1)) + C\,|s - 1|,$$

by the same scaling argument as in lemma 5.12. Applying this with $W = \Xi^{\mathrm{asy,RR}}_{n,n_0}(u)$ and using theorem 5.4 gives the already proved $\operatorname{polylog}(n) n^{-1/4}$ term. The upper burn-in bound in (5.81) and $\alpha = cn^{-1/2}$ give

$$|s_{n,n_0} - 1| \leq \frac{2C_0}{ca} \frac{\log^2 n}{n^{1/2}},$$

which is lower order and is absorbed into the same balanced-scale rate. □

5.13 Chapter Summary and Results

The chapter transfers the stationary augmented-chain result to deterministic initial conditions. It controls the burned-in deterministic weights, transient terms, random initial products, startup discrepancies, and the finite-window normalization, culminating in the balanced burn-in Berry–Esseen bound.

Chapter 6

Numerical Experiments

6.1 Goals and scope

The purpose of this chapter is to compare the finite-sample behavior of the constant-stepsize Richardson–Romberg estimator with standard alternatives for Markovian linear stochastic approximation. The experiments are not used in the proofs of theorems 4.3 and 5.4. Instead, they serve three complementary roles:

- They check whether stepsize Richardson–Romberg extrapolation reduces the point-estimator bias at the sample sizes where confidence intervals are constructed.
- They compare the resulting confidence intervals with diminishing-stepsize and Polyak–Ruppert baselines.
- They isolate the effect of practical long-run variance estimators, in particular overlapping batch means (OBM) and its lugsail variant.

The last point is deliberately experimental in this thesis. The theoretical chapters above prove Berry–Esseen and burn-in transfer results for the RR-averaged estimator itself. A detailed non-asymptotic theory for OBM and lugsail covariance estimation along RR-averaged constant-stepsize LSA trajectories is left for future work.

6.2 Experimental setup

The main comparison experiment follows the finite-state Markovian LSA setup used in Huo, Chen, and Xie (2024). For each randomly generated problem, the Markov chain has 10 states and the parameter dimension is $d = 5$. The reported numbers aggregate 100 independent problems, with 100 Monte Carlo trajectories per problem and trajectory length $T = 10^6$.

The methods compared in the main table differ in two ways: the estimator used for the center of the interval and the estimator used for the long-run variance.

The main comparison reports three metrics. The L2 error is the Euclidean norm of the final estimation error, aggregated over problems; it reflects both finite-sample variance and bias. The CI width is the length of the two-sided scalar confidence interval in the sampled projection direction. Coverage is the empirical probability that this scalar interval contains the projected target $u^\top \theta^*$. Thus the coverage numbers are one-dimensional random-direction coverages; they are not coordinatewise coverages and not simultaneous multivariate coverages. L2 errors and CI widths are shown in units of 10^{-3}.

Table 6.1. Main comparison experiment setup.

Quantity	**Value used in the main comparison**
Problem class	Finite-state Markovian linear stochastic approximation
Number of problems	100 independently generated problems
Trajectories per problem	100
Dimension and states	$d = 5$ and 10 Markov states
Trajectory length	$T = 10^6$
Problem generation	`eig_min = 0.25`, `eig_max = 0.60`, `noise_target = 0.35`, `A_norm` cap `1.0`
Diminishing-stepsize schedule	$c_0 = 200$, $k_0 = 20000$, $\gamma = 0.65$
Constant stepsizes for RR	$\alpha \in \{0.2, 0.02\}$
OBM and OBM-LW block sizes	$b_n = \lfloor T^{0.6} \rfloor = 3981$; OBM-LW uses $\lambda = 2$, hence the two windows $b_n = 3981$ and $2b_n = 7962$
Projection for confidence intervals	One random unit direction is sampled for each problem; coverage is reported for this scalar direction, not for the first coordinate.

Table 6.2. Methods compared in the main experiment.

Method	**Description**
$\alpha = 0.2$ constant	Constant-stepsize averaged LSA with the larger stepsize branch.
$\alpha = 0.02$ constant	Constant-stepsize averaged LSA with the smaller stepsize branch.
RR constant stepsizes	Richardson–Romberg combination of the two constant-stepsize averages, using $\alpha \in \{0.2, 0.02\}$.
Diminishing $0.2/\sqrt{k}$	Classical diminishing-stepsize baseline with $\alpha_k = 0.2/\sqrt{k}$.
PR + OBM	Polyak–Ruppert averaging with OBM long-run variance estimation.
RR + OBM	RR point estimator with OBM long-run variance estimation.
RR + OBM-LW	RR point estimator with OBM long-run variance estimation using a lugsail window.

6.3 Long-run variance estimation: OBM and lugsail

The main theorems above are stated either with a deterministic finite-window variance proxy or with the asymptotic variance $\sigma^2(u) = u^\top \Sigma_\infty u$. In an actual confidence interval this quantity must be estimated from the observed dependent trajectory. The naive sample variance of the iterates estimates the marginal variance $\mathrm{Var}(Y_t)$, not the long-run variance $\sigma^2(u)$, and therefore misses the serial-correlation correction. OBM is used precisely because it targets this long-run variance.

For a scalar projected trajectory $Y_t = u^\top \theta_t$, the overlapping batch means (OBM) estimator with block size b is

$$\widehat{\sigma}^2_{\mathrm{OBM}}(b) = \frac{b}{T-b+1} \sum_{s=0}^{T-b} \left(\overline{Y}_{s,b} - \overline{Y}_T\right)^2, \qquad \overline{Y}_{s,b} = \frac{1}{b} \sum_{j=s}^{s+b-1} Y_j. \tag{6.1}$$

Equivalently, OBM is the batch-means form of the Bartlett-window estimator of the spectral density at frequency zero. This is the appropriate object for estimating the time-average covariance in Markov chain CLTs; see Flegal and Jones (2010) and Liu, Vats, and Flegal (2022).

The price is a window bias. For Bartlett/OBM estimators the standard asymptotic template is

$$\mathbb{E}\widehat{\sigma}^2_{\mathrm{OBM}}(b) = \sigma^2(u) + \frac{c_1(u)}{b} + \text{lower order terms}, \tag{6.2}$$

and the MSE has the characteristic bias–variance tradeoff $c_1(u)^2/b^2 + C(u)b/T$. For positively correlated output, the leading bias is typically negative, so confidence intervals based on the raw OBM estimator can be too narrow.

Lugsail windows are designed to reduce this leading window bias of the variance estimator (Vats and Flegal, 2022). In the implementation used here, the lugsail-window OBM estimator, abbreviated OBM-LW, is

$$\widehat{\sigma}^2_{\text{OBM-LW}}(b, \lambda) = \frac{\lambda}{\lambda - 1} \widehat{\sigma}^2_{\mathrm{OBM}}(\lambda b) - \frac{1}{\lambda - 1} \widehat{\sigma}^2_{\mathrm{OBM}}(b), \qquad \lambda > 1. \tag{6.3}$$

For the Bartlett kernel, $\lambda = 2$ cancels the leading $1/b$ bias term. This correction is conceptually separate from Richardson–Romberg extrapolation in the stepsize. RR in α targets the stochastic approximation bias of the point estimator $\overline{\theta}^{(\mathrm{RR},\alpha)}$; the lugsail window in b targets the window bias of the long-run variance estimator. Thus stepsize extrapolation and lugsail-window variance estimation operate on different quantities and are not substitutes for each other.

The lugsail estimator is a signed linear combination of two OBM estimators. Consequently, in finite samples the scalar estimate can be negative, and the matrix version can fail to be positive semidefinite. Any implementation that uses lugsail covariance matrices must therefore report how often this occurs or specify a clipping/projection convention.

In this thesis, OBM and lugsail are investigated experimentally only. A theorem for OBM/lugsail covariance estimation along RR-averaged constant-stepsize LSA trajectories is not proved here and is left for future work. That future analysis should combine the OBM and lugsail theory above with finite-sample spectral-density corrections in the spirit of Ng and Perron (1996) and with recent batch-means methodology for stochastic-gradient inference such as Singh, Shukla, and Vats (2025).

6.4 Main comparison

The table below gives a compact summary of the completed main comparison run. The entries are medians over the 100 generated problems and focus on the methods most directly relevant to the thesis message.

The expected behavior is as follows. A single large constant stepsize should produce short intervals but can leave substantial steady-state bias. A single small constant stepsize reduces this bias only partially at a fixed horizon. The Richardson–Romberg combination is designed to cancel the leading stepsize bias while keeping the long-run variance target essentially the same. Therefore the desirable empirical signature is lower L2 error and near-nominal coverage without a major increase in interval width.

The main comparison confirms this pattern.

Table 6.3. Main comparison results at $T = 10^6$.

Method	L2	CI width	Coverage
$\alpha = 0.2$ constant	26.67	8.36	0.5%
$\alpha = 0.02$ constant	13.93	8.27	40.5%
RR constant stepsizes	4.52	8.22	94.0%
Diminishing $0.2/\sqrt{k}$	6.80	10.34	90.0%
PR + OBM	5.35	8.04	92.0%
RR + OBM	4.52	8.23	95.0%
RR + OBM-LW	4.52	8.22	95.0%

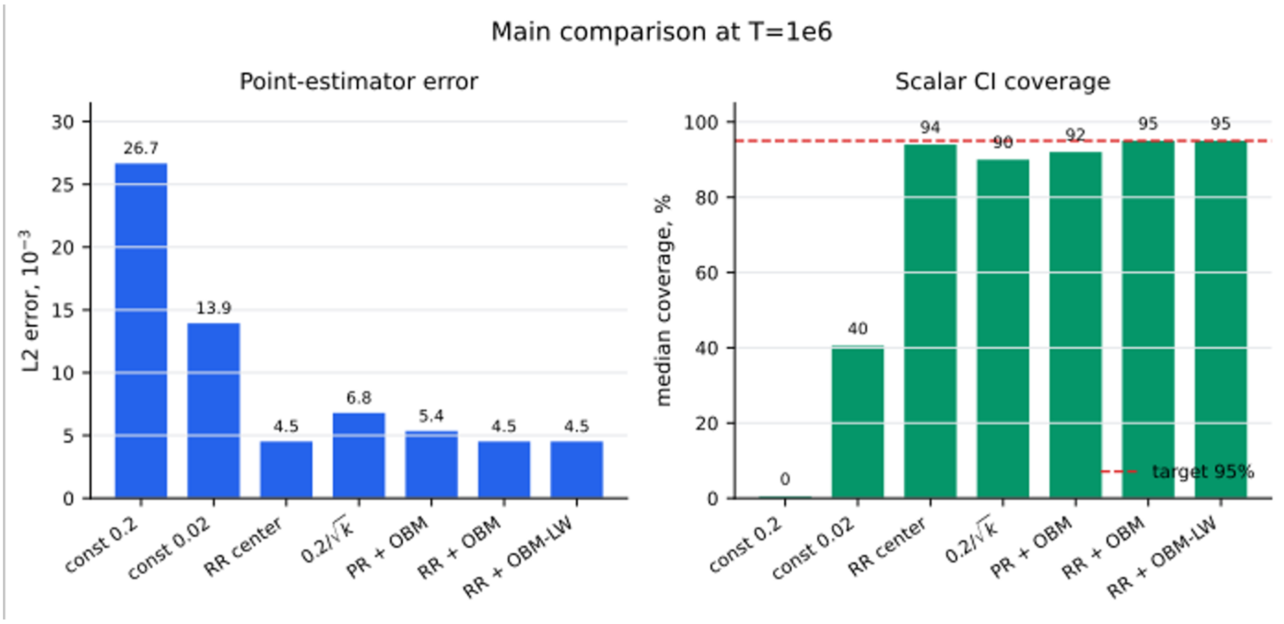

Figure 6.1. Main comparison at $T = 10^6$. The left panel shows median Euclidean error and the right panel shows median scalar coverage across the same 100 generated problems.

The RR combination reduces the median L2 error to $4.52 \cdot 10^{-3}$ and brings the median scalar coverage close to the nominal 95% level. Its interval widths are comparable to the single-stepsize intervals, so the coverage gain is not obtained by simply making intervals wider.

The single constant-stepsize branches undercover because the centers of the intervals are biased. This is most visible for $\alpha = 0.2$, but it remains substantial for $\alpha = 0.02$. Changing the variance estimator cannot repair this failure: OBM, MSB, or OBM-LW can adjust the estimated uncertainty around the center, but they do not move the biased center itself.

In this long-horizon run, OBM-LW is essentially neutral. At $T = 10^6$, the OBM window bias is already small enough that the lugsail correction does not visibly improve coverage. The remaining differences in CI width between RR+OBM and RR+OBM-LW are within the scale of Monte Carlo variation in this comparison.

6.5 Theory-aligned RR stepsize sweep

The main comparison above uses the wide practical pair $\alpha \in \{0.2, 0.02\}$. To check that the effect is not specific to this wide pair, a separate sweep repeats the comparison for the adjacent two-level pairs $(2\alpha, \alpha)$ with $\alpha \in \{0.02, 0.05, 0.10\}$. This is the stepsize geometry closest to the first-order Richardson–Romberg expansion used in the theory.

The setup is the same finite-state LSA experiment as in the main comparison: 100 problems, 100 trajectories per problem, $T = 10^6$, $d = 5$, 10 Markov states, and a random scalar projection direction per problem. The same problem seeds are used across the three rows. The table reports medians over problems; L2 errors and CI widths are again in units of 10^{-3}.

Table 6.4. Theory-aligned adjacent RR stepsize sweep.

RR pair	**Single-α coverage**	**RR L2**	**RR width**	**RR coverage**
$(0.04, 0.02)$	0.04: 92.0%; 0.02: 94.0%	2.97	5.36	94.0%
$(0.10, 0.05)$	0.10: 86.0%; 0.05: 91.5%	2.97	5.37	94.0%
$(0.20, 0.10)$	0.20: 67.5%; 0.10: 86.0%	2.97	5.38	94.0%

The sweep shows that the RR gain is not an artifact of one particular wide-pair tuning. As the larger single-alpha branch moves from 0.04 to 0.20, its median coverage deteriorates from 92.0% to 67.5%. After Richardson–Romberg extrapolation, however, the median L2 error, CI width, and coverage are essentially unchanged across the three adjacent pairs.

The OBM and OBM-LW versions of the RR intervals give the same qualitative message. In this sweep, RR+OBM and RR+OBM-LW have median coverage between 94.0% and 95.0%, with width changes below about one percent. Thus this experiment reinforces the separation between the two corrections: RR in α removes the point-estimator bias, while OBM and OBM-LW in b only modify the estimated long-run variance.

6.6 Oracle-variance diagnostic

The theory-aligned sweep still uses estimated long-run variances. To separate variance-estimation error from point-estimator bias and normal-approximation error, we repeat the largest adjacent-pair experiment, $(0.20, 0.10)$, with an oracle interval based on the analytic finite-state value $\sigma^2(u) = u^\top \Sigma_\infty u$. The interval center is the same RR-averaged estimator in every row.

Table 6.5. Oracle-variance diagnostic for the $(0.20, 0.10)$ RR pair.

Method	**L2**	**CI width**	**Coverage**	**Width/oracle**
RR + batch means	2.97	5.38	94.0%	0.990
RR + oracle variance	2.97	5.43	95.0%	1.000
RR + OBM	2.97	5.42	95.0%	0.998
RR + OBM-LW	2.97	5.39	95.0%	0.993
RR + MSB	2.97	5.36	95.0%	0.987

The oracle interval is nearly indistinguishable from the practical variance-estimator intervals. OBM is only about 0.2% narrower than the oracle interval at the median, and OBM-LW is about 0.7% narrower. The median coverage remains 95% for the oracle, OBM, OBM-LW, and MSB rows. Thus, at $T = 10^6$, long-run-variance estimation is not the bottleneck for the RR confidence intervals. The remaining coverage error is already at the scale of Monte Carlo variation across the 100 generated problems.

6.7 Coverage over trajectory length

We next repeat the oracle-variance comparison over $T \in \{2 \cdot 10^4, 5 \cdot 10^4, 10^5, 3 \cdot 10^5, 10^6\}$, using the same RR pair $(0.20, 0.10)$ and the same problem seeds across horizons. The goal is to identify whether undercoverage at shorter horizons comes from the RR center and normal approximation, or from estimating the long-run variance.

Table 6.6. Oracle and data-driven coverage over trajectory length.

T	**Oracle cov.**	**OBM cov.**	**OBM-LW cov.**	**OBM/oracle**	**OBM-LW/oracle**
$2 \cdot 10^4$	95.5%	94.0%	93.0%	0.942	0.944
$5 \cdot 10^4$	95.0%	94.0%	93.0%	0.967	0.958
10^5	95.0%	94.0%	94.0%	0.978	0.972
$3 \cdot 10^5$	95.0%	94.0%	94.0%	0.989	0.989
10^6	95.0%	95.0%	95.0%	0.998	0.993

The oracle coverage is already close to the nominal level at all horizons. This suggests that, in this problem class, the RR center and the normal approximation are not the main cause of the short-horizon coverage loss. The gap is instead in the practical variance estimators: for example, at $T = 2 \cdot 10^4$ the OBM interval is about 5.8% narrower than the oracle interval, and its median coverage is 94.0% instead of 95.5%. This gap shrinks monotonically in the width ratio as T grows.

With the default block-size rule $b = \lfloor T^{0.6} \rfloor$, OBM-LW does not improve coverage over OBM in this sweep. It is neutral at the largest horizon and slightly lower at the smaller horizons. Therefore lugsail should not be presented as an automatic coverage fix. Rather, together with the bias–variance diagnostic below, the evidence says that lugsail is a variance-estimator bias-reduction device whose practical benefit depends on the block-size regime.

6.8 Block-size sweep for coverage

To test this block-size dependence directly, we repeat the RR coverage experiment for $b = \lfloor T^{\eta} \rfloor$, $\eta \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$, and $T \in \{2 \cdot 10^4, 10^5, 10^6\}$. The RR center is fixed at the adjacent pair $(0.20, 0.10)$, so the differences across rows are caused by the long-run variance estimator. The figure plots the most informative slice of this grid.

This sweep resolves the apparent tension between the default-rule coverage sweep and the lugsail bias–variance diagnostic. Lugsail improves coverage when OBM is still dominated by negative Bartlett-window bias, as shown in figure 6.2. For example, at $T = 2 \cdot 10^4$ and $\eta = 0.5$, OBM-LW changes the median relative bias from -0.222 to -0.022, the width/oracle ratio from 0.878 to 0.979, and median coverage from 91.5% to 95.0%. At $T = 10^5$, OBM-LW with $\eta = 0.4$ or $\eta = 0.5$ reaches 95.0% median coverage, while OBM at the same block sizes remains too narrow.

At the production rule $\eta = 0.6$, OBM is already close to the oracle width in this problem class. The lugsail correction is therefore neutral rather than beneficial. For very large blocks, however, OBM-LW becomes unstable: at $\eta = 0.8$ the signed lugsail estimate is negative in 14.77% of trajectory-level estimates at $T = 2 \cdot 10^4$, 6.71% at $T = 10^5$, and 1.74% at $T = 10^6$. Thus lugsail should be reported together with negative/clamped-estimate diagnostics, not only with the final clamped CI width.

6.9 Conditioning and noise stress test

The conditioning/noise stress test checks whether the block-size conclusion survives moderate changes in the problem generator. The run keeps the same RR pair $(0.20, 0.10)$ and the same random finite-state Markov-chain generator, but varies the eigenvalue range of $-\overline{A}$ and the target norm of the state-dependent matrix noise. Thus this is a conditioning and matrix-noise stress test, not yet a mixing-time stress test.

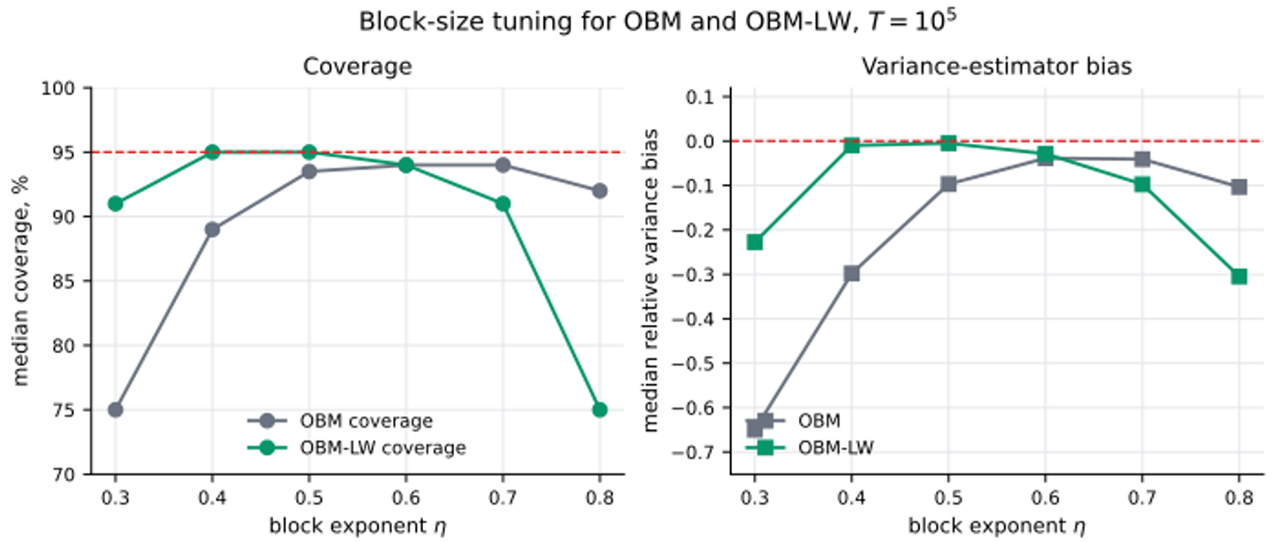

Figure 6.2. Block-size sensitivity of OBM and OBM-LW at $T = 10^5$. The left panel shows median scalar coverage, and the right panel shows median relative bias of the corresponding long-run variance estimator.

The main conclusion is short: the oracle coverage is 95% in every moderate conditioning/noise scenario, which indicates that the RR center and the normal approximation remain adequate for these stress levels. Weaker mean contraction increases the scale of the problem: at $T = 10^6$, the median L2 error is $6.04 \cdot 10^{-3}$ in the weak-mean scenario, compared with $2.97 \cdot 10^{-3}$ at baseline. Nevertheless, OBM-LW at $\eta = 0.5$ stays near the oracle width and has median coverage 95%.

The smaller block-size row $\eta = 0.4$ makes the OBM window bias more visible. At $T = 10^6$, OBM coverage is 91% in the weak-mean scenario and 92% in the weak-plus-noise scenario, while OBM-LW restores 95% coverage in both cases. The corresponding raw variance-estimator bias changes from -0.252 to -0.004 in the weak-mean scenario and from -0.208 to approximately zero in the weak-plus-noise scenario. This supports the same interpretation as the block-size sweep: lugsail helps when OBM is too narrow because of negative Bartlett-window bias, but the useful block-size range must still be checked.

6.10 Lugsail bias–variance diagnostic

A separate lugsail bias–variance experiment isolates the covariance estimator itself. It compares OBM and OBM-LW over a grid of block sizes and uses the analytic value of $\sigma^2(u)$ as ground truth.

In the short-horizon regime, lugsail helps because the raw OBM estimator has visible negative Bartlett-window bias. For PR iterates, OBM-LW with $\lambda = 2$ reduces the leading window bias and shifts the useful block-size range downward. At $T = 10^5$, the best OBM point has relative bias about -2.5% and MSE 0.72, while OBM-LW with $\lambda = 2$ has relative bias about -0.7% and MSE 0.34. Across $T \in \{20000, 50000, 100000\}$, the observed MSE reduction is roughly 40–55%.

For constant-stepsize and RR iterate paths, the improvement is more limited because part of the error is SA transient or point-estimator bias, not Bartlett-window bias. Lugsail can correct the leading window bias of the variance estimator, but it cannot correct this separate stochastic approximation bias component.

These experiments support the following interpretation. Step-size RR is the primary tool for improving the center of the confidence interval. OBM is the baseline practical estimator of Σ_∞. OBM-LW is useful when the long-run-variance estimator has visible negative window bias, mainly in shorter-horizon or more persistent regimes.

6.11 Limitations and future diagnostics

The current experiments support the qualitative message of the theory, but they do not yet cover all diagnostics needed for a complete empirical study. The following diagnostics would make the empirical study more complete by separating theorem validation, variance-estimation accuracy, and robustness to problem conditioning.

Table 6.7. Suggested extensions for the empirical study.

Extension	**Purpose**
Burn-in and initialization sweep	Varying n_0, θ_0, and the initial law of Z_0 would test the deterministic-start transfer and the practical size of the $(\alpha a)^{-1} \log^2 n$ burn-in window.
Mixing-aware tuning	Additional runs should vary horizon, burn-in, and stepsize as functions of the spectral gap or t_{mix} to check how slow mixing affects oracle and data-driven coverage.
OBM and OBM-LW theory and tuning	A separate study should analyze OBM and lugsail-window OBM along RR-averaged constant-stepsize LSA trajectories, including non-asymptotic window-bias bounds, block-size selection, and the regimes where the signed OBM-LW correction improves or destabilizes coverage.
Matrix covariance diagnostics	The present coverage results are scalar random-direction diagnostics. Additional runs should check several directions and the full estimated covariance matrix, including positive-semidefiniteness of lugsail corrections.

The present experiment set already includes a theory-aligned stepsize sweep for three adjacent pairs, an oracle T-sweep, a block-size coverage sweep, and conditioning/noise stress tests for the largest adjacent pair. The most important remaining diagnostics are therefore mixing-aware tuning and matrix-valued covariance checks: the current evidence is still scalar-directional.

6.12 Chapter Summary and Results

The experiments support the theoretical interpretation: step-size Richardson–Romberg extrapolation primarily improves the interval center by reducing bias, while OBM and lugsail OBM affect the estimated long-run variance. The diagnostics also show that the benefit of lugsail correction is block-size and dependence-regime dependent.

Chapter 7

Conclusion

The thesis develops a distributional analysis for Richardson–Romberg averaged constant-stepsize linear stochastic approximation under Markovian noise. The work starts from the usual LSA recursion and isolates the effect of coupling two step sizes, α and 2α, in the Polyak–Ruppert average. The main mathematical contribution is a proof strategy that combines deterministic RR weight bounds, Poisson martingale reduction for Markovian noise, predictable-variance comparison, and explicit control of the misadjustment terms.

The stationary part of the analysis gives a Berry–Esseen assembly for scalar RR statistics under the augmented-chain convention. At the balanced scale $\alpha = cn^{-1/2}$, the result identifies the covariance target $\Sigma_\infty = \overline{A}^{-1}\Sigma_\varepsilon^{(M)}\overline{A}^{-\top}$ and gives a non-asymptotic normal approximation up to logarithmic factors. The deterministic-start part then transfers the stationary theorem to the practical burned-in estimator by controlling deterministic transients, initial random products, startup discrepancies, and finite-window normalization errors.

The numerical experiments are consistent with the theoretical picture. Step-size Richardson–Romberg extrapolation mainly improves the center of the confidence interval by reducing constant-stepsize bias, whereas OBM and lugsail OBM affect the long-run variance estimate. The experiments also show that lugsail corrections should be interpreted as variance-estimator bias-reduction tools whose effect depends on the block size and dependence regime.

Further work should complete three directions. First, the current scalar Berry–Esseen theory should be extended to more explicitly multivariate confidence regions. Second, the OBM and lugsail covariance estimators should be analyzed non-asymptotically along RR-averaged constant-stepsize LSA trajectories. Third, the dependence on the mixing time and the Lyapunov contraction constants should be sharpened so that the theory gives more direct practical guidance for choosing burn-in lengths, step sizes, and block sizes.

Chapter 8

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Appendix A

Key Quantities and Basic Objects

The *Markovian noise covariance matrix* captures both the marginal variance and the temporal correlations of the noise:

$$\Sigma_{\varepsilon}^{(\mathrm{M})} = \mathbb{E}_{\pi}[\varepsilon(Z_0)\varepsilon(Z_0)^{\mathsf{T}}] + \sum_{\ell=1}^{\infty} \left(\mathbb{E}_{\pi}[\varepsilon(Z_0)\varepsilon(Z_\ell)^{\mathsf{T}}] + \mathbb{E}_{\pi}[\varepsilon(Z_\ell)\varepsilon(Z_0)^{\mathsf{T}}]\right). \tag{A.1}$$

The series is absolutely convergent under Assumptions 1 and 3. Indeed, for centered bounded ε, $\|\mathsf{Q}^{\ell}\varepsilon\|_\infty \leq 2\|\varepsilon\|_\infty (1/4)^{\lfloor \ell/t_{\mathrm{mix}}\rfloor}$, and hence

$$\|\mathbb{E}_{\pi}[\varepsilon(Z_0)\varepsilon(Z_\ell)^{\mathsf{T}}]\| \leq \|\varepsilon\|_\infty \|\mathsf{Q}^{\ell}\varepsilon\|_\infty \leq 2\|\varepsilon\|_\infty^2 (1/4)^{\lfloor \ell/t_{\mathrm{mix}}\rfloor}.$$

The same bound applies to the transposed covariance term, so the covariance series converges absolutely in operator norm. This matrix is the limiting covariance in the Markov chain CLT for the partial sums $n^{-1/2}\sum_{t=0}^{n-1}\varepsilon(Z_t)$ (cf. Douc et al., 2018, Theorem 21.2.10).

The *asymptotically optimal covariance matrix* is given by

$$\Sigma_\infty = \overline{A}^{-1}\Sigma_{\varepsilon}^{(\mathrm{M})}\overline{A}^{-\mathsf{T}}. \tag{A.2}$$

This is the covariance target attained by the averaged linearized recursion. We call it optimal in the usual averaged-SA sense; a full Hájek–Le Cam optimality statement would require an additional local-asymptotic experiment argument, which is not part of this thesis.

The *Lyapunov equation* plays a central role in the contraction analysis. For any $P = P^{\mathsf{T}} \succ 0$, there exists a unique $Q = Q^{\mathsf{T}} \succ 0$ satisfying $\overline{A}^{\mathsf{T}}Q + Q\overline{A} = P$. Defining $a = \lambda_{\min}(P)/(2\|Q\|)$ and $\kappa_Q = \lambda_{\max}(Q)/\lambda_{\min}(Q)$, the key contraction property holds: for all $\alpha \in [0, \alpha_\infty]$,

$$\|I - \alpha\overline{A}\|_Q^2 \leq 1 - \alpha a. \tag{A.3}$$

Since the iterates $\theta_k^{(\alpha)}$ alone are generally not Markovian (due to the Markovian noise), we consider the *joint process* $(\theta_k^{(\alpha)}, Z_{k+1})$ with kernel

$$\overline{\mathsf{P}}_\alpha f(\theta, z) = \int_{\mathsf{Z}} \mathsf{Q}(z, dz') f(F_z(\theta), z'),$$

where $F_z(\theta) = (I - \alpha A(z))\theta + \alpha b(z)$. Thus the current second coordinate z is the observation used to update θ, and the next coordinate z' is carried forward to the following step. Under Assumptions 1–3, this joint chain admits a unique invariant distribution Π_α for sufficiently small $\alpha > 0$ (Levin et al., 2025).

Appendix B

External Inputs and Local Extensions

B.1 Direct Imported Working Forms

This appendix records the external statements in the exact working forms used in the thesis. Items explicitly marked as direct citations are imported from the cited papers after translating to the sign convention of this thesis. Items marked as local extensions are proved in the main text and should not be read as separate external theorems.

Lemma B.1. ***(Direct citation: Levin et al. (2025, Lemma 11) – Markov concentration.)*** *Under **UGE 1**, there is a universal constant C_{MC} such that, for any bounded measurable functions g_i with $\pi(g_i) = 0$ and $\|g_i\|_\infty \le c_i$, every initial distribution ξ, and every $p \ge 2$,*

$$\|\sum_{i=1}^{N} g_i(Z_i)\|_{L_p(\xi)} \le C_{\mathrm{MC}} \sqrt{p\, t_{\mathrm{mix}} \sum_{i=1}^{N} c_i^2}. \tag{B.1}$$

This is the scalar time-inhomogeneous consequence used in the predictable-variation estimates and in the preliminary last-iterate bounds.

Lemma B.2. ***(Direct citation: Samsonov et al. (2025, Lemma 21) – martingale Berry–Esseen.)*** *Let $(X_l, \mathcal{F}_l)_{l=1}^N$ be scalar martingale differences such that $|X_l| \le \kappa$ almost surely. Define $S_N := \sum_{l=1}^N X_l$ and $V_N^2 := \sum_{l=1}^N \mathbb{E}[X_l^2 | \mathcal{F}_{l-1}]$. Then, for every deterministic scale $v_N^2 > 0$ and every $p \ge 1$,*

$$\begin{aligned} d_K\left(S_N/v_N, \mathcal{N}(0,1)\right) \le L_B(\kappa) \frac{(2N+1)\log(2N+1)}{v_N^3} &+ C_1 \sqrt{p} v_N^{-2p/(2p+1)} (\mathbb{E}|V_N^2 - v_N^2|^p)^{1/(2p+1)} \\ &+ C_2 p v_N^{-2p/(2p+1)} \kappa^{2p/(2p+1)}. \end{aligned} \tag{B.2}$$

Here $L_B(\kappa) < \infty$ and C_1, C_2 are the constants in that martingale theorem.

Lemma B.3. ***(Direct citation: Levin et al. (2025, Proposition 2) – stationary bias of $J^{(1)}$.)*** *Under stationarity for the augmented chain $(Z_{t+1}, J_t^{(0,w)}, J_t^{(1,w)})$ and for $w \le \alpha_{\mathrm{L,P2}}(q, t_{\mathrm{mix}})$,*

$$\mathbb{E}_\pi[J_\infty^{(1,w)}] = w\,\Delta + R(w), \quad \|R(w)\| \le 12\, \|\overline{A}^{-1}\|\, C_A^2\, t_{\mathrm{mix}}^2\, w^2\, \|\varepsilon\|_\infty,$$

with

$$\Delta := \overline{A}^{-1} \sum_{k \ge 1} \mathbb{E}_\pi[(\mathsf{Q}^k \tilde{A})(Z_0)\, \varepsilon(Z_0)].$$

Lemma B.4. ***(Direct citation: Levin et al. (2025, Corollary 6) – centered bilinear L_p bound.)*** *Define*

$$\overline{\psi}_w(j,z) := \tilde{A}(z) j - \mathbb{E}_{\Pi_{J^{(0)},w}}[\tilde{A}(Z_1)\, J_0^{(0,w)}].$$

For $w \le \alpha_{\mathrm{L,C6}}(q, t_{\mathrm{mix}})$, every initial distribution, every $r \ge 1$, and every $p \ge 2$,

$$\left(\| \sum_{t=0}^{r-1} \overline{\psi}_w(J_t^{(0,w)}, Z_{t+1}) \| \right)_{L_p} \le c_{W,1}\, p^{3/2} \sqrt{wr} + c_{W,2}\, p^3\, w^{-1/2}\, \log^{1/2}(1/(wa)),$$

with $c_{W,1}, c_{W,2}$ depending only on $C_A, \kappa_Q, t_{\mathrm{mix}}, \|\varepsilon\|_\infty$. The logarithmic factor is the one displayed in Levin et al. (2025, Corollary 6, Eq. (67)); it is absorbed into $\operatorname{polylog}(n)$ *in the working-scale corollaries.*

Lemma B.5. ***(Direct citation: Levin et al. (2025, Proposition 8) – depth-two $J^{(2)}$ moment bound.)*** *For every $q \ge 2$ and every p satisfying $2 \le p \le q/2$, if $w \le \alpha_{\mathrm{L,P8}}(q, t_{\mathrm{mix}})$, then*

$$\|J_n^{(2,w)}\|_{L_p} \le D_J\, t_{\mathrm{mix}}^{5/2}\, p^{7/2}\, w^{3/2}\, \log^{3/2}(1/(wa)),$$

uniformly in n, with D_J depending only on $C_A, \kappa_Q, \|\overline{A}^{-1}\|, \|\varepsilon\|_\infty$.

Lemma B.6. ***(Direct citation/extraction: Durmus et al. (2025, Proposition 7) – random-product stability.)*** *Under the stability and bounded-noise assumptions of the depth-two setup, there are product-stability ceilings $\alpha_{\mathrm{prod}}(r)$, $r \ge 2$, and constants $C_{\mathrm{prod}} < \infty$, $c_{\mathrm{prod}} > 0$, independent of p, w, s, k, such that, for $p \ge 2$, $w \le \alpha_{\mathrm{prod}}(p)$, every $0 \le s \le k$, and every deterministic vector v,*

$$\mathbb{E}^{\frac{1}{p}}\left[\|\Gamma_{s+1:k}^{(w)} v\|^p | Z_s \right] \le C_{\mathrm{prod}} \exp\left(-\frac{c_{\mathrm{prod}} wa(k-s)}{p} \right) \|v\|. \tag{B.3}$$

This is the constant-step deterministic-vector form used in Levin et al. (2025, Appendix D.1, Proposition 9); the displayed exponent is weaker than the source bound and is sufficient for the burn-in transfer.

Lemma B.7. ***(Direct citation: Levin et al. (2025, Proposition 9) – one-trajectory $H^{(2)}$ moment bound.)*** *For every $q \ge 2$ and every p satisfying $2 \le p \le q/2$, if $w \le \alpha_{\mathrm{L,P9}}(q, t_{\mathrm{mix}})$, then*

$$\|H_n^{(2,w)}\|_{L_p} \le D_H\, d^{1/q}\, t_{\mathrm{mix}}^{5/2}\, p^{7/2}\, w^{3/2}\, \log^{3/2}(1/(wa)),$$

uniformly in n, with D_H depending only on $C_A, \kappa_Q, \|\overline{A}^{-1}\|, \|\varepsilon\|_\infty$. The proof uses the representation

$$H_k^{(2,w)} = -w \sum_{l=1}^{k} \Gamma_{l+1:k}^{(w)} \tilde{A}(Z_l) J_{l-1}^{(2,w)}. \tag{B.4}$$

Lemma B.8. ***(Direct citation/extraction: Levin et al. (2025, Appendix B.2, Proposition 5) – depth-two startup contraction for the J coordinates.)*** *For*

$$Y_k^{(w)} := (Z_{k+1}, J_k^{(0,w)}, J_k^{(1,w)}, J_k^{(2,w)})$$

and $y = (z, j_0, j_1, j_2)$, $y' = (z', j_0', j_1', j_2')$, define the depth-two augmented-chain cost from Levin et al. (2025, Appendix B.2, Eq. (49)):

$$\begin{aligned} c_{J,2}^{(w)}(y, y') := {} & \|j_0 - j_0'\| + \|j_1 - j_1'\| + \|j_2 - j_2'\| \\ & + \left(\|j_0\| + \|j_0'\| + \|j_1\| + \|j_1'\| + \|j_2\| + \|j_2'\| + \sqrt{wa}\|\varepsilon\|_\infty \right) 1_{z \neq z'}. \end{aligned} \tag{B.5}$$

Under the Proposition-5 step-size restriction $w \le \alpha_{\mathrm{L,P5}}(p, t_{\mathrm{mix}})$, two copies of $Y_k^{(w)}$ started from deterministic states y, y' can be coupled so that, for $\ell = 0, 1, 2$,

$$\|J_k^{(\ell,w)}(y) - J_k^{(\ell,w)}(y')\|_{L_p} \le C_W\, p^{7/2}\, t_{\mathrm{mix}}^{5/2}\, \log^{3/2}(1/(wa)) \exp\left(-wak/(12p)\right) c_{J,2}^{(w)}(y, y'). \tag{B.6}$$

The displayed componentwise estimate is the coordinate projection of the Wasserstein contraction, with constants as in Levin et al. (2025, Eq. (55)).

Lemma B.9. ***(Direct citation: Levin et al. (2025, Corollary 4) – invariant depth-two J law.)*** *Under the same stationary depth-two ceilings as above, the Markov chain $Y_k^{(w)}$ admits a unique invariant law $\Pi_{J,2,w}$. Its stationary version has finite moments at the orders used in lemma B.5, and the contraction in lemma B.8 gives convergence to this law from every deterministic initial augmented state.*

B.2 Local Extensions Used in This Thesis

The burn-in transfer also uses one product-stability input and two local consequences:

- lemma 5.3 derives the conditional/adapted-vector version needed in this thesis from the deterministic-vector estimate lemma B.6. Its conditional display (5.24) is the only product-stability form used at random coupling times; (5.25) gives the deterministic-time corollary used for finite-past and initial-product estimates.

The following statements are local extensions or assemblies, not direct citations from Levin et al. or Samsonov et al.:

- lemma 4.6 constructs the stationary full depth-two augmented state including $H^{(2)}$ by a finite-past limit. It uses lemmas B.7 to B.9.
- lemma 5.8 proves conditional product stability at a random coupling time from the deterministic product-stability working form lemma 5.3 and the coupling-time tail.
- lemma 5.9 proves the full-state startup contraction for $J^{(1)} + J^{(2)} + H^{(2)}$ by combining the J contraction lemma B.8 with the local random-time product estimate.

Whenever the main proof invokes these local statements, the invoked result is the local lemma named above, not an unquoted external proposition.